Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
3	Emotions	593	72	6	1.48	0.16
4	Scene	2407	294	6	1.25	0.11
5	PlantPseAAC	978	440	12	6.69	0.68
6	Water-quality	1060	16	14	1.77	0.29
7	Yeast	2417	103	14	7.20	1.82
8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 1: (Table 3) Datasets used in the revised experiments. N, P, K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **LGBM run on `enron`.** Sequentially training (2–3h per task due to K=53 pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}_{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form <order>-<target>-<heig
order $\in$ {PA = partial order, PR = pre-order}; *target* $\in$ {H = Hamming, S = Subset 0/1}; *height* $\in$ {2, ∞ (omitted in the label)}.

- 1 = **PA-H-2**: partial order, Hamming target, height ≤ 2.

- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2.
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2.
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2.
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper's Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey "x" on the axis when the method has no data at all for that panel).

3 Paper results (RF base learner)

Reproduces the layout of the original paper's main experiment tables (Tables 5–7): 3 datasets across, 4 metrics down, one table per prediction type (binary vector, partial abstention, score vector). Per the revision, $f^{\mathrm{ham}}_{\mathrm{MLC}}$ and $f^{\mathrm{sub}}_{\mathrm{MLC}}$ are dropped in favour of $f^{\mathrm{Jacc}}_{\mathrm{MLC}}$.

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
3	Emotions	593	72	6	1.48	0.16
4	Scene	2407	294	6	1.25	0.11
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7	Yeast	2417	103	14	7.20	1.82
8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 2: (Table 3) Datasets used in the revised experiments. N, P, K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

3.1 Standard MLC predictions (BinaryVector)

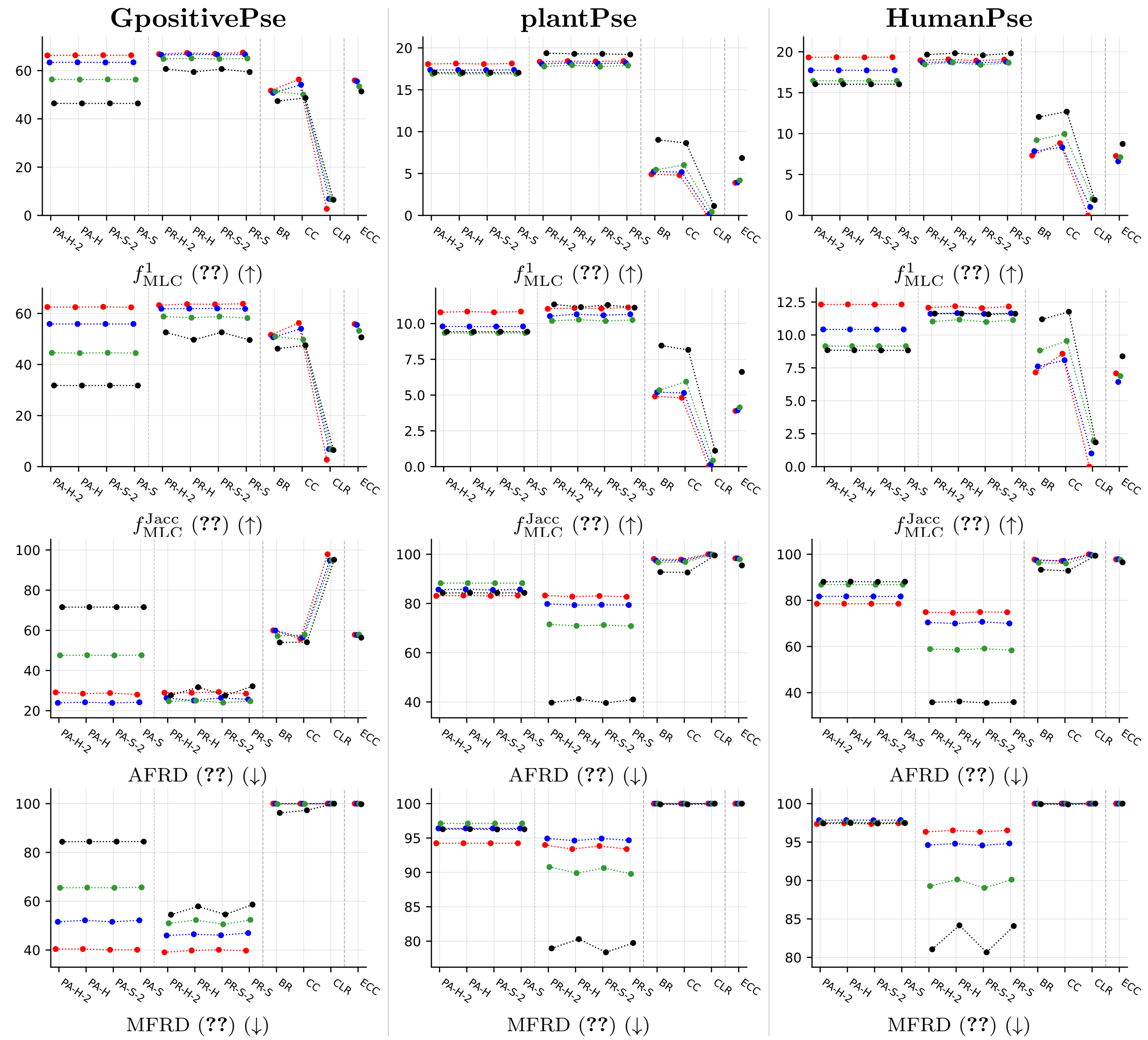


Table 3: (Table 4) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the classifiers (x-axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S), preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S), and standard MLC baselines (BR, CC, CLR). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.2 Partial abstention (PartialAbstention)

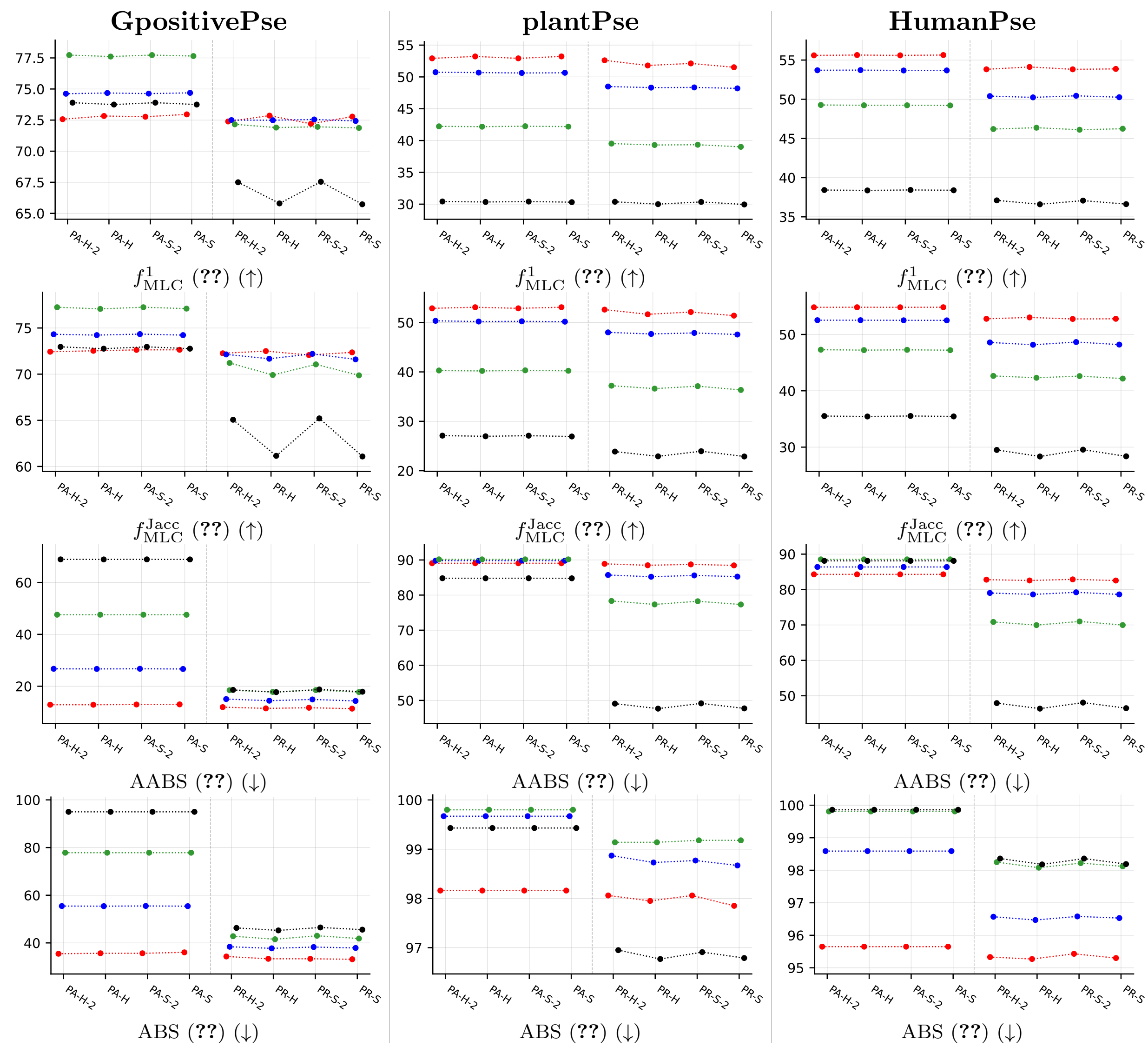
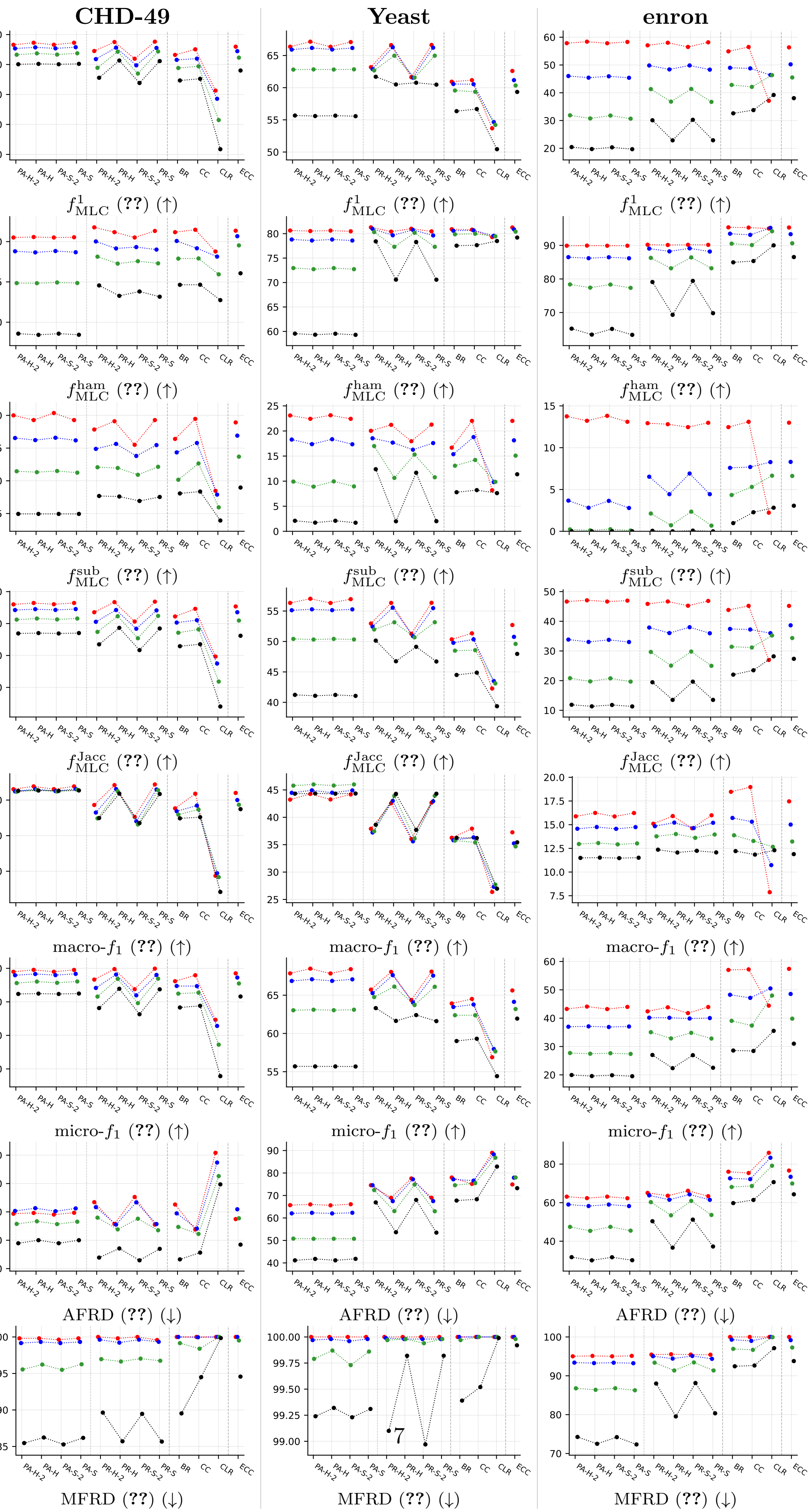


Table 4: (Table 5) Average scores (in %, y-axis) over 5×5 cross-validation folds, plotted against the classifiers (x-axis): partial-order predictors (PA-H-2, PA-H, PA-S-2, PA-S) and preorder predictors (PR-H-2, PR-H, PR-S-2, PR-S). Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

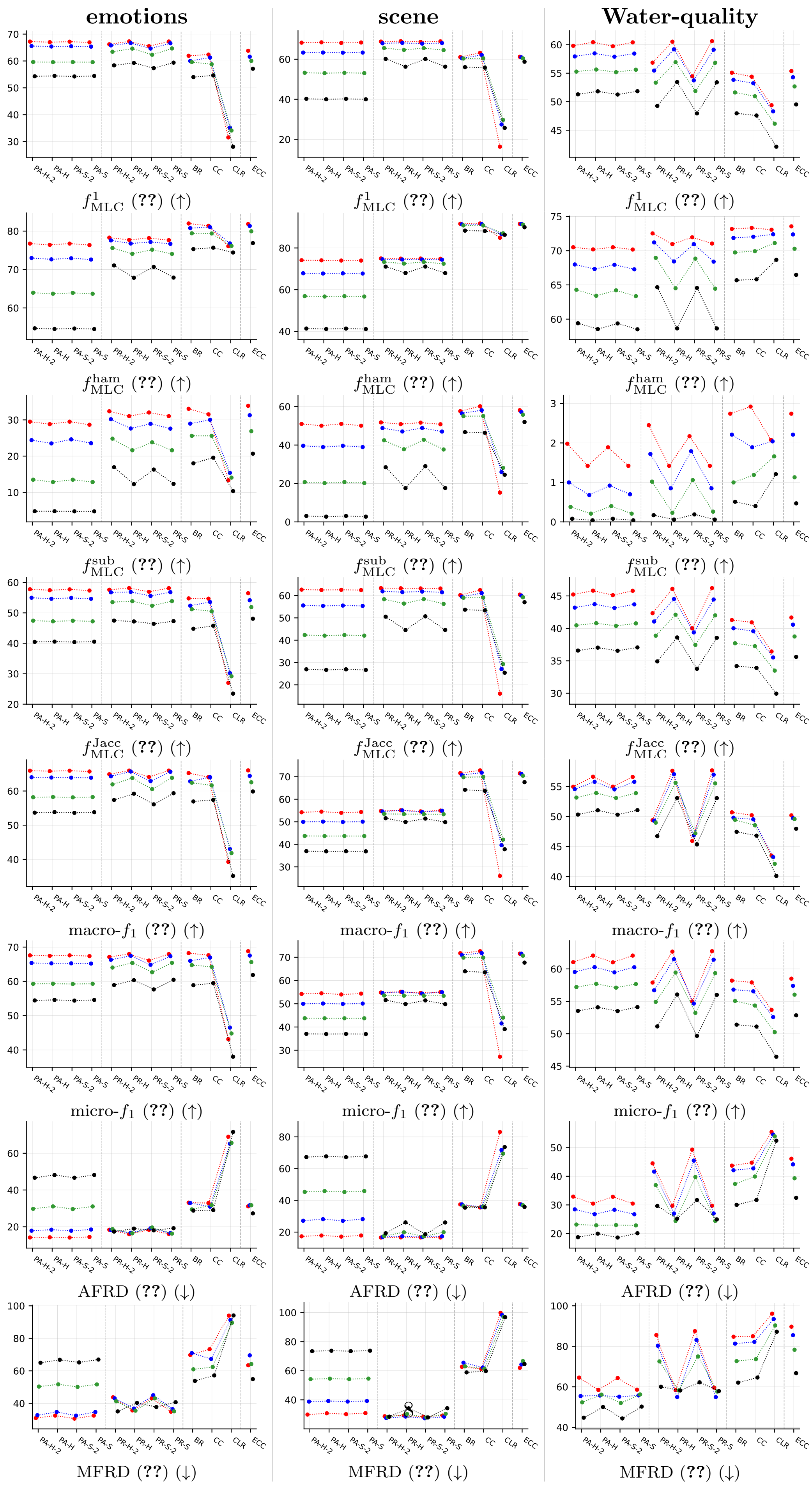
3.3 Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables All tables share the same 8-metric set (original 5 + 3 revision additions: $f^{\mathrm{Jacc}}_{\mathrm{MLC}}$, macro-$f_1$, micro-$f_1$). G2/G4 = imbalanced datasets, G3/G5 = balanced datasets (split follows the original paper's appendix). G6/G7 = main-paper datasets (GpositivePse, PlantPse, HumanPse) shown with the full metric set; the main body only carries the revised compact Tables 4/5.

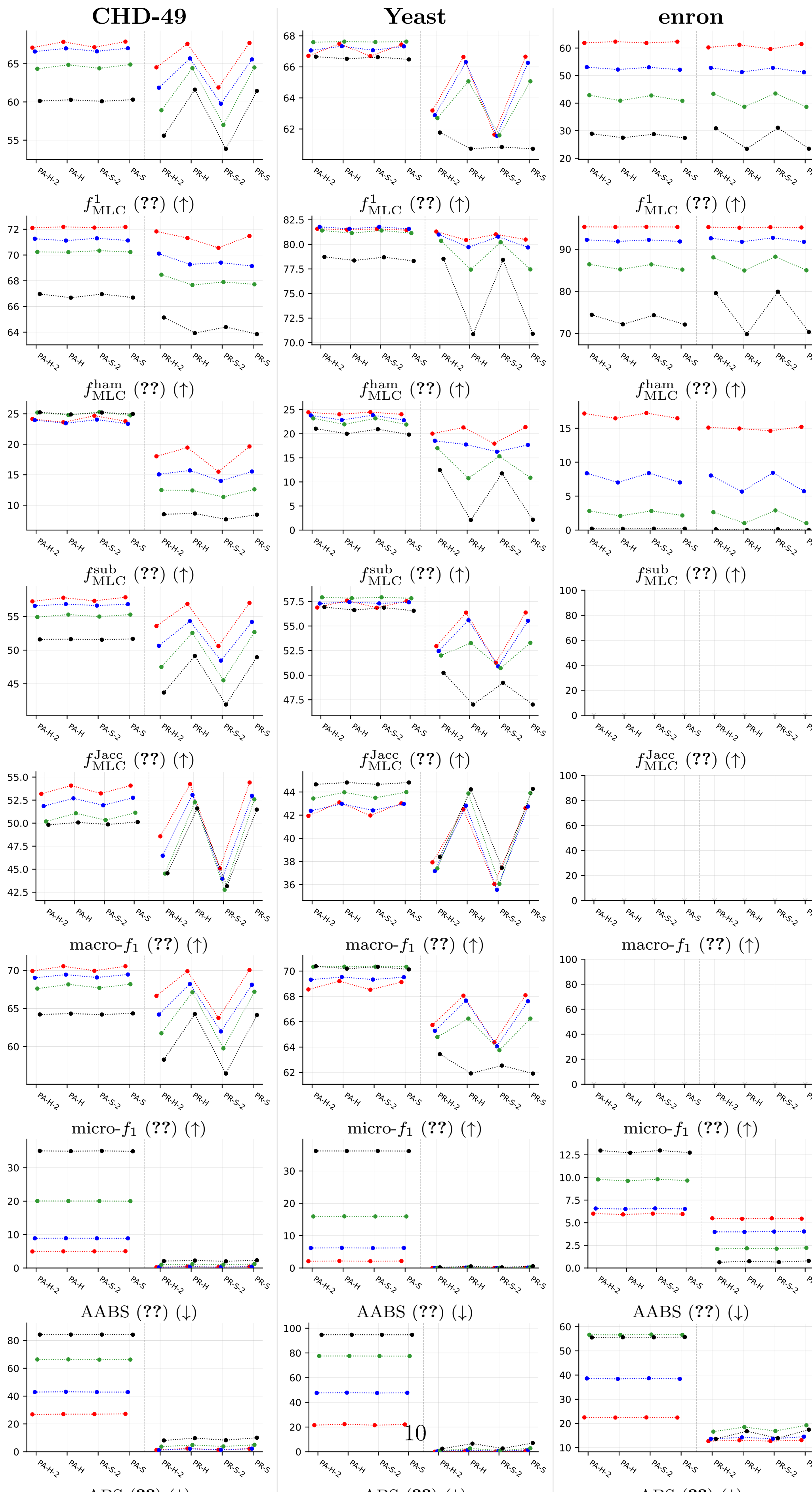
3.3.1 Standard MLC predictions (BinaryVector)

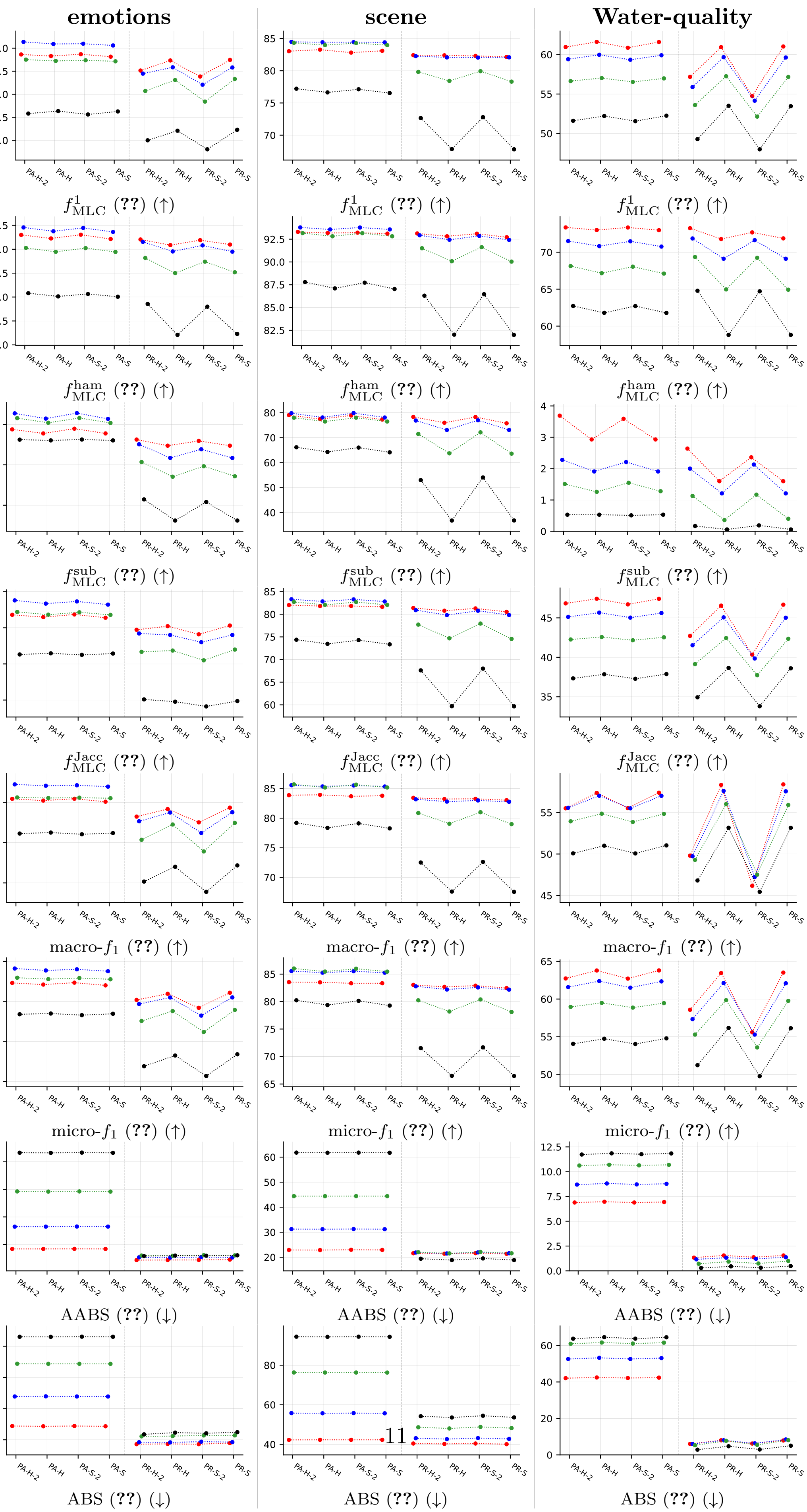


CHD-49
Yeast
enron
f^1_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC

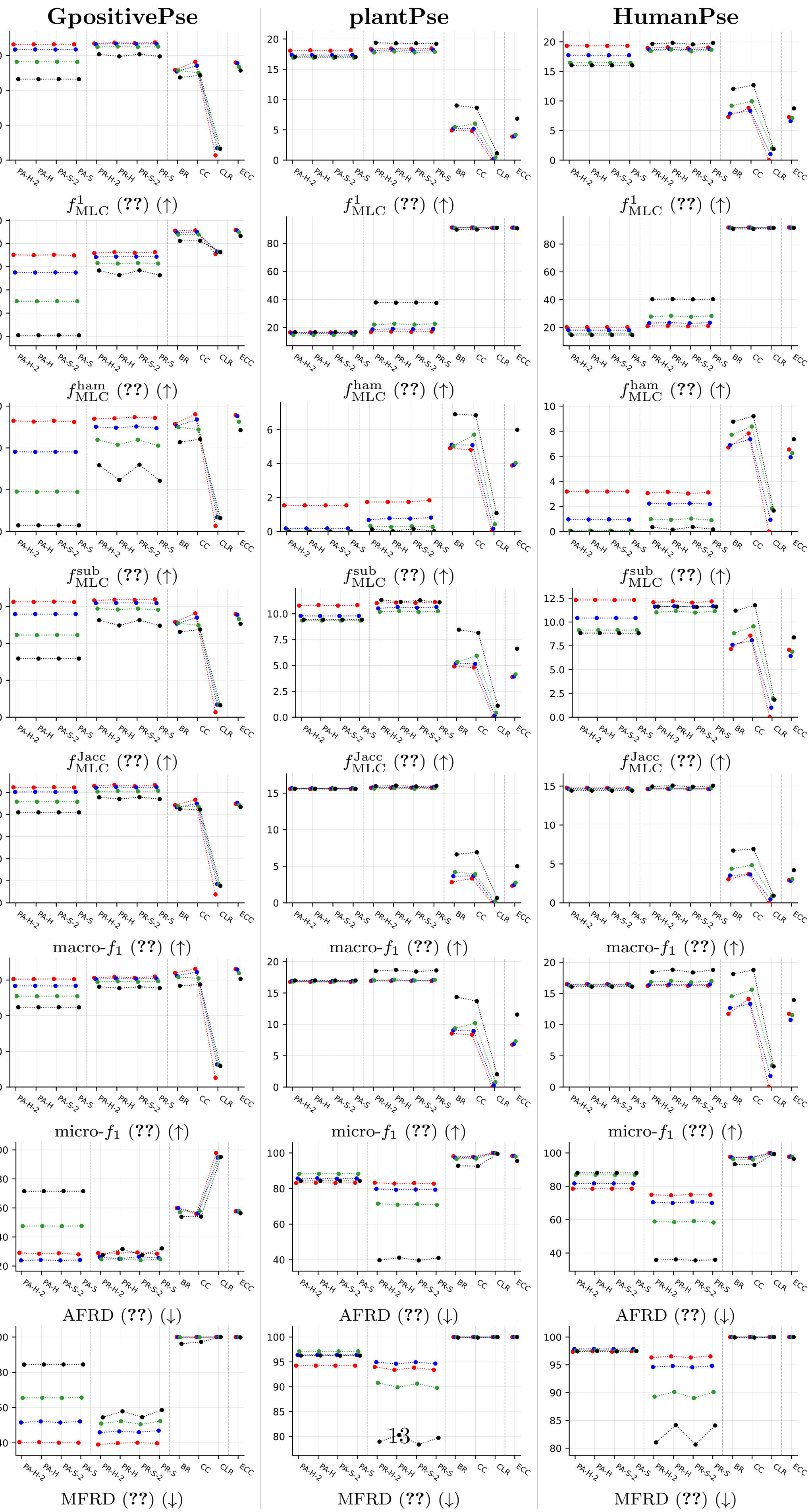

emotions
scene
Water-quality
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC
f^1_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)

3.3.2 Partial abstention (PartialAbstention)

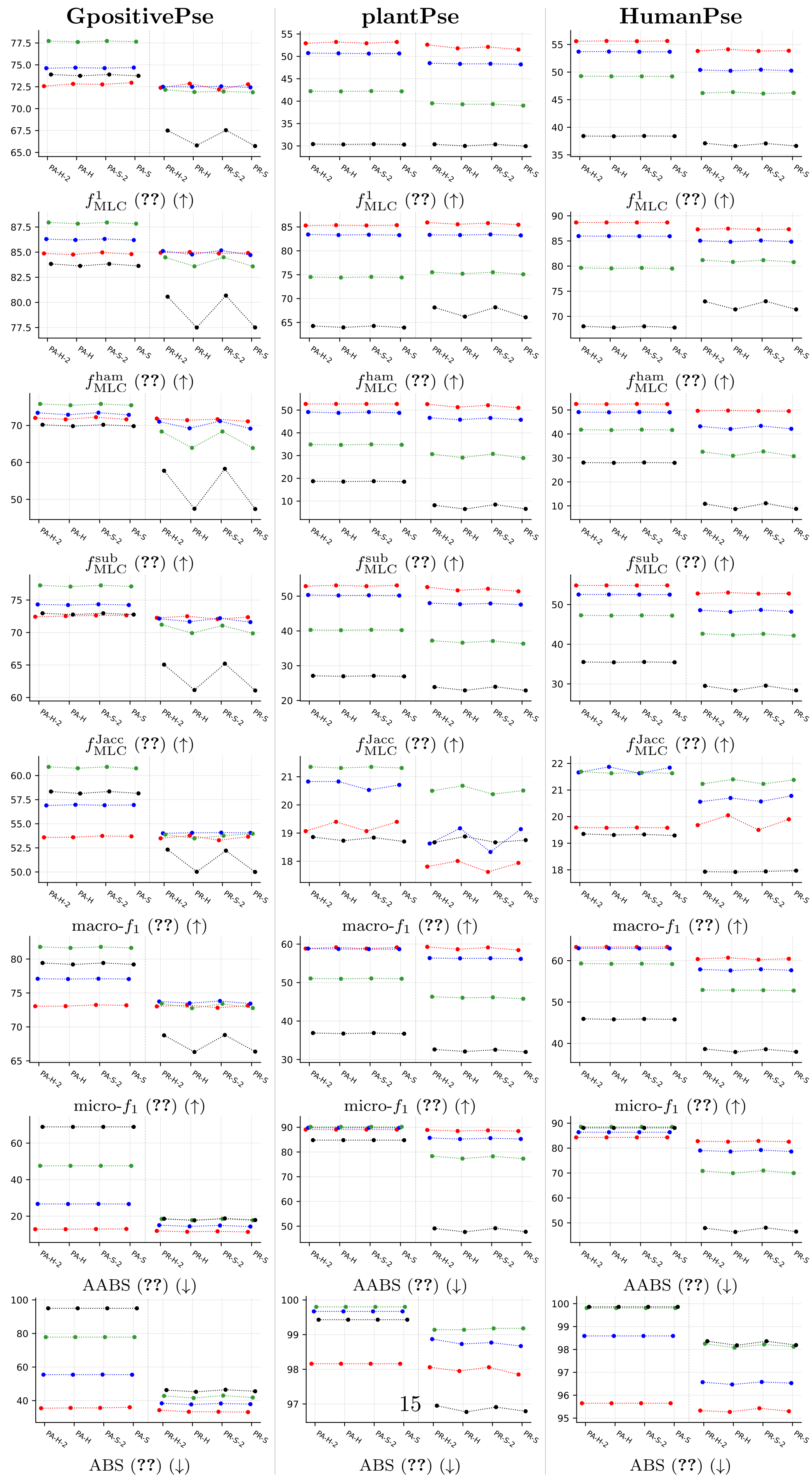

CHD-49
Yeast
enron
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)


emotions
scene
Water-quality
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)

3.3.3 Standard MLC predictions (BinaryVector)


GpositivePse
plantPse
HumanPse
PA-H-2 PA-H PA-S-2 PA-S PR-H-2 PR-H PR-S-2 PR-S BR CC CLR ECC
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)

3.3.4 Partial abstention (PartialAbstention)


GpositivePse
plantPse
HumanPse
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

3.4 Abstention overview: aggregate trends

Purpose. This compact section shows the headline abstention-vs-standard-MLC trend across noise levels, averaged over all datasets, for both base learners (RF, LGBM) and both retained quality metrics (f_1, Jaccard). Use it as a quick visual summary before diving into the detailed per-method / per-dataset figures in Section 5.

How to read. In each panel: blue bar = standard MLC (no abstention) on its native quality metric; coloured bar = with-abstention quality on retained labels. Green "$+x.y$" label = score-point gain of abstention over the standard-MLC bar of the same method. The four sub-panels per figure correspond to $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

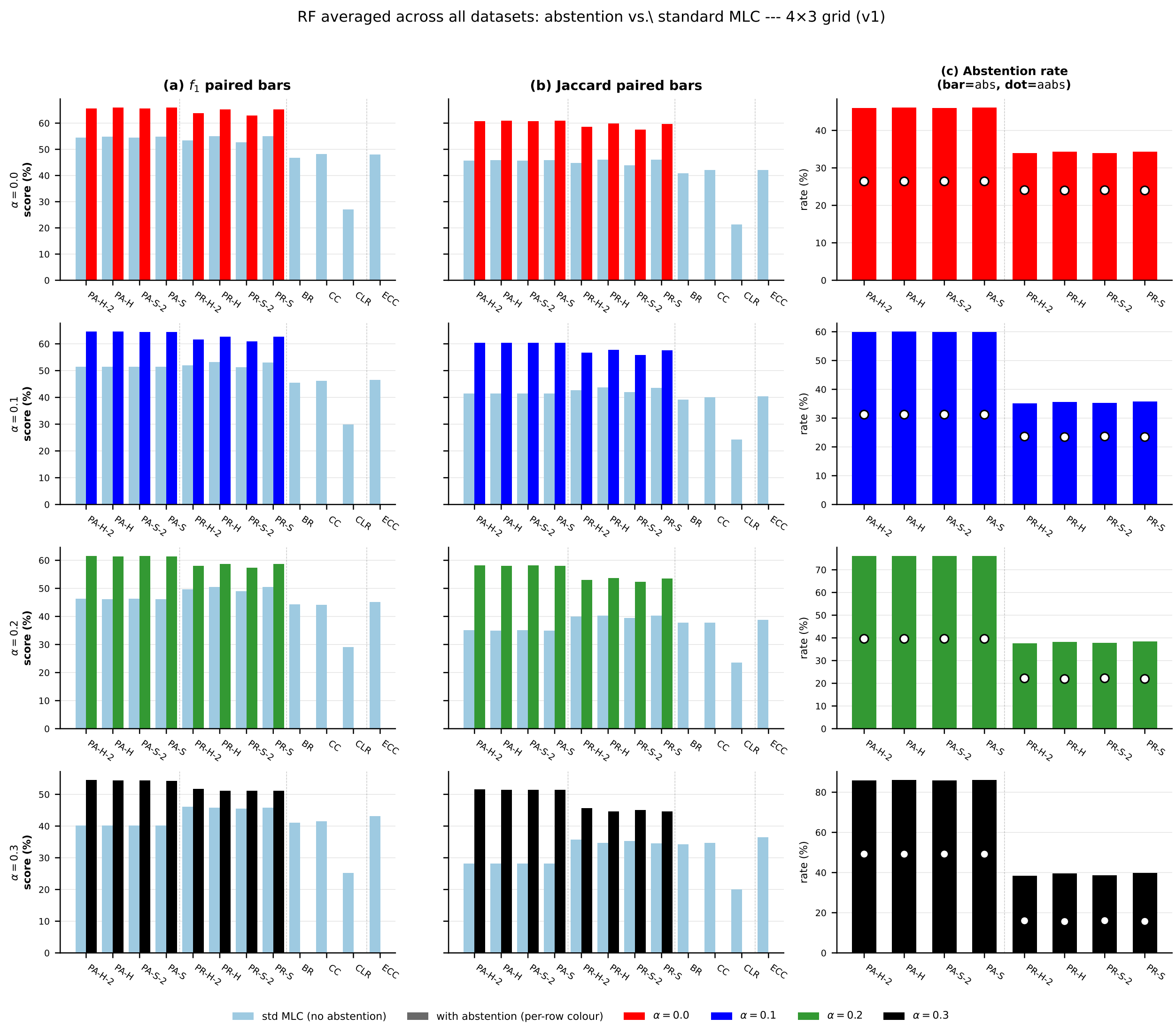


Figure 1: RF averaged across all datasets, 4×3 grid (rows = α, cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x-axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α-coloured = with-abstention; in (c), bar = `abs` (instance-level), dot = `aabs` (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

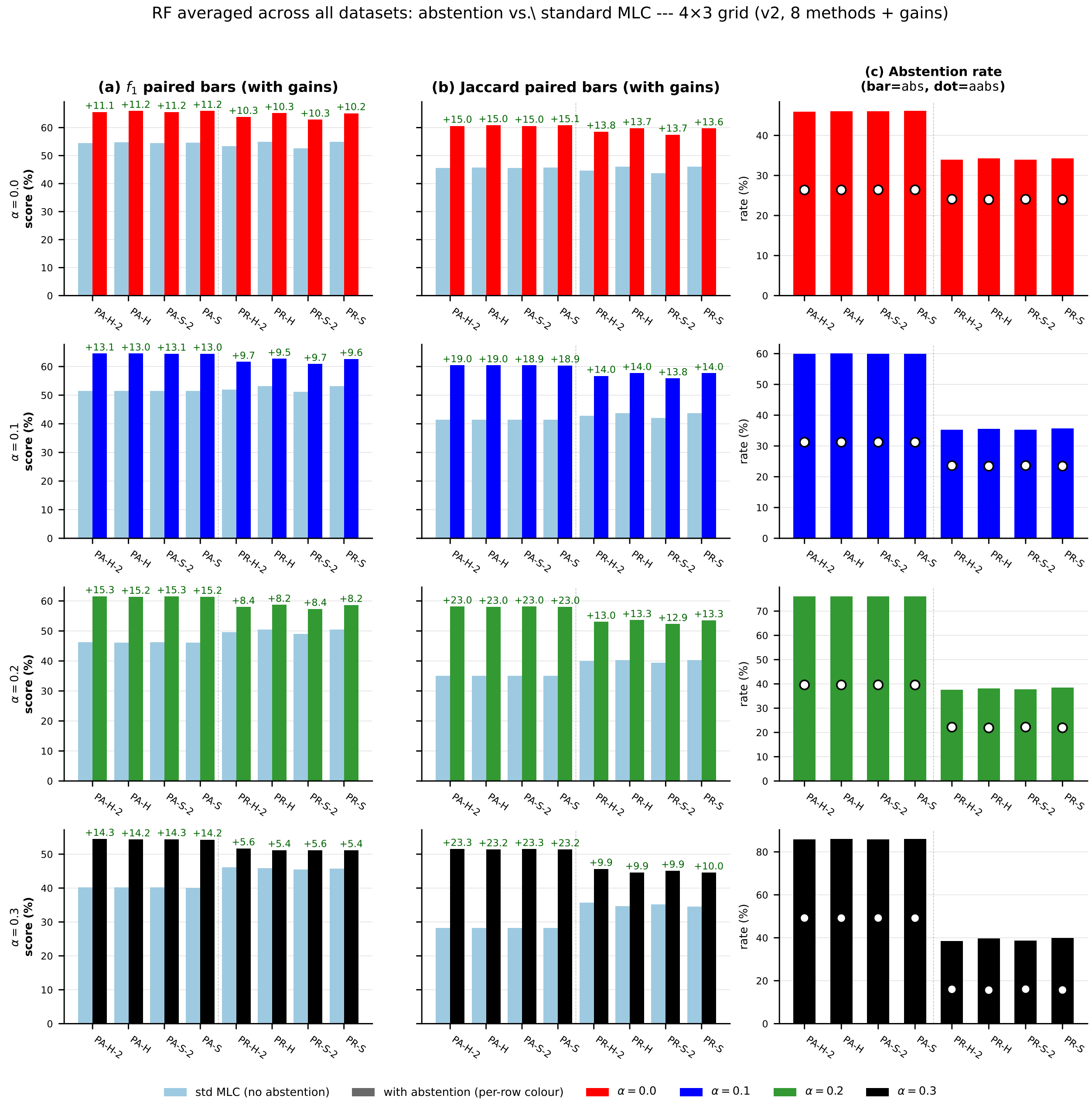


Figure 2: RF averaged across all datasets, 4×3 grid (rows $= \alpha$, cols $= f_1$ / Jaccard / abstention rate) — (version 2) only the 8 PA/PR methods that can abstain; gain labels $+x.y$ printed above each with-abstention bar so the score-point benefit over the std-MLC bar at the same method is visible at a glance. Same colour / layout conventions as version 1. Color $= \alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

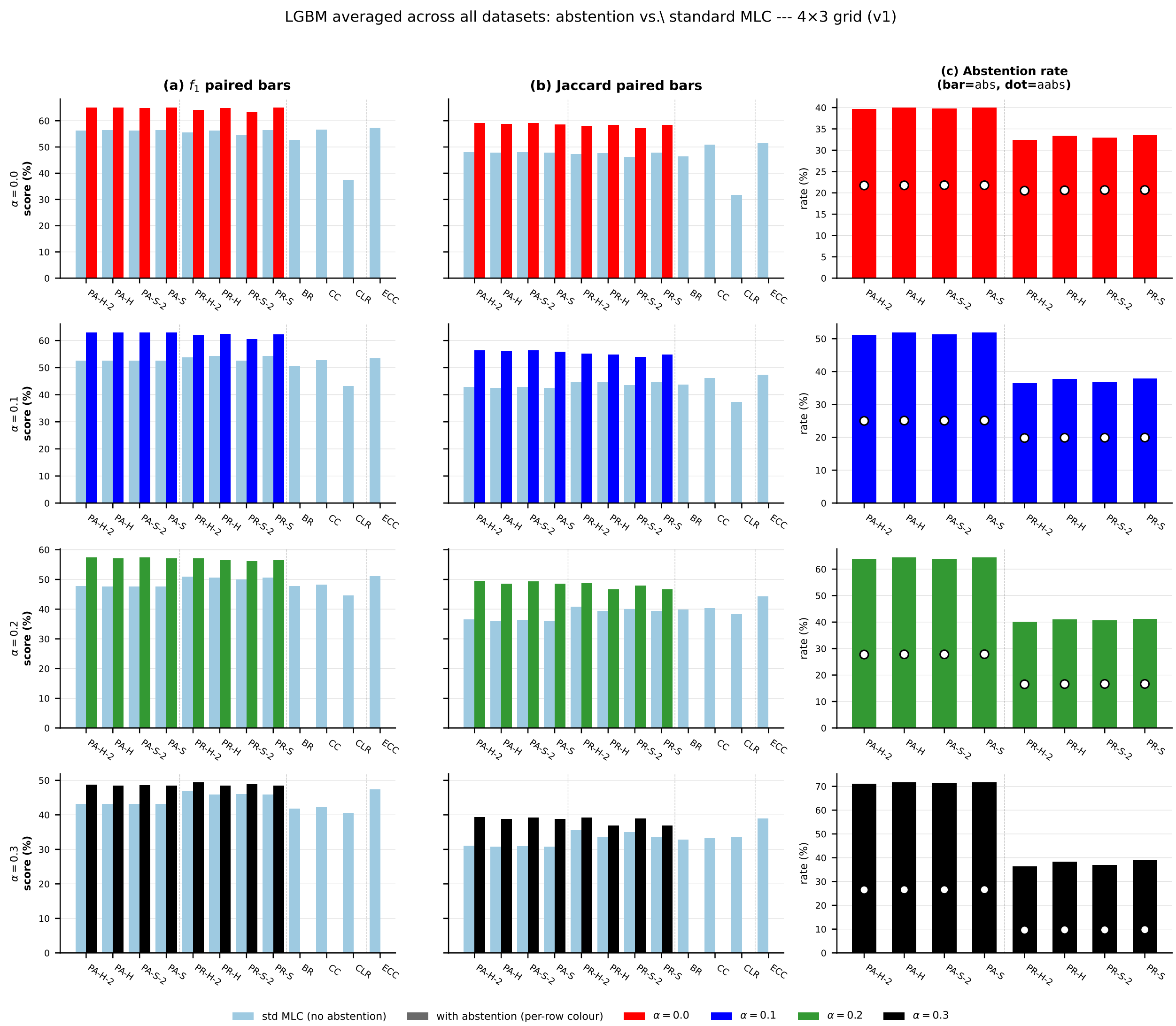


Figure 3: LGBM averaged across all datasets, 4×3 grid (rows = α, cols = f_1 / Jaccard / abstention rate) — (version 1) all 12 methods on the x-axis (PA + PR + baselines BR/CC/CLR/ECC); baselines plot only the std-MLC bar since they cannot abstain. Light blue = standard MLC, α-coloured = with-abstention; in (c), bar = `abs` (instance-level), dot = `aabs` (per-cell). Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

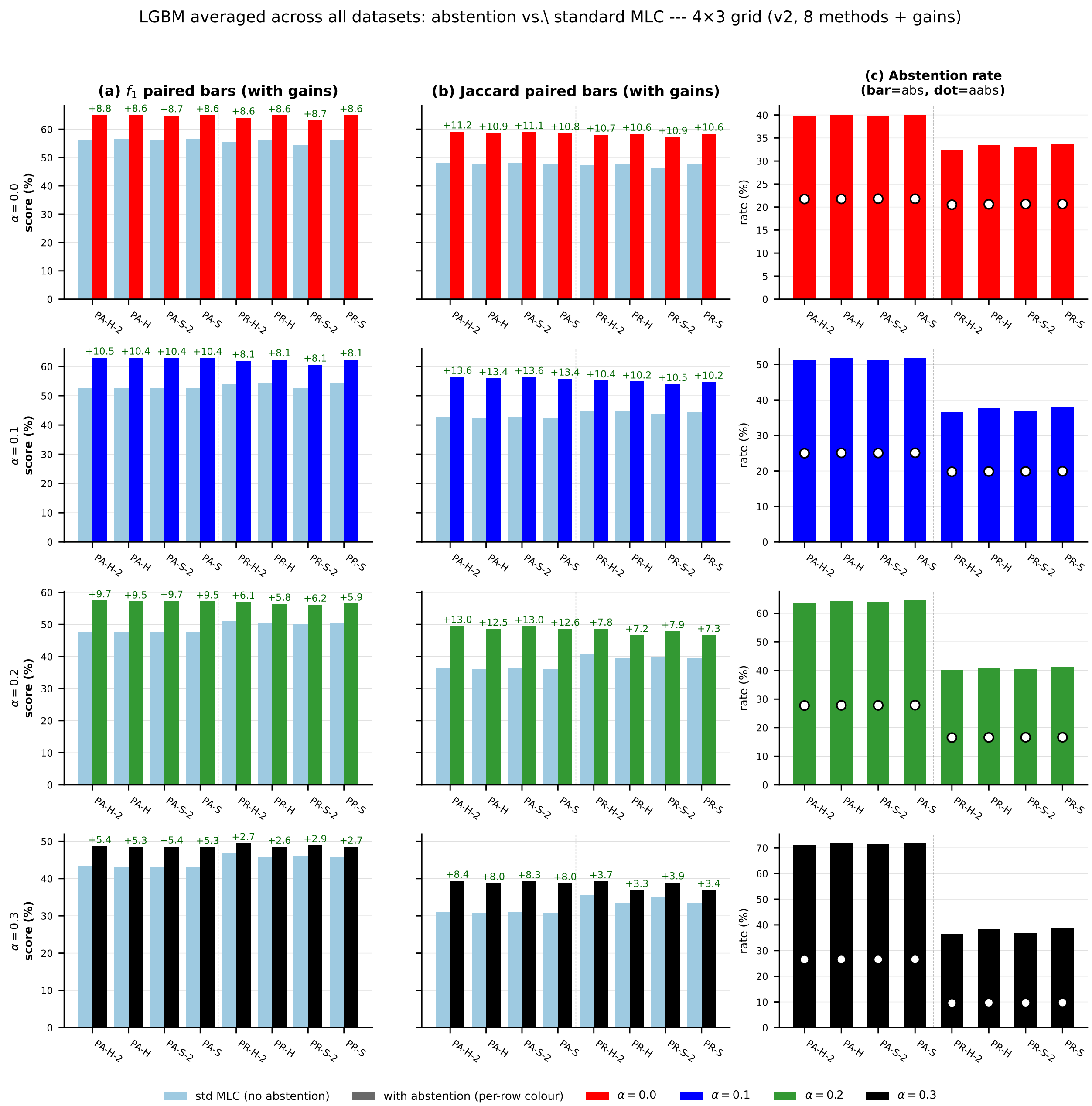


Figure 4: LGBM averaged across all datasets, 4×3 grid (rows = α, cols = f_1 / Jaccard / abstention rate) — (version 2) only the 8 PA/PR methods that can abstain; gain labels $+x.y$ printed above each with-abstention bar so the score-point benefit over the std-MLC bar at the same method is visible at a glance. Same colour / layout conventions as version 1. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

4 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

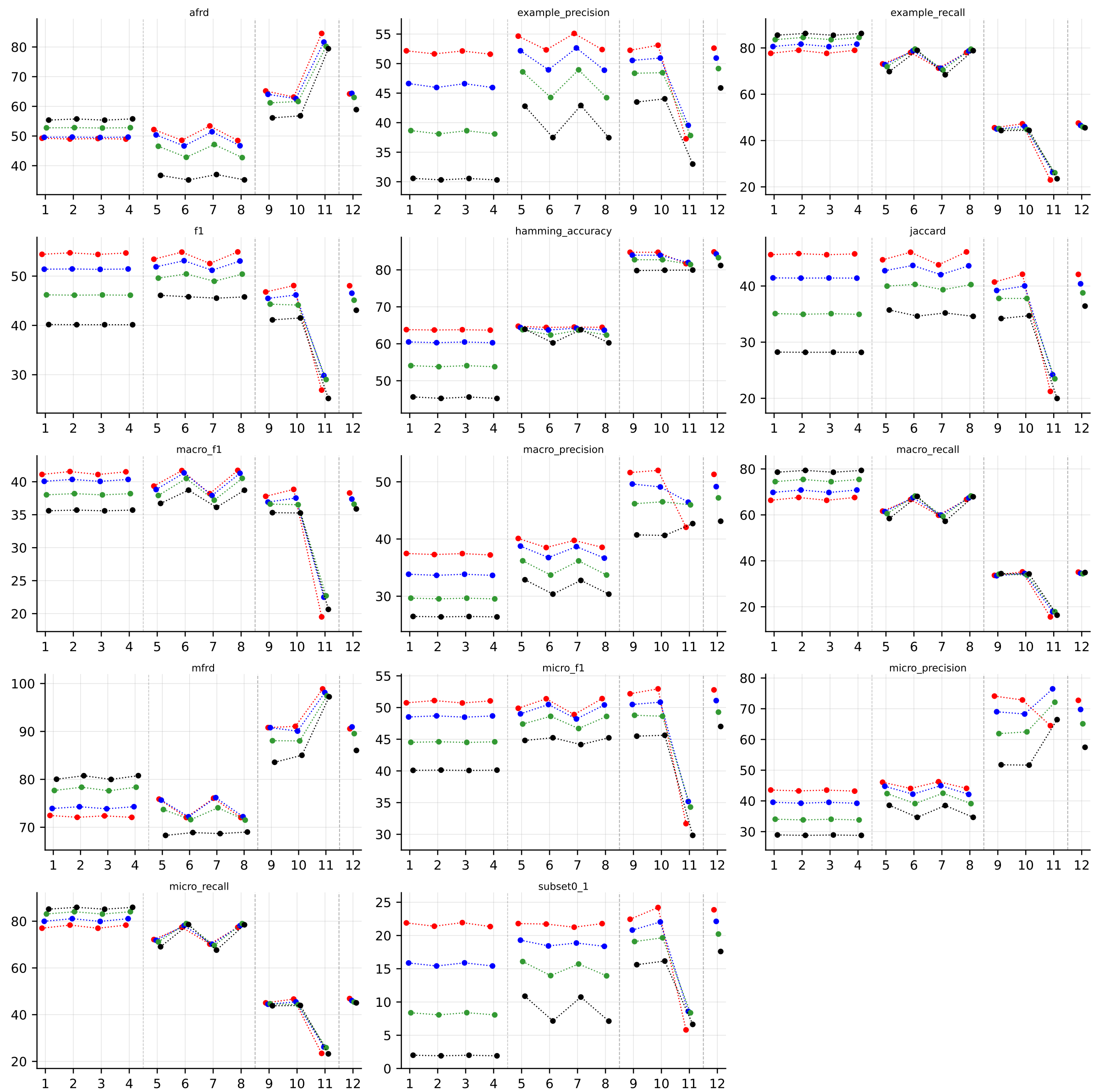


Table 11: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

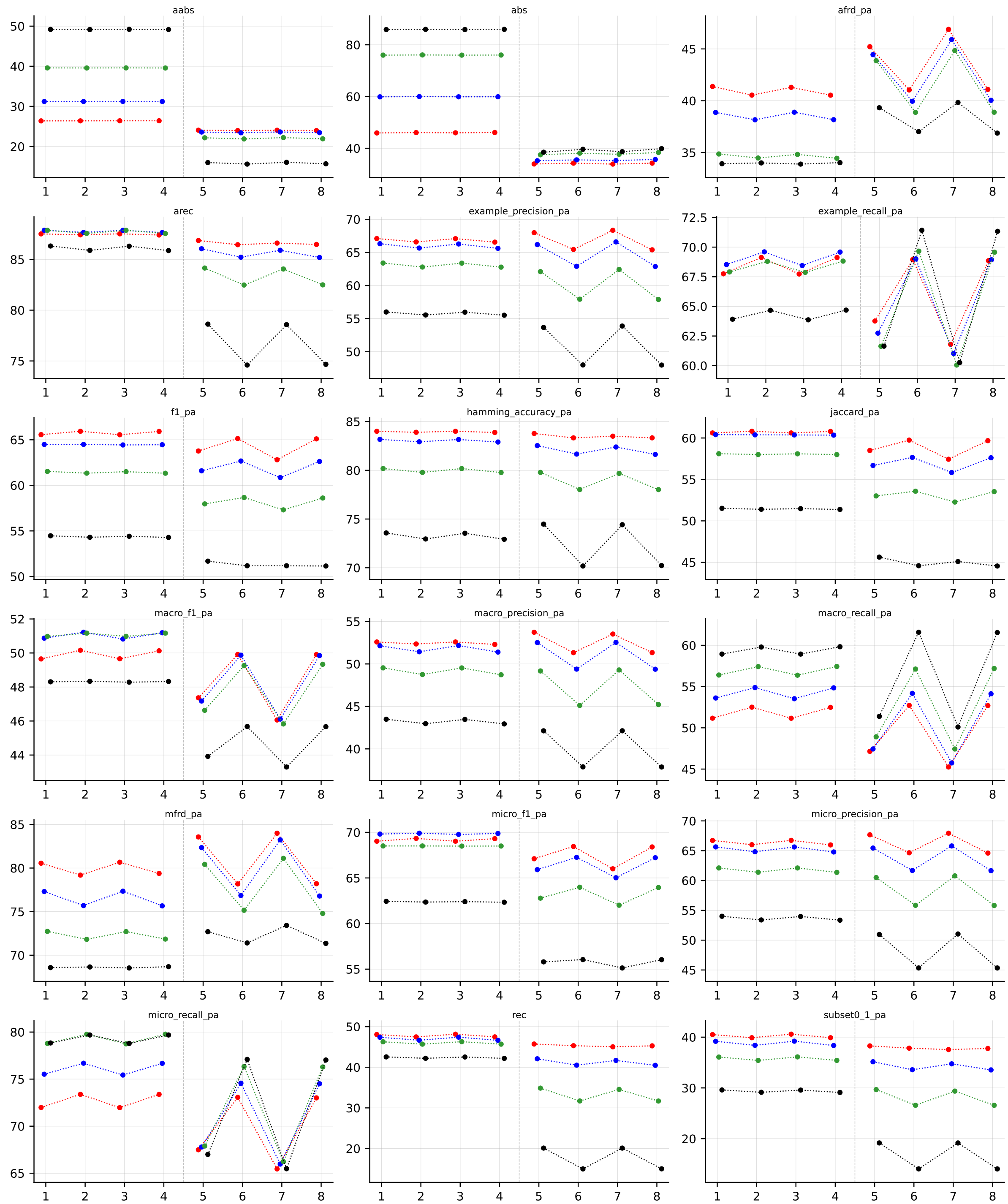


Table 12: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

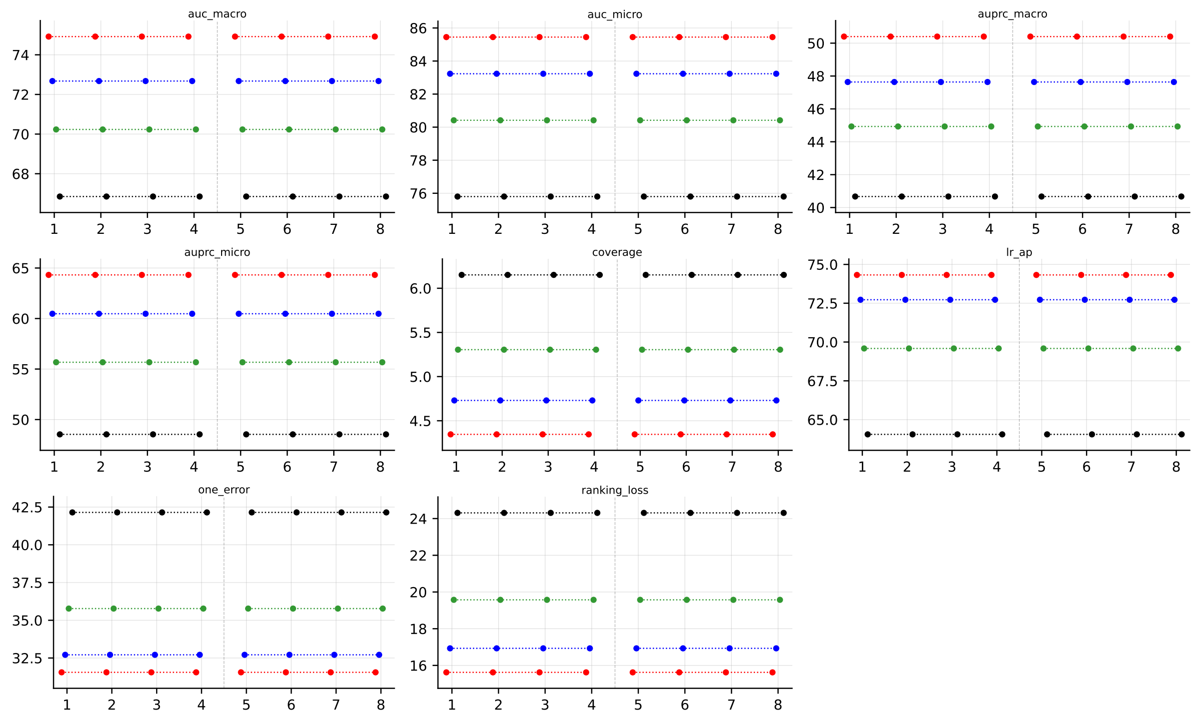


Table 13: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

5 Abstention vs standard MLC

Why this section exists. The paper's central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. '*Standard MLC*' (a.k.a. `BinaryVector` / BV) means each method outputs a 0/1 prediction for every one of the K labels. '*With abstention*' (a.k.a. `PartialAbstention` / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (`f1_pa`, `jaccard_pa`, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that "no abstention" has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (`f1`, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green "$+x.y$" label above each warm bar is the gain in score-points over that method's own BV bar.

5.1 RF, metric `f1_pa` — aggregate

Figure 5: RF (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

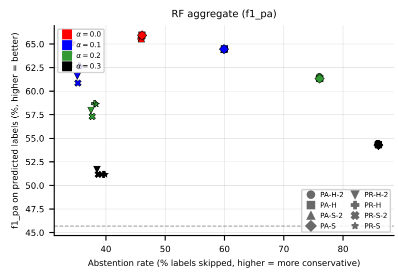


Figure 6: RF (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 7: RF (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 8: RF (f1_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1-\text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.2 RF, metric `f1_pa` — per dataset

Figure 9: RF (f1_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

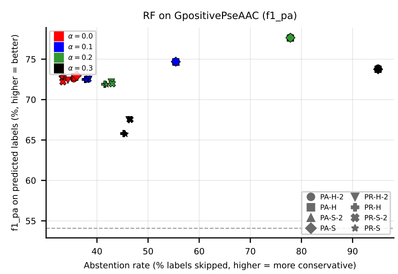


Figure 10: RF (f1_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 11: RF (f1_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa − f_1) heatmap (right).

Figure 12: RF (f1_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 13: RF (f1_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

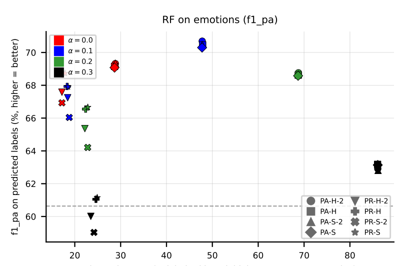


Figure 14: RF (f1_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 15: RF (f1_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 16: RF (f1_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 17: RF (f1_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

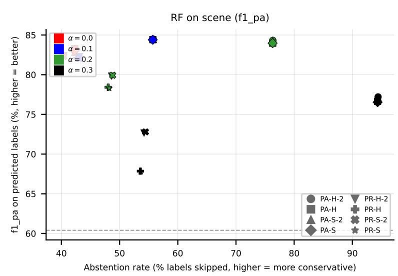


Figure 18: RF (f1_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 19: RF (f1_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa − f_1) heatmap (right).

Figure 20: RF (f1_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = (1 − abs) · $f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 21: RF (f1_pa) on `PlantPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

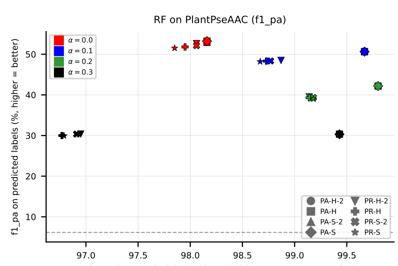


Figure 22: RF (f1_pa) on `PlantPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 23: RF (f1_pa) on `PlantPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 24: RF (f1_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 25: RF (f1_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

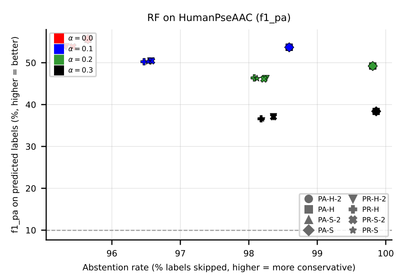


Figure 26: RF (f1_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 27: RF (f1_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 28: RF (f1_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = (1 − abs) · $f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 29: RF (f1_pa) on Yeast: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = abs (% of instances with at least one abstained label, instance-level); dot = aabs (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

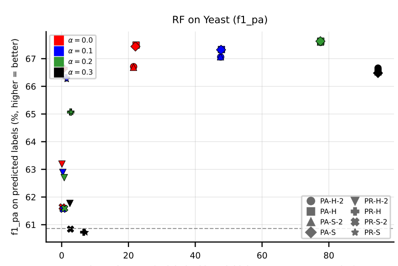


Figure 30: RF (f1_pa) on Yeast: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 31: RF (f1_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 32: RF (f1_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 33: RF (f1_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

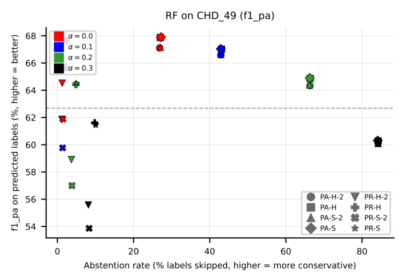


Figure 34: RF (f1_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 35: RF (f1_pa) on `CHD_49`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 36: RF (f1_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter (x = coverage $= 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa $= (1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 37: RF (f1_pa) on Water-quality: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = abs (% of instances with at least one abstained label, instance-level); dot = aabs (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

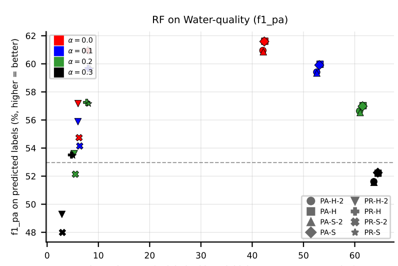


Figure 38: RF (f1_pa) on Water-quality: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 39: RF (f1_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 40: RF (f1_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 41: RF (f1_pa) on `enron`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

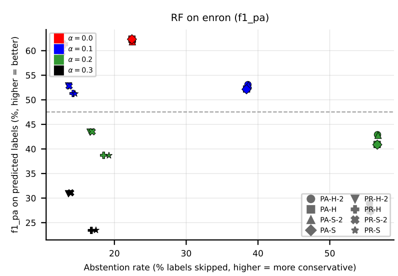


Figure 42: RF (f1_pa) on `enron`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 43: RF (f1_pa) on `enron`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 44: RF (f1_pa) on `enron` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.3 RF, metric `jaccard_pa` — aggregate

Figure 45: RF (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

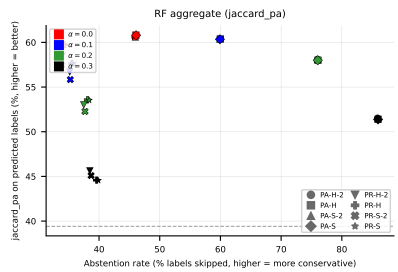


Figure 46: RF (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 47: RF (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 48: RF (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.4 RF, metric `jaccard_pa` — per dataset

Figure 49: RF (jaccard_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

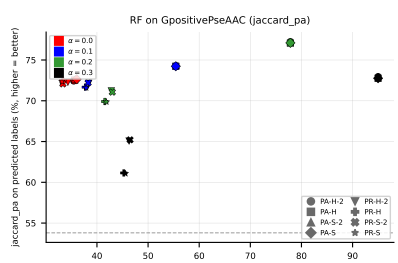


Figure 50: RF (jaccard_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 51: RF (jaccard_pa) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 52: RF (jaccard_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 53: RF (jaccard_pa) on `emotions`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

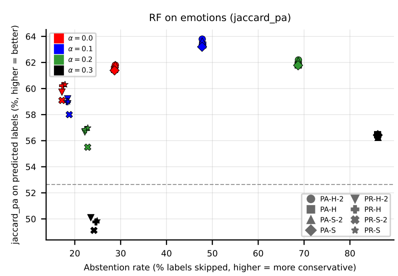


Figure 54: RF (jaccard_pa) on `emotions`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 55: RF (jaccard_pa) on `emotions`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 56: RF (jaccard_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 57: RF (jaccard_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

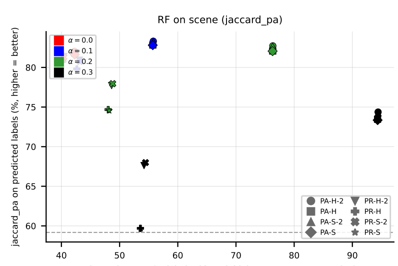


Figure 58: RF (jaccard_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 59: RF (jaccard_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 60: RF (jaccard_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 61: RF (jaccard_pa) on `PlantPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

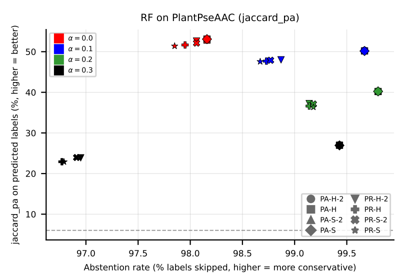


Figure 62: RF (jaccard_pa) on `PlantPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 63: RF (jaccard_pa) on `PlantPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 64: RF (jaccard_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 65: RF (jaccard_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

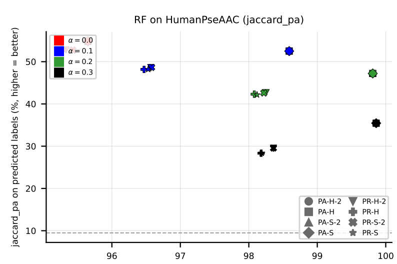


Figure 66: RF (jaccard_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 67: RF (jaccard_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 68: RF (jaccard_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 69: RF (jaccard_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

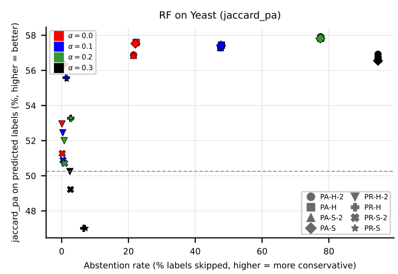


Figure 70: RF (jaccard_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 71: RF (jaccard_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 72: RF (jaccard_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 73: RF (jaccard_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

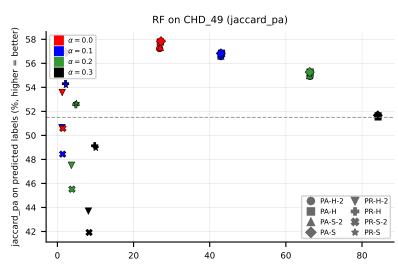
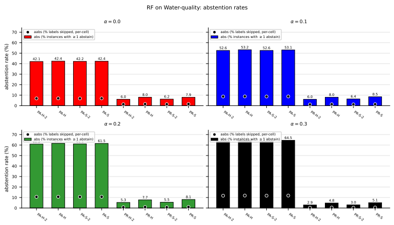


Figure 74: RF (jaccard_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 75: RF (jaccard_pa) on `CHD_49`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 76: RF (jaccard_pa) on `CHD_49` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



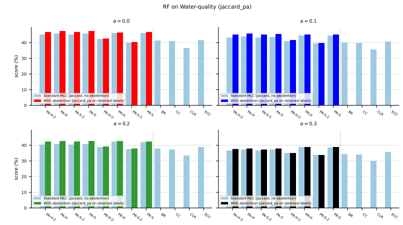


Figure 77: RF (jaccard_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

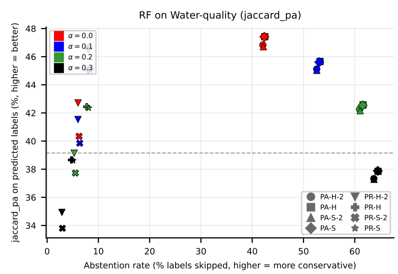
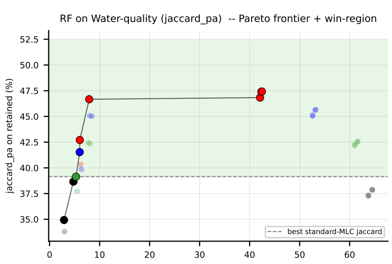


Figure 78: RF (jaccard_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



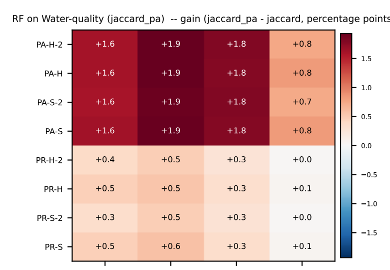


Figure 79: RF (jaccard_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 80: RF (jaccard_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1−abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 81: RF (jaccard_pa) on `enron`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

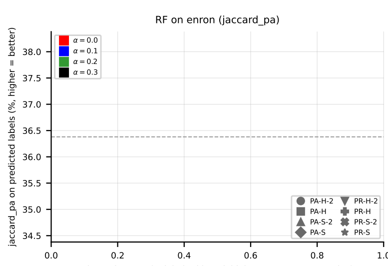
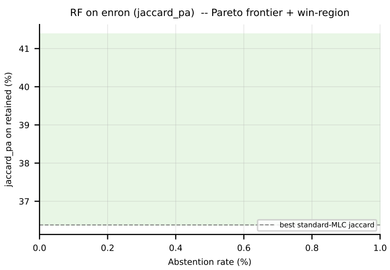


Figure 82: RF (jaccard_pa) on `enron`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



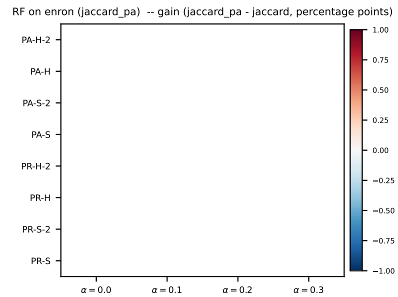
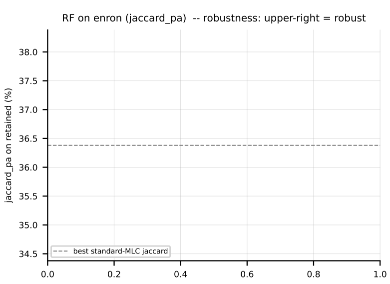


Figure 83: RF (jaccard_pa) on `enron`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).



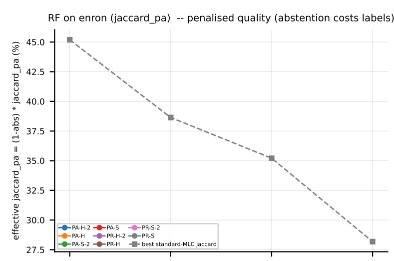
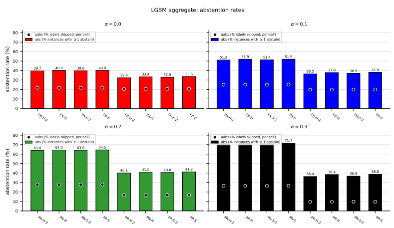


Figure 84: RF (jaccard_pa) on `enron` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.5 LGBM, metric `f1_pa` — aggregate



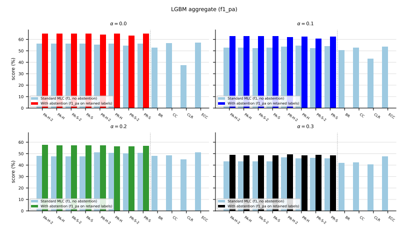


Figure 85: LGBM (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

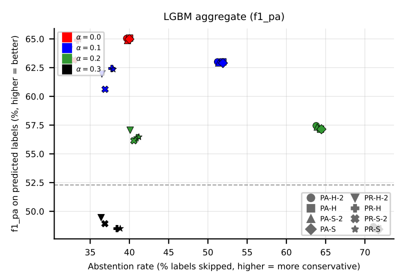
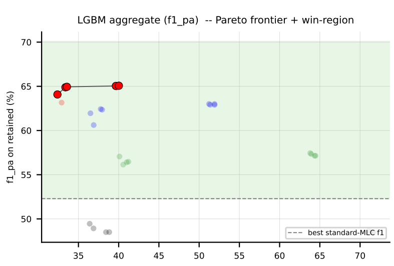


Figure 86: LGBM (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



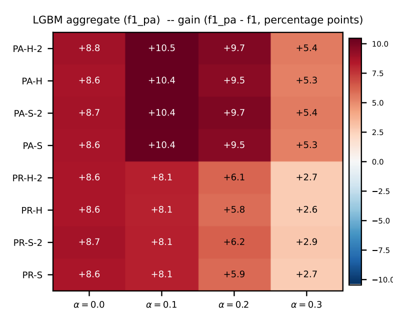


Figure 87: LGBM (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 88: LGBM (f1_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.6 LGBM, metric `f1_pa` — per dataset

Figure 89: LGBM (f1_pa) on **`GpositivePseAAC`**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **`abs`** (% of instances with at least one abstained label, instance-level); dot = **`aabs`** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

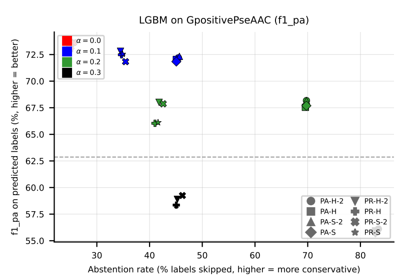
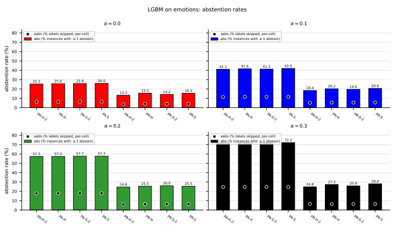


Figure 90: LGBM (f1_pa) on **`GpositivePseAAC`**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 91: LGBM (f1_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 92: LGBM (f1_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



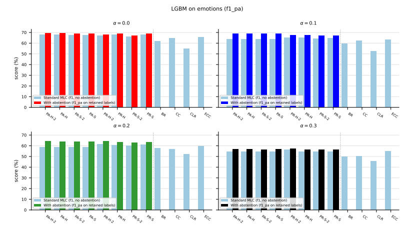


Figure 93: LGBM (f1_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

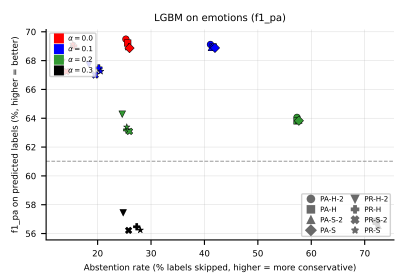
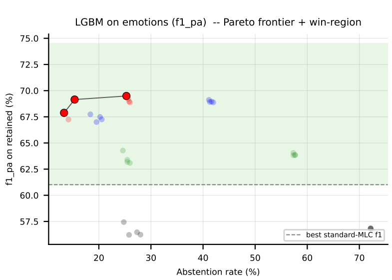


Figure 94: LGBM (f1_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



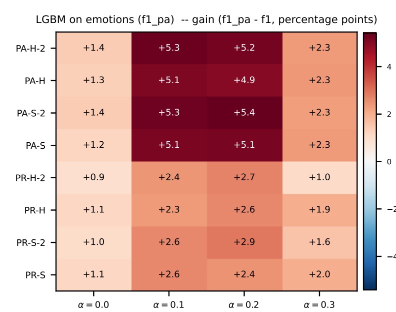


Figure 95: LGBM (f1_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 96: LGBM (f1_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 97: LGBM (f1_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

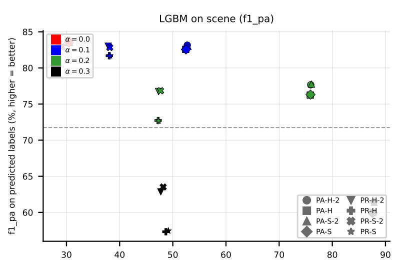
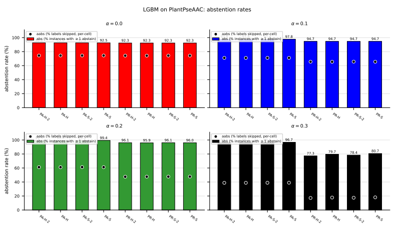


Figure 98: LGBM (f1_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 99: LGBM (f1_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 100: LGBM (f1_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



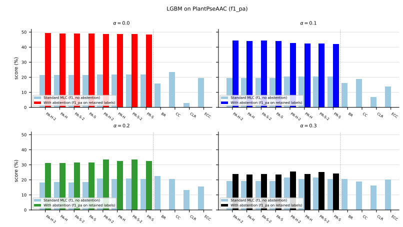


Figure 101: LGBM (f1_pa) on **PlantPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

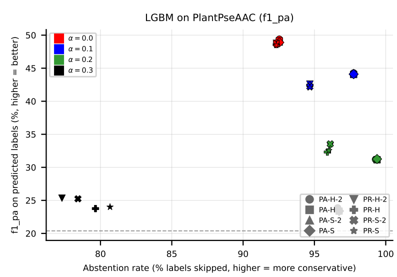
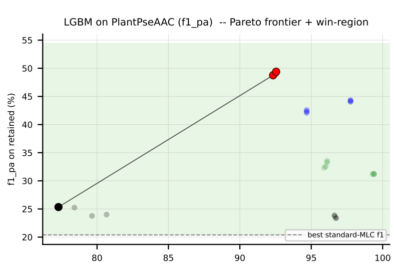


Figure 102: LGBM (f1_pa) on **PlantPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



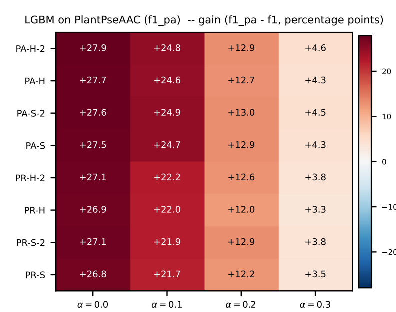
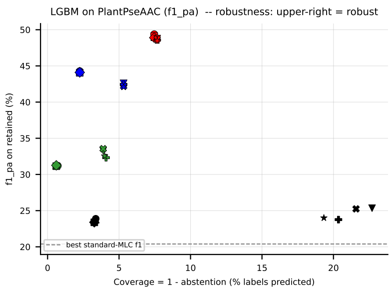


Figure 103: LGBM (f1_pa) on **PlantPseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).



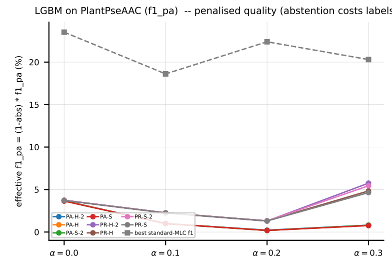
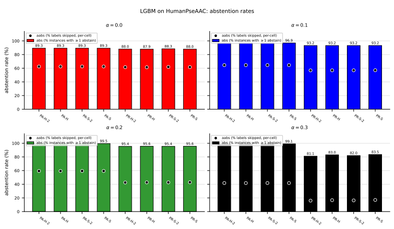


Figure 104: LGBM (f1_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



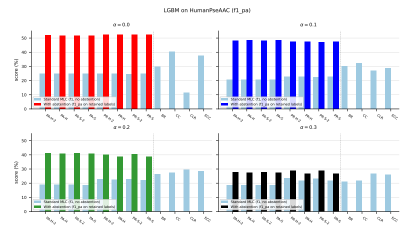


Figure 105: LGBM (f1_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

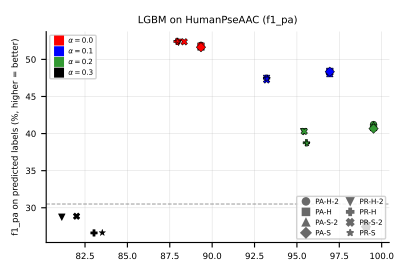
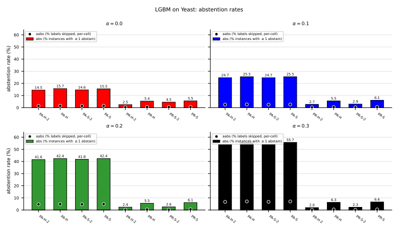


Figure 106: LGBM (f1_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 107: LGBM (f1_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 108: LGBM (f1_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



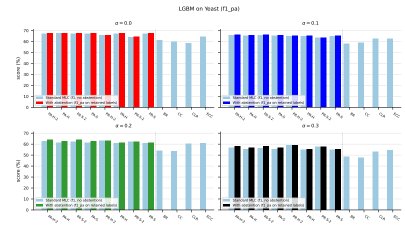


Figure 109: LGBM (f1_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

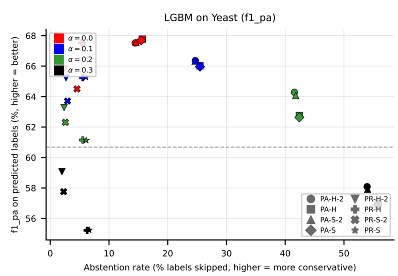
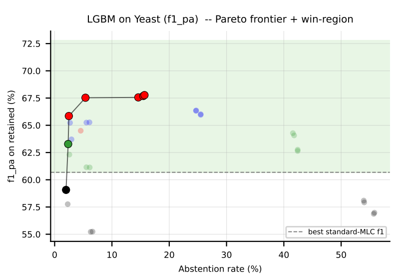


Figure 110: LGBM (f1_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



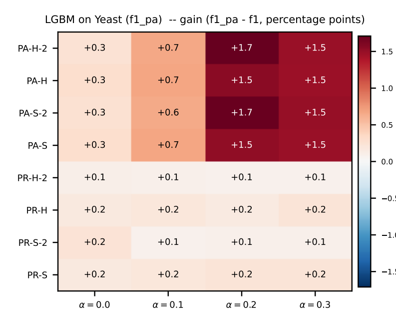


Figure 111: LGBM (f1_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).

Figure 112: LGBM (f1_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 113: LGBM (f1_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

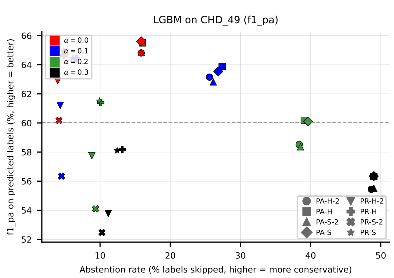


Figure 114: LGBM (f1_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 115: LGBM (f1_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $-$ f_1) heatmap (right).

Figure 116: LGBM (f1_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 117: LGBM (f1_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

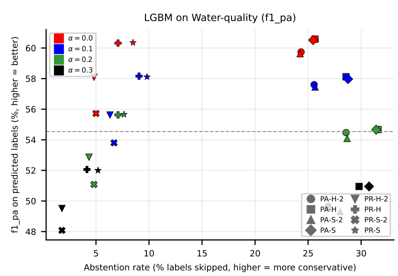
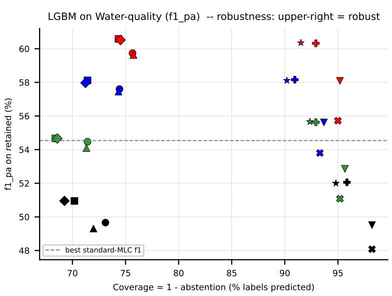


Figure 118: LGBM (f1_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 119: LGBM (f1_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (f1_pa $- f_1$) heatmap (right).



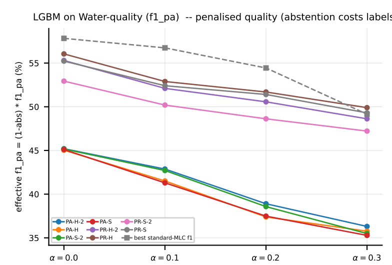
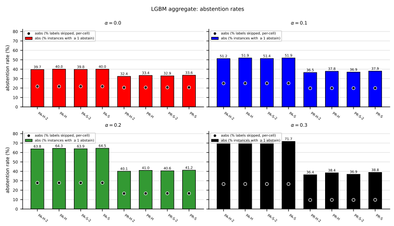


Figure 120: LGBM (f1_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.7 LGBM, metric `jaccard_pa` — aggregate



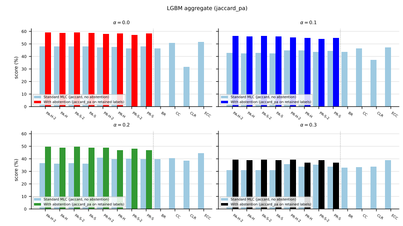


Figure 121: LGBM (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **`abs`** (% of instances with at least one abstained label, instance-level); dot = **`aabs`** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

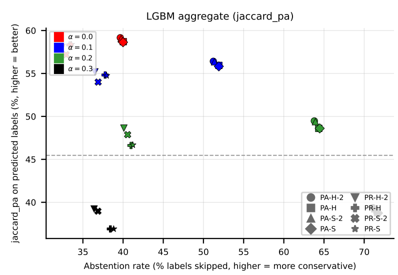
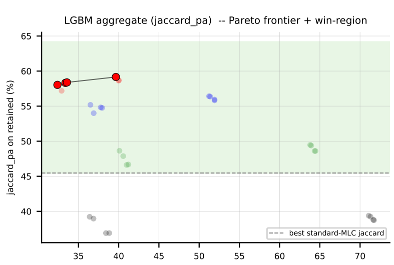


Figure 122: LGBM (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



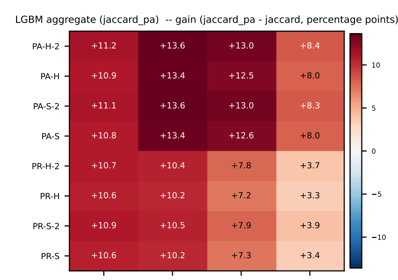


Figure 123: LGBM (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 124: LGBM (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.8 LGBM, metric `jaccard_pa` — per dataset

Figure 125: LGBM (jaccard_pa) on `GpositivePseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

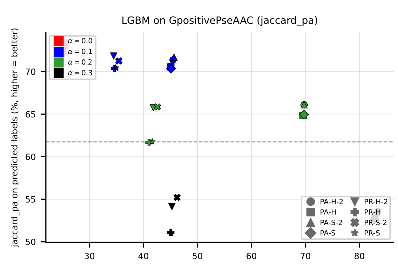


Figure 126: LGBM (jaccard_pa) on `GpositivePseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 127: LGBM (jaccard_pa) on `GpositivePseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 128: LGBM (jaccard_pa) on `GpositivePseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 $-$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs}) \cdot \textit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 129: LGBM (jaccard_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

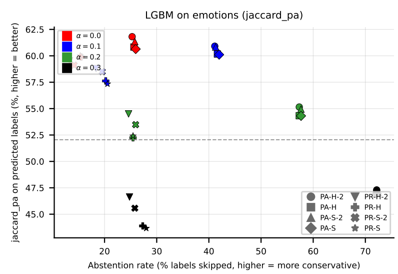


Figure 130: LGBM (jaccard_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 131: LGBM (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa − f_1) heatmap (right).

Figure 132: LGBM (jaccard_pa) on `emotions` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 133: LGBM (jaccard_pa) on `scene`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

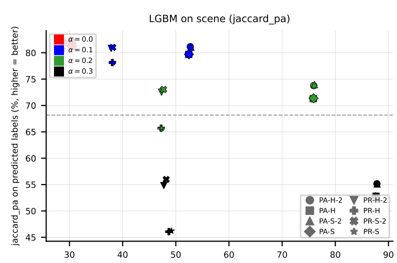


Figure 134: LGBM (jaccard_pa) on `scene`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 135: LGBM (jaccard_pa) on `scene`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 136: LGBM (jaccard_pa) on `scene` — robustness view. Left: efficiency-quality scatter (x = coverage = $1 -$ abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 137: LGBM (jaccard_pa) on **PlantPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

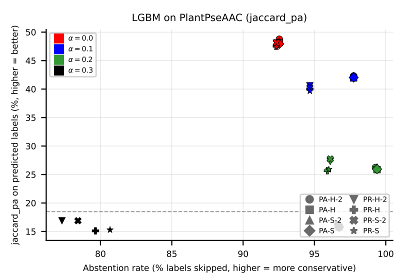


Figure 138: LGBM (jaccard_pa) on **PlantPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 139: LGBM (jaccard_pa) on **PlantPseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).

Figure 140: LGBM (jaccard_pa) on `PlantPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs})\cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 141: LGBM (jaccard_pa) on `HumanPseAAC`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

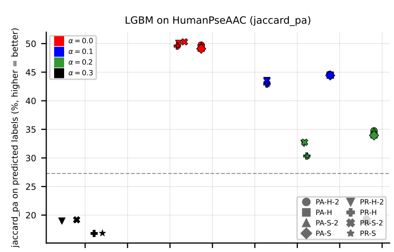
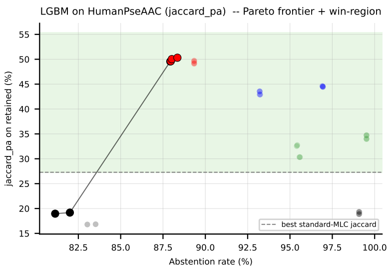


Figure 142: LGBM (jaccard_pa) on `HumanPseAAC`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



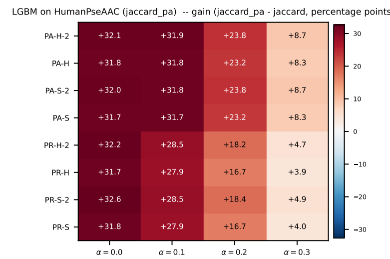
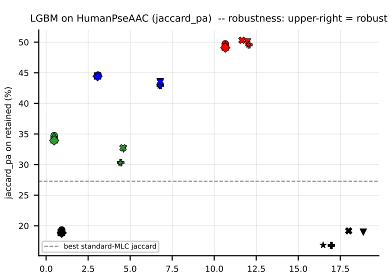


Figure 143: LGBM (jaccard_pa) on `HumanPseAAC`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).



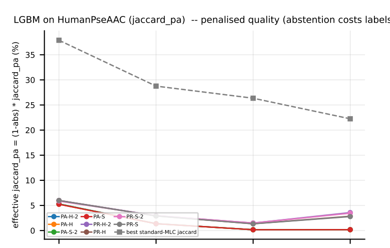


Figure 144: LGBM (jaccard_pa) on `HumanPseAAC` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa $= (1-\text{abs}) \cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 145: LGBM (jaccard_pa) on `Yeast`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = `abs` (% of instances with at least one abstained label, instance-level); dot = `aabs` (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

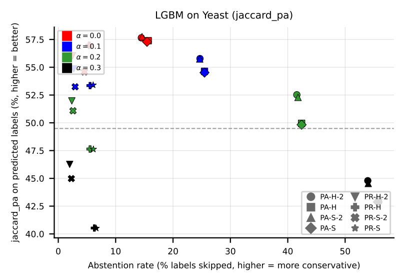


Figure 146: LGBM (jaccard_pa) on `Yeast`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

Figure 147: LGBM (jaccard_pa) on `Yeast`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).

Figure 148: LGBM (jaccard_pa) on `Yeast` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \textit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

Figure 149: LGBM (jaccard_pa) on `CHD_49`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

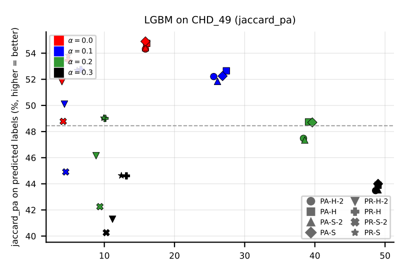
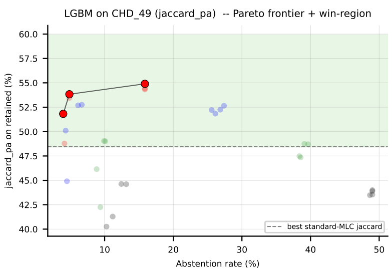


Figure 150: LGBM (jaccard_pa) on `CHD_49`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



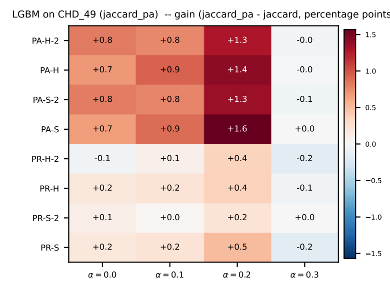
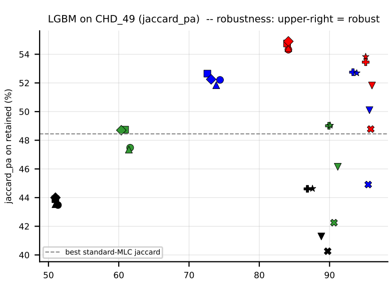


Figure 151: LGBM (jaccard_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $- f_1$) heatmap (right).



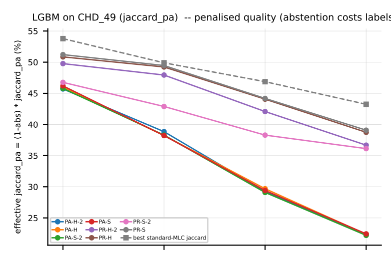
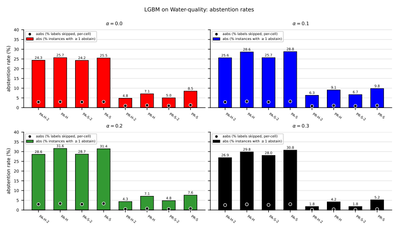


Figure 152: LGBM (jaccard_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \mathit{jaccard\_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.



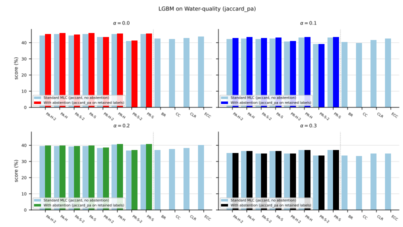


Figure 153: LGBM (jaccard_pa) on `Water-quality`: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

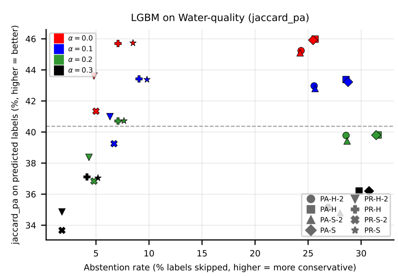
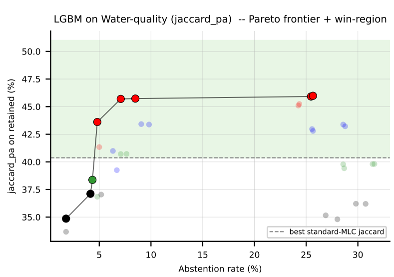


Figure 154: LGBM (jaccard_pa) on `Water-quality`: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



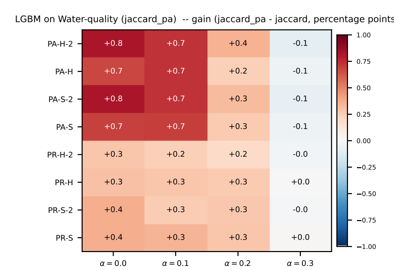
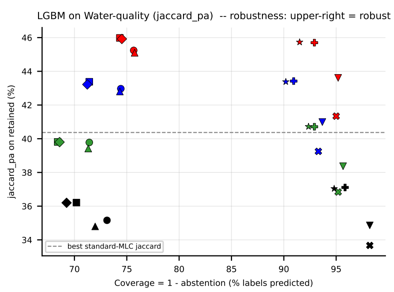


Figure 155: LGBM (jaccard_pa) on `Water-quality`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa $-$ f_1) heatmap (right).



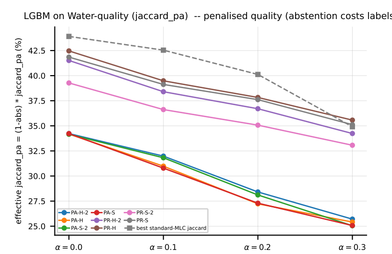
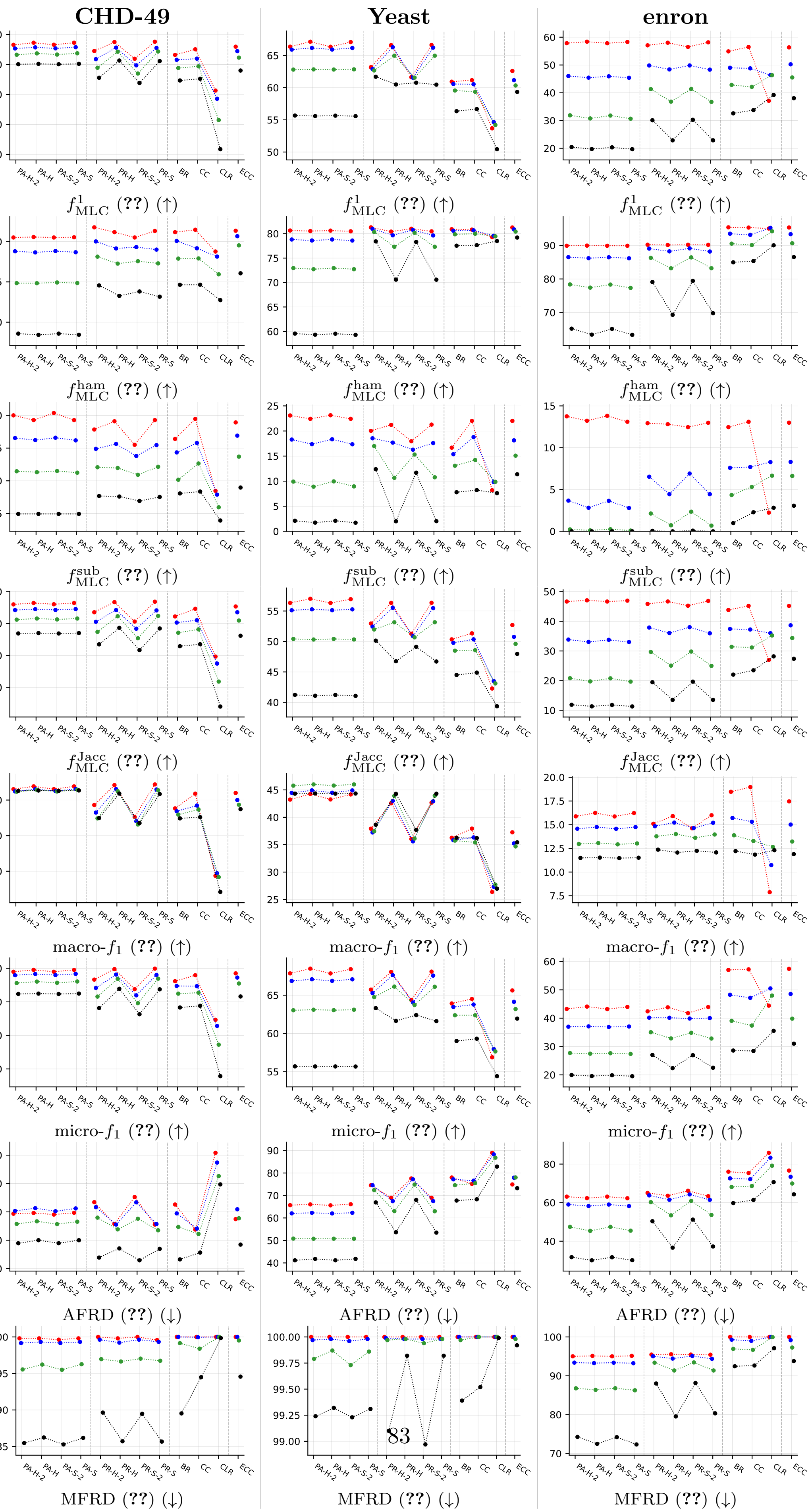


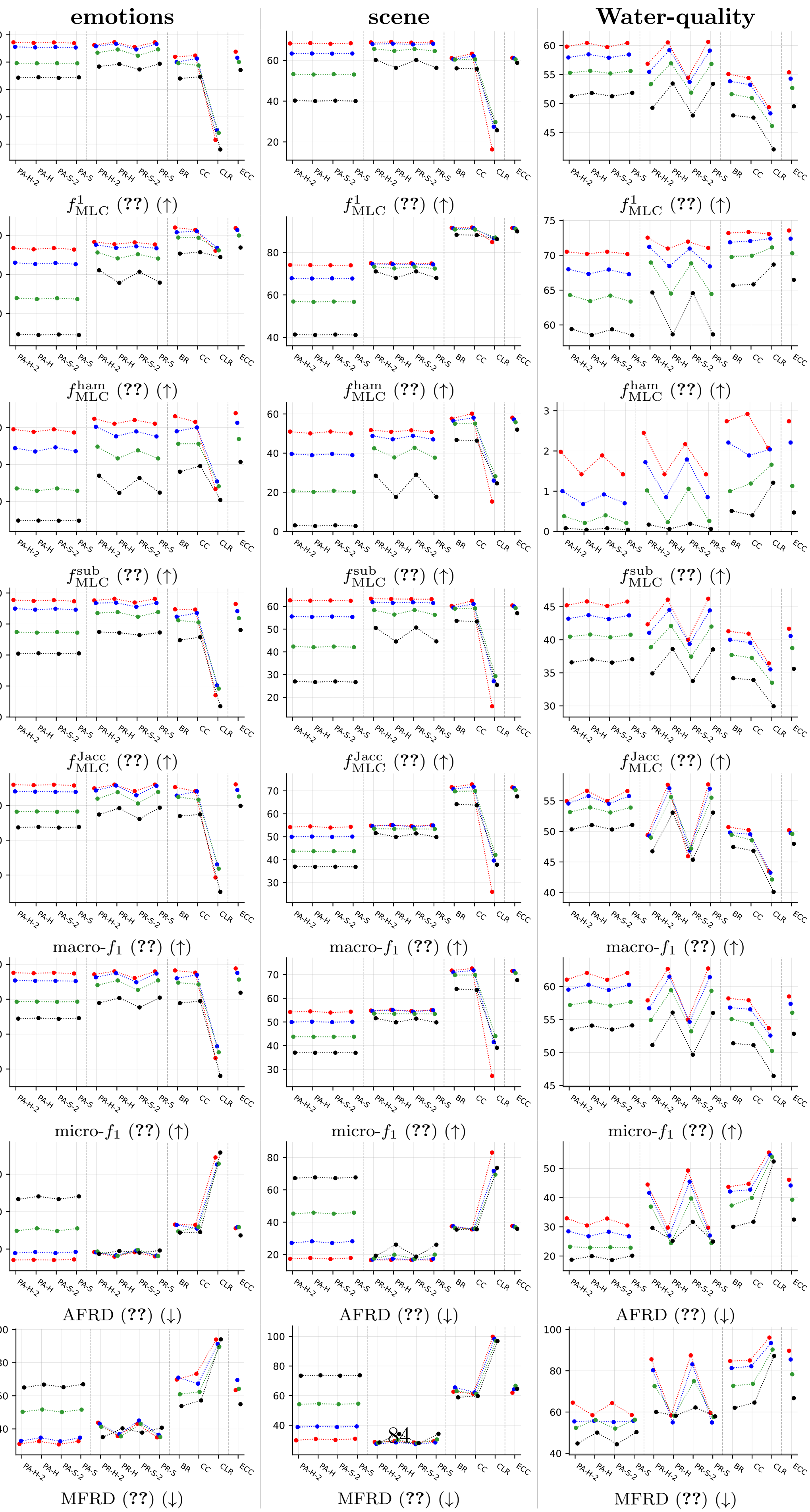
Figure 156: LGBM (jaccard_pa) on `Water-quality` — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1-\text{abs})\cdot jaccard\_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

A Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables All tables share the same 8-metric set (original 5 + 3 revision additions: $f_{\mathrm{MLC}}^{\mathrm{Jacc}}$, macro-$f_1$, micro-$f_1$). G2/G4 = imbalanced datasets, G3/G5 = balanced datasets (split follows the original paper's appendix). G6/G7 = main-paper datasets (GpositivePse, PlantPse, HumanPse) shown with the full metric set; the main body only carries the revised compact Tables 4/5.

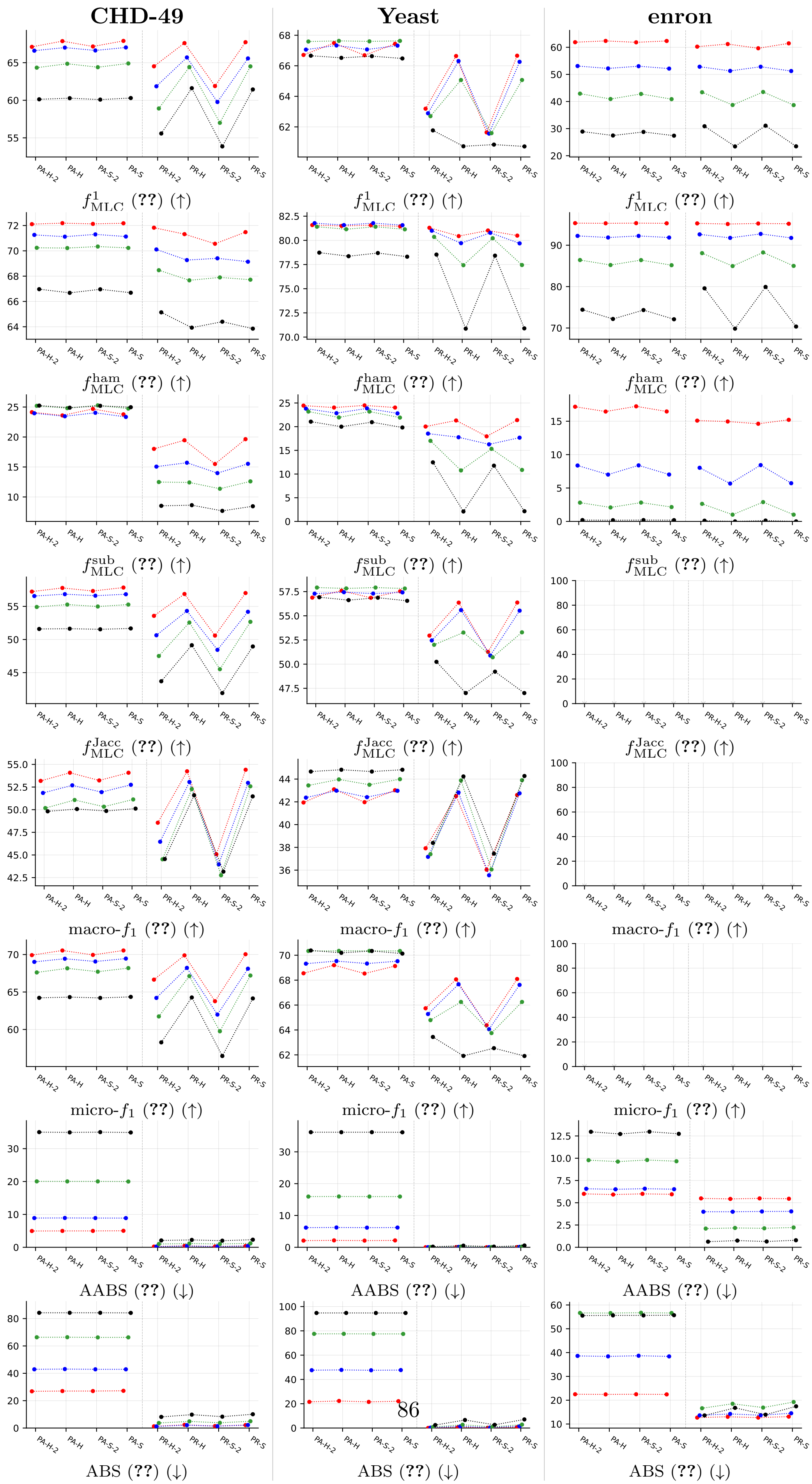
A.1 Standard MLC predictions (BinaryVector)

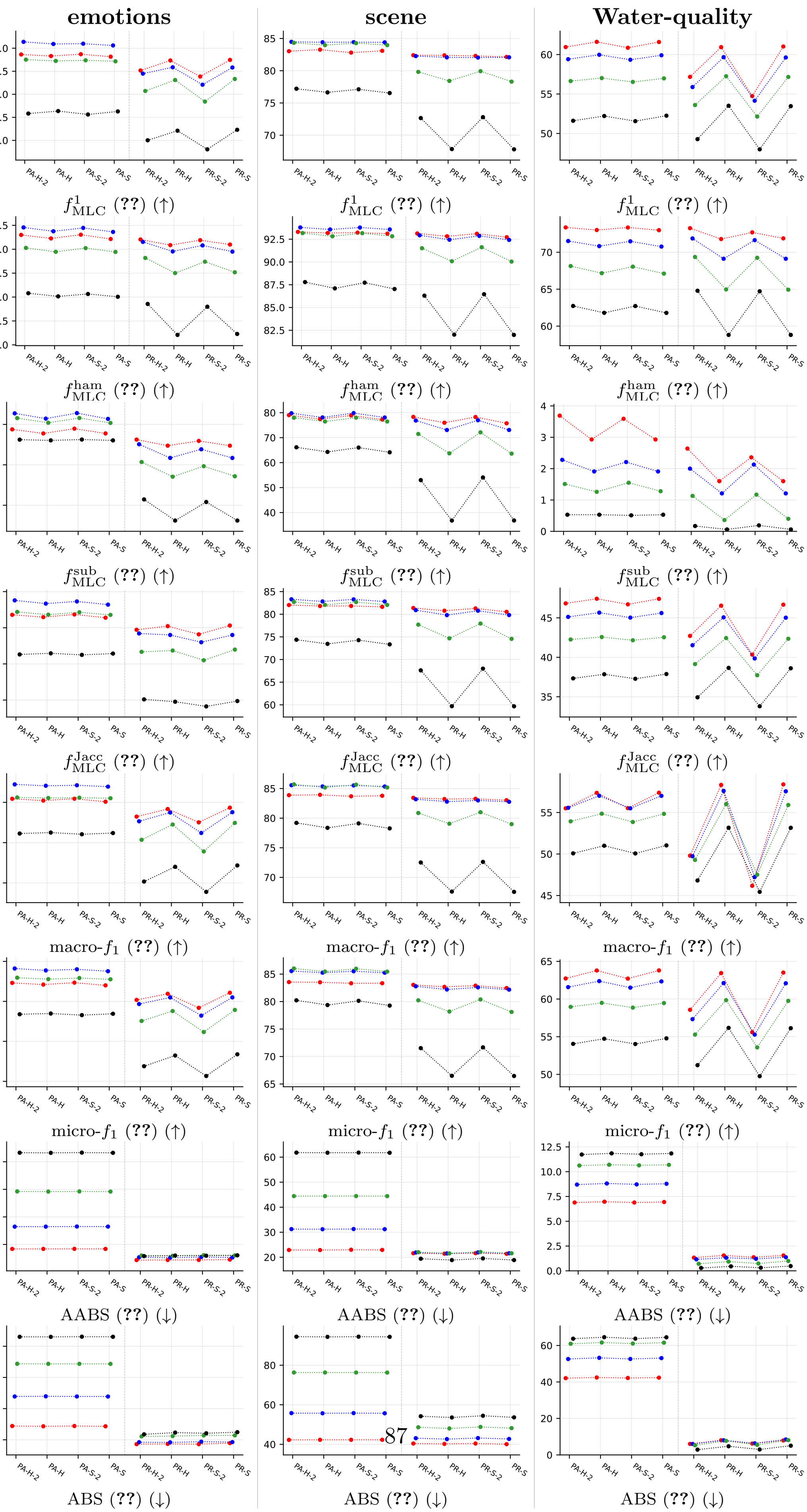

CHD-49
Yeast
enron
f^1_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC



emotions
scene
Water-quality
PA-H-2 PA-H PA-S-2 PA-S PR-H-2 PR-H PR-S-2 PR-S BR CC CLR ECC
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)

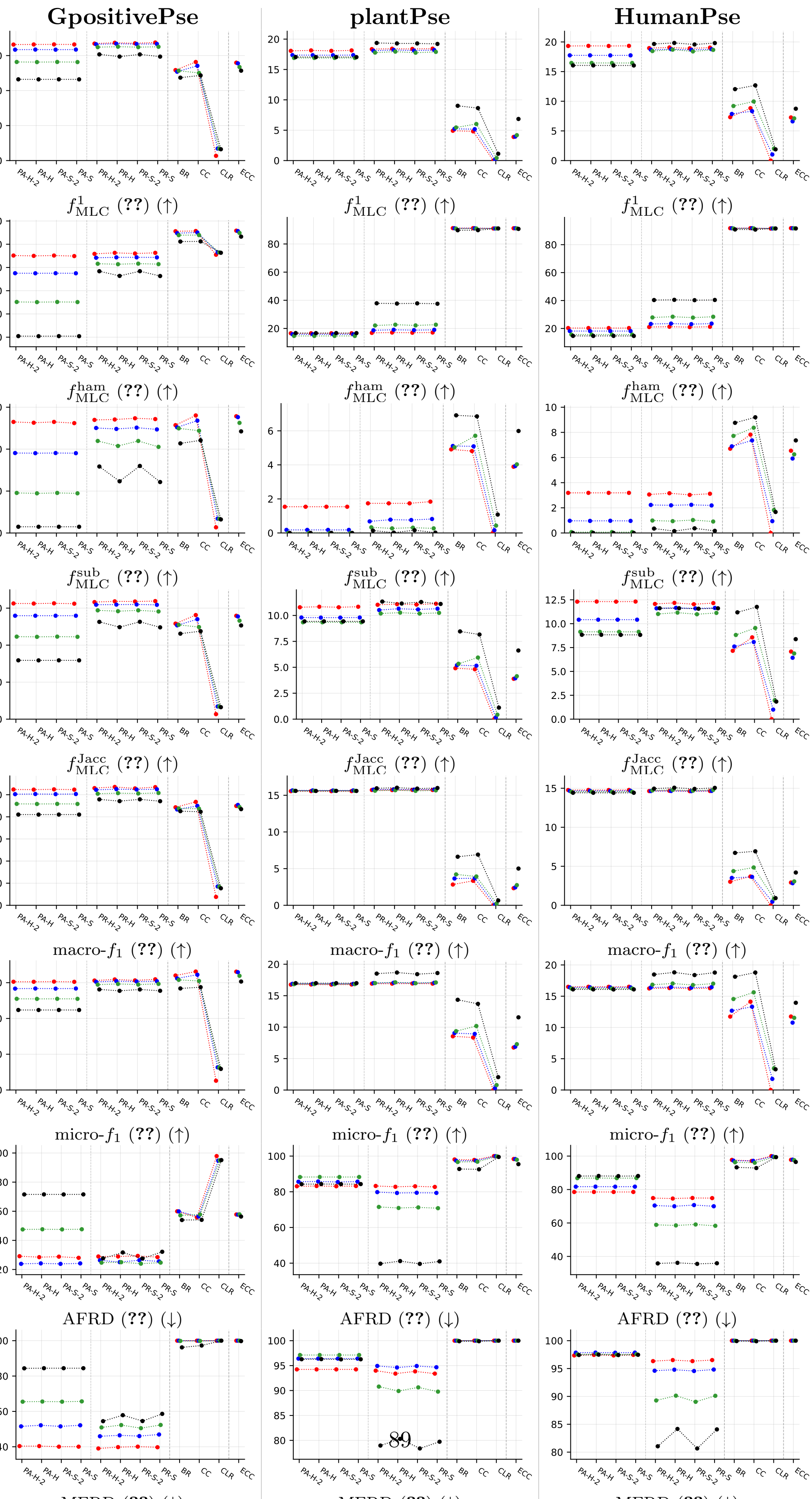
A.2 Partial abstention (PartialAbstention)


CHD-49
Yeast
enron
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)



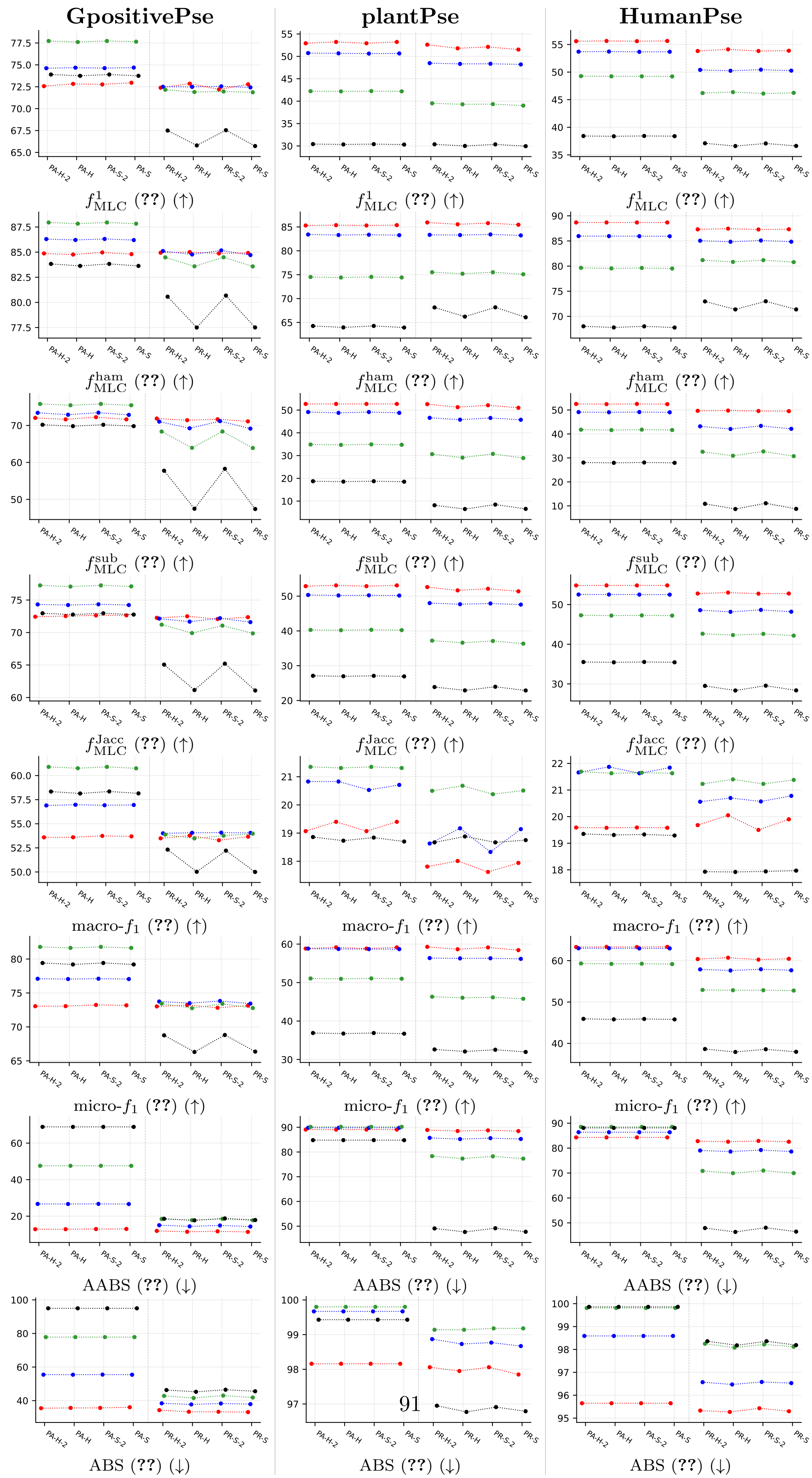
emotions
scene
Water-quality
PA-H-2 PA-H PA-S-2 PA-S PR-H-2 PR-H PR-S-2 PR-S
f^1_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)

A.3 Standard MLC predictions (BinaryVector)



GpositivePse
plantPse
HumanPse
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AFRD (??) (↓)
MFRD (??) (↓)
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC

A.4 Partial abstention (PartialAbstention)


GpositivePse
plantPse
HumanPse
f^{1}_{MLC} (??) (↑)
$f^{\mathrm{ham}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{sub}}_{\mathrm{MLC}}$ (??) (↑)
$f^{\mathrm{Jacc}}_{\mathrm{MLC}}$ (??) (↑)
macro-f_1 (??) (↑)
micro-f_1 (??) (↑)
AABS (??) (↓)
ABS (??) (↓)
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

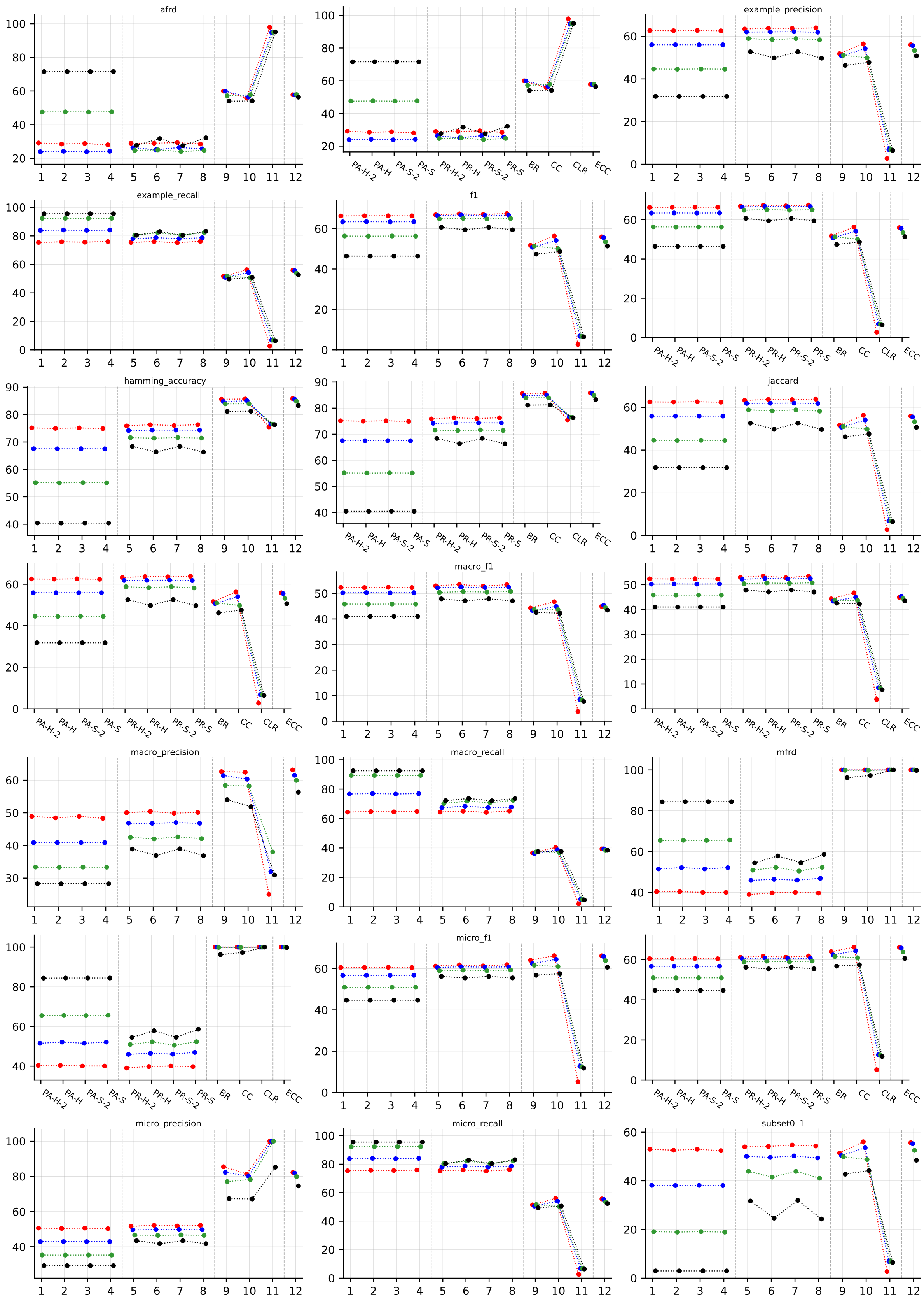
B Per-dataset results (RF base learner)

B.1 GpositivePseAAC ($K = 4$)

RF • `GpositivePseAAC` • BinaryVector — Standard MLC predictions

RF • `GpositivePseAAC` • PartialAbstention — Partial abstention

RF • `GpositivePseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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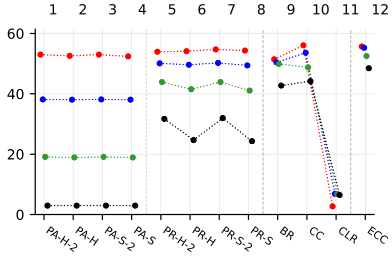
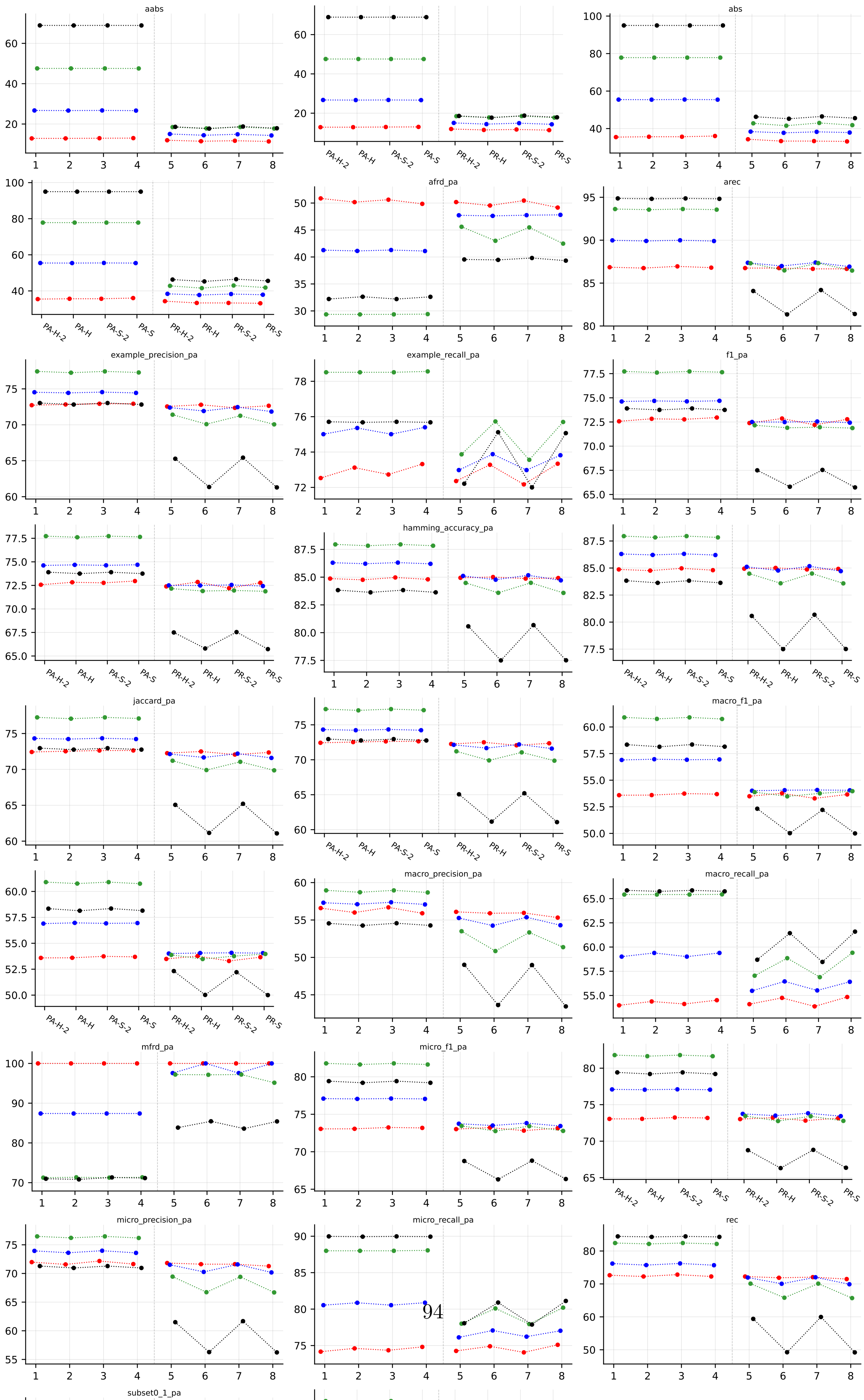
example_recall
f1
hamming_accuracy
jaccard
macro_f1
macro_precision
macro_recall
mfrd
micro_f1
micro_precision
micro_recall
subset0_1
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC

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f1_pa
hamming_accuracy_pa
jaccard_pa
macro_f1_pa
macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
rec
subset0_1_pa
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

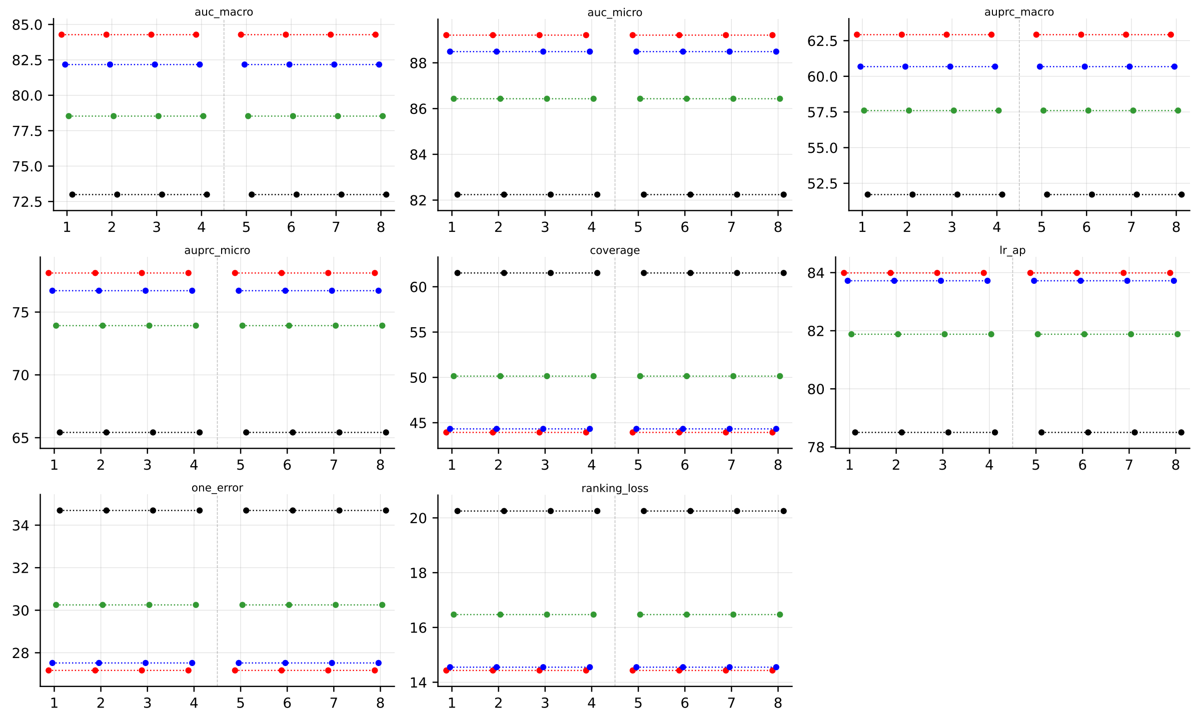


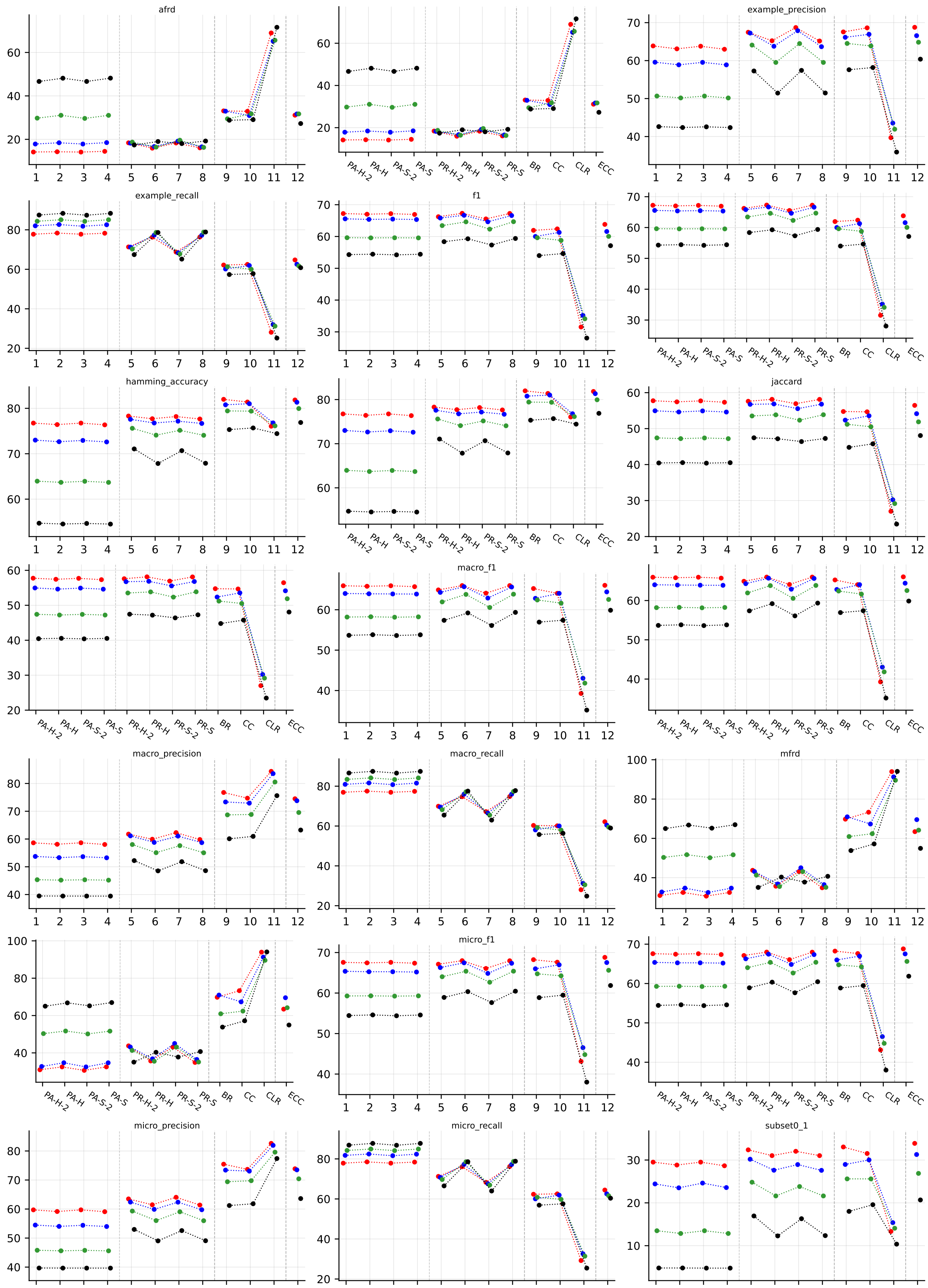
Table 22: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `GpositivePseAAC` (RF base learner).

B.2 Emotions ($K = 6$)

RF • `emotions` • BinaryVector — Standard MLC predictions

RF • `emotions` • PartialAbstention — Partial abstention

RF • `emotions` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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example_recall
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jaccard
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macro_recall
mfrd
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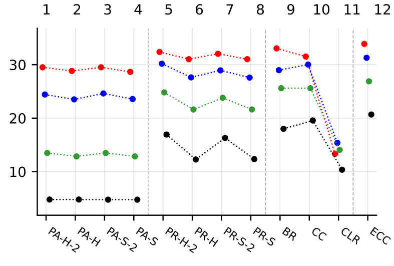
micro_precision
micro_recall
subset0_1
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC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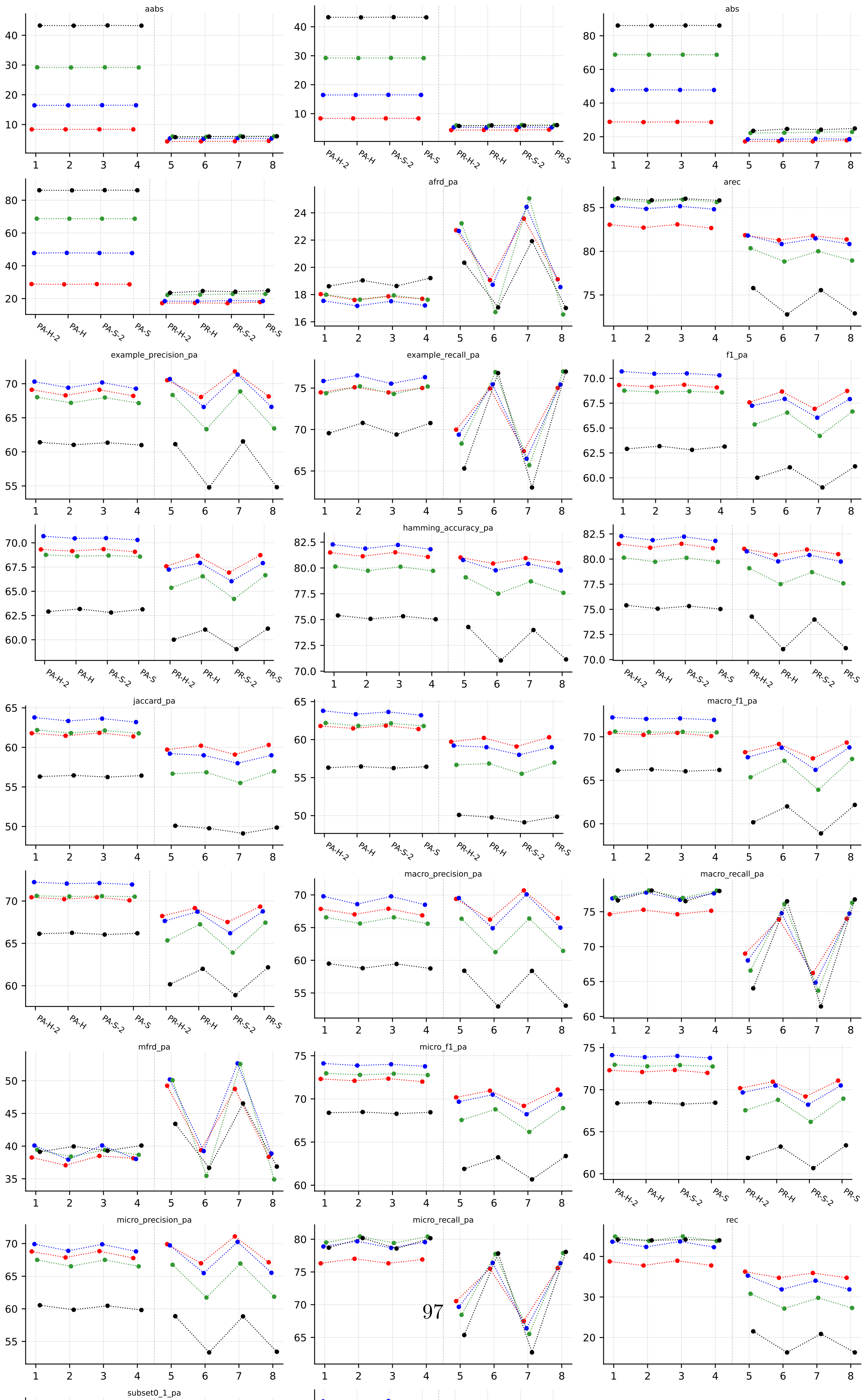
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC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jaccard_pa
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macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
rec
subset0_1_pa
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

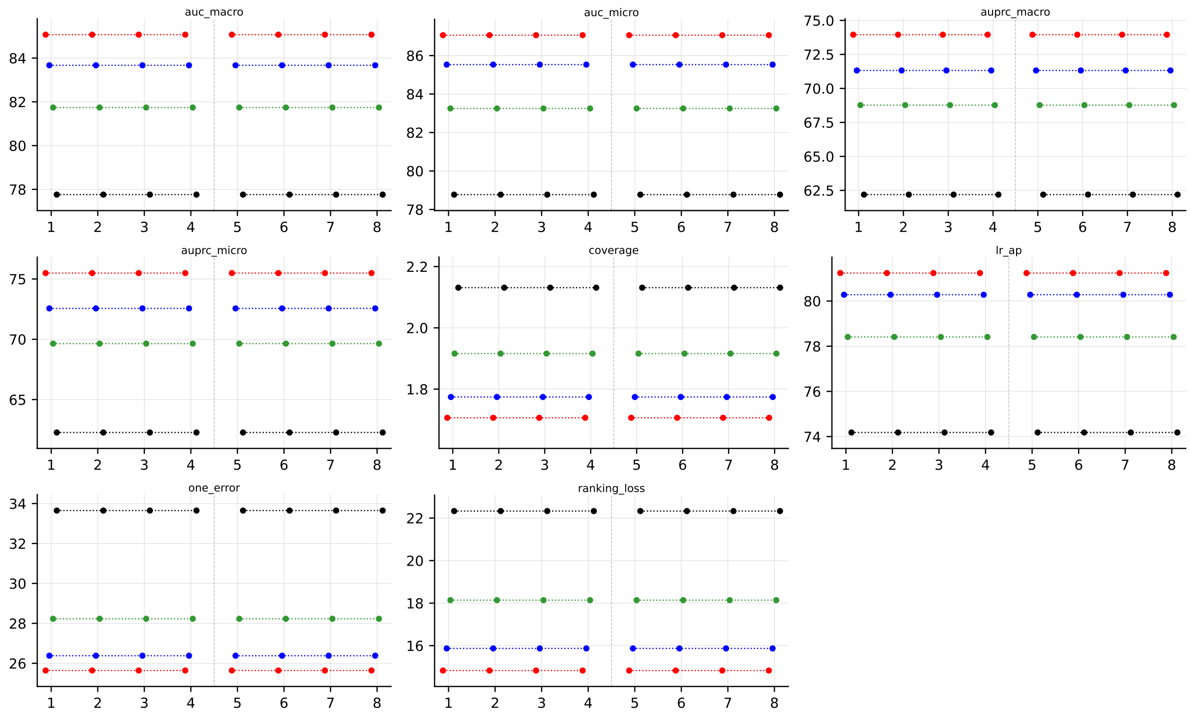


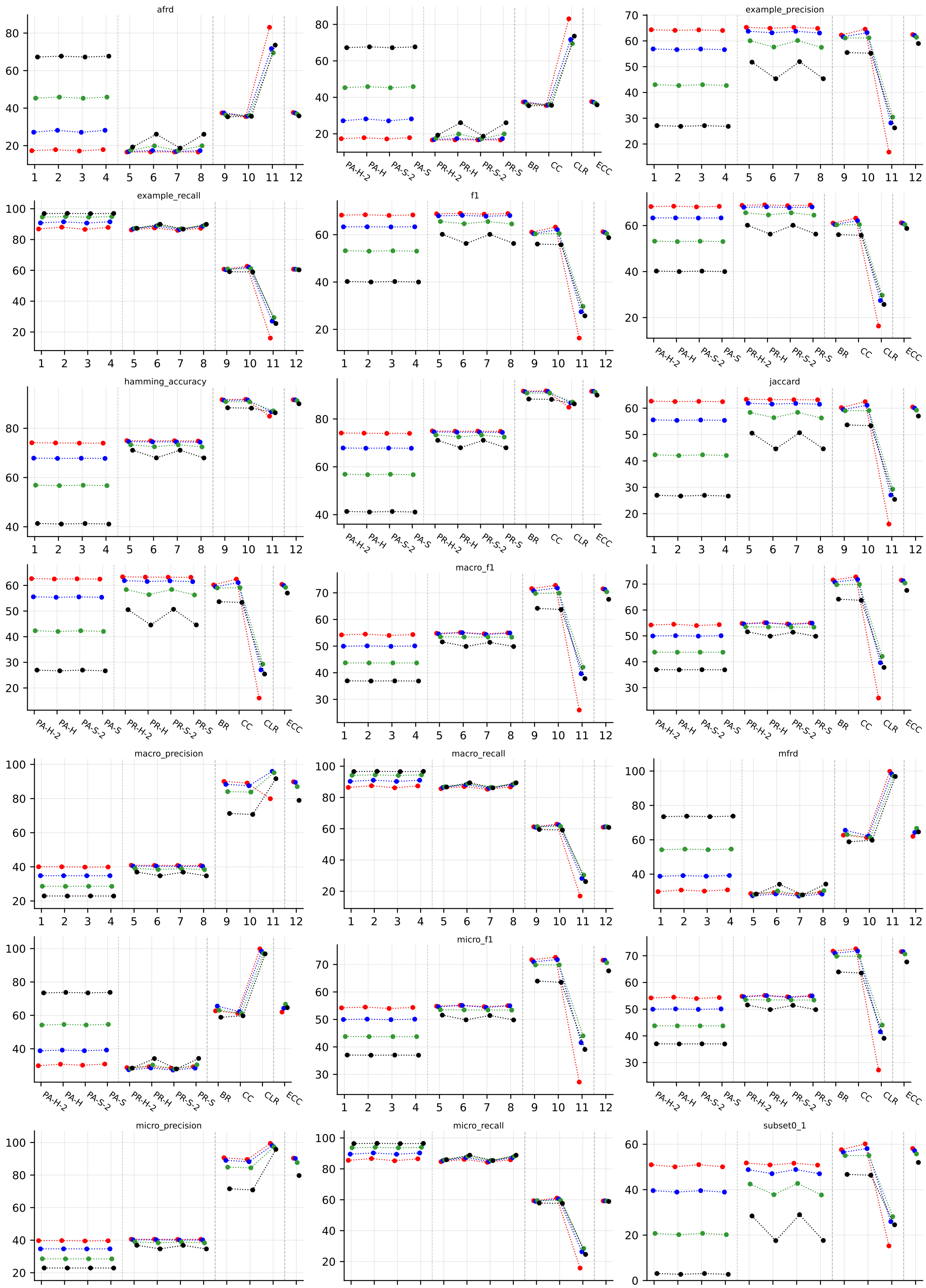
Table 25: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `emotions` (RF base learner).

B.3 Scene ($K = 6$)

RF • `scene` • BinaryVector — Standard MLC predictions

RF • `scene` • PartialAbstention — Partial abstention

RF • `scene` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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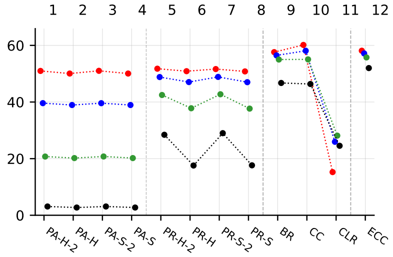
example_recall
f1
hamming_accuracy
jaccard
macro_f1
macro_precision
macro_recall
mfrd
micro_f1
micro_precision
micro_recall
subset0_1
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC


PA-H-2
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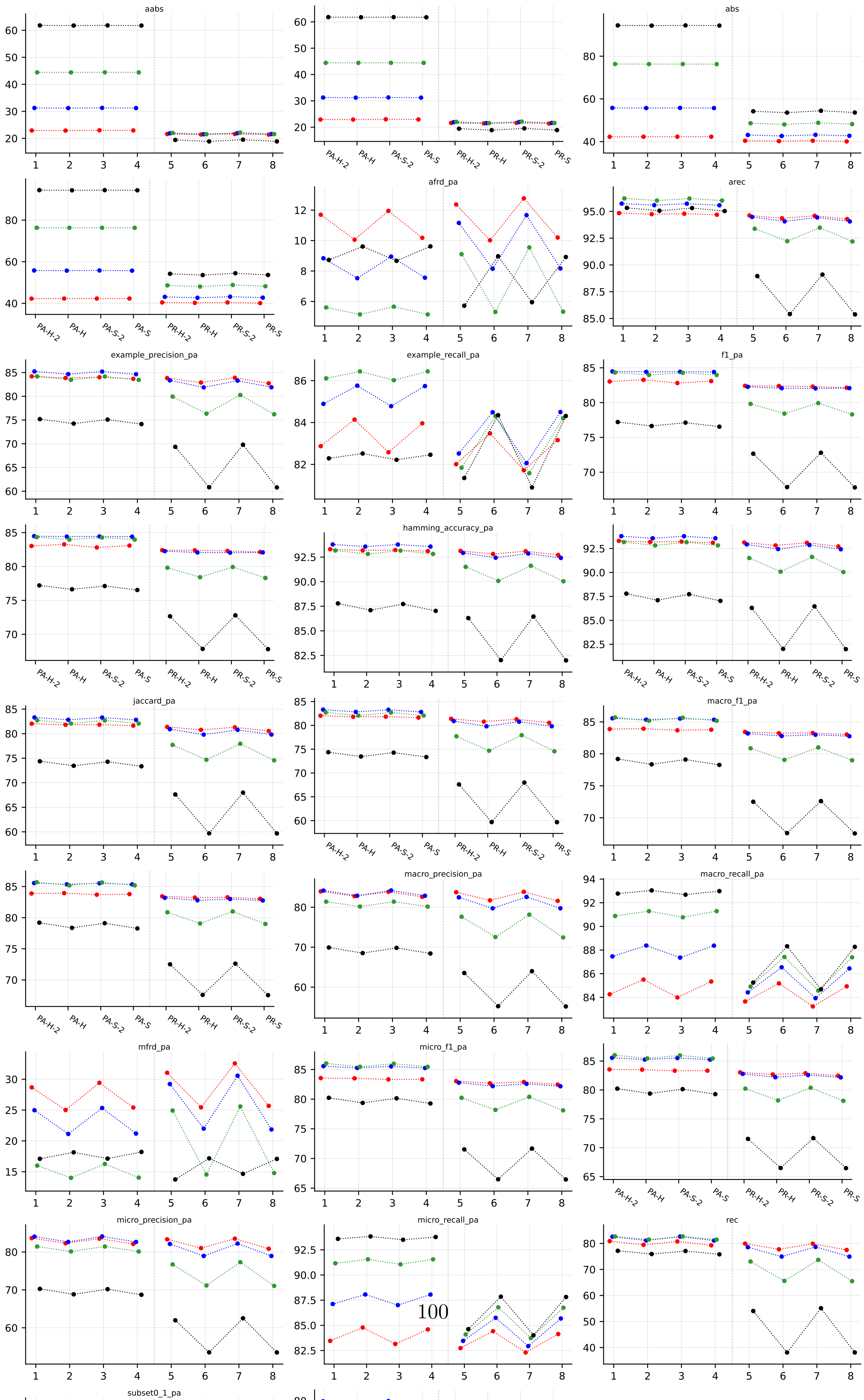
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC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hamming_accuracy_pa
jaccard_pa
macro_f1_pa
macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
rec
subset0_1_pa
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

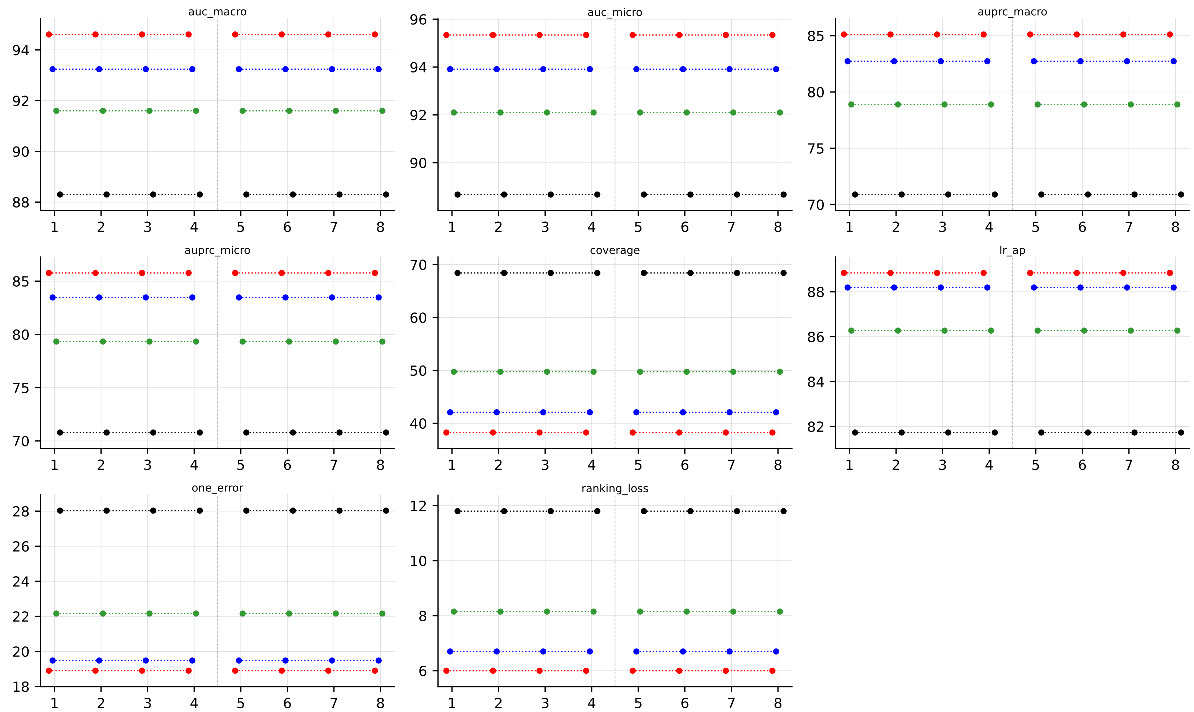


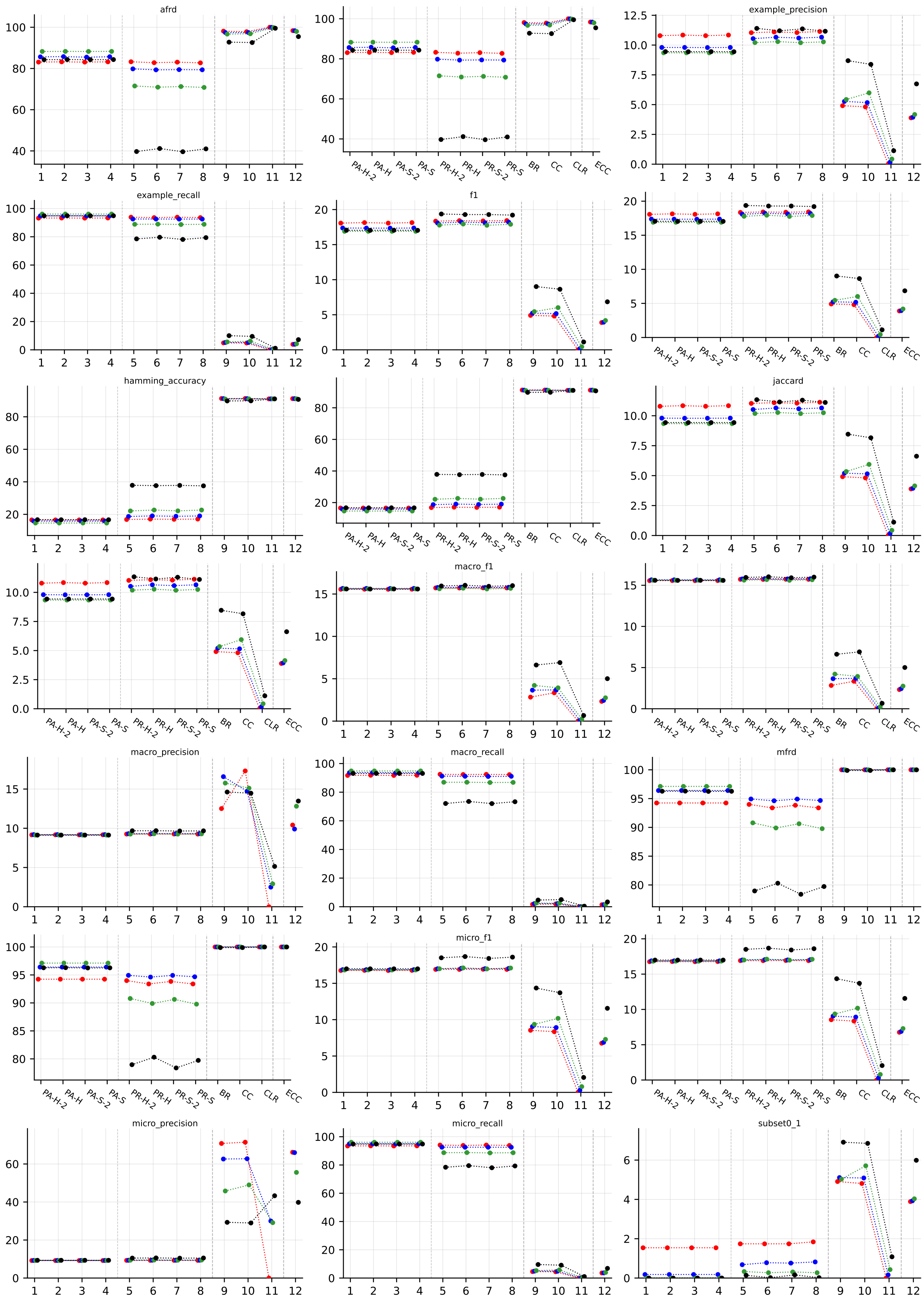
Table 28: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `scene` (RF base learner).

B.4 PlantPseAAC ($K = 12$)

RF • `PlantPseAAC` • BinaryVector — Standard MLC predictions

RF • `PlantPseAAC` • PartialAbstention — Partial abstention

RF • `PlantPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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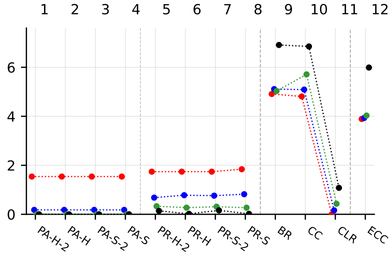
PR-S
BR
CC
CLR
ECC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PR-S
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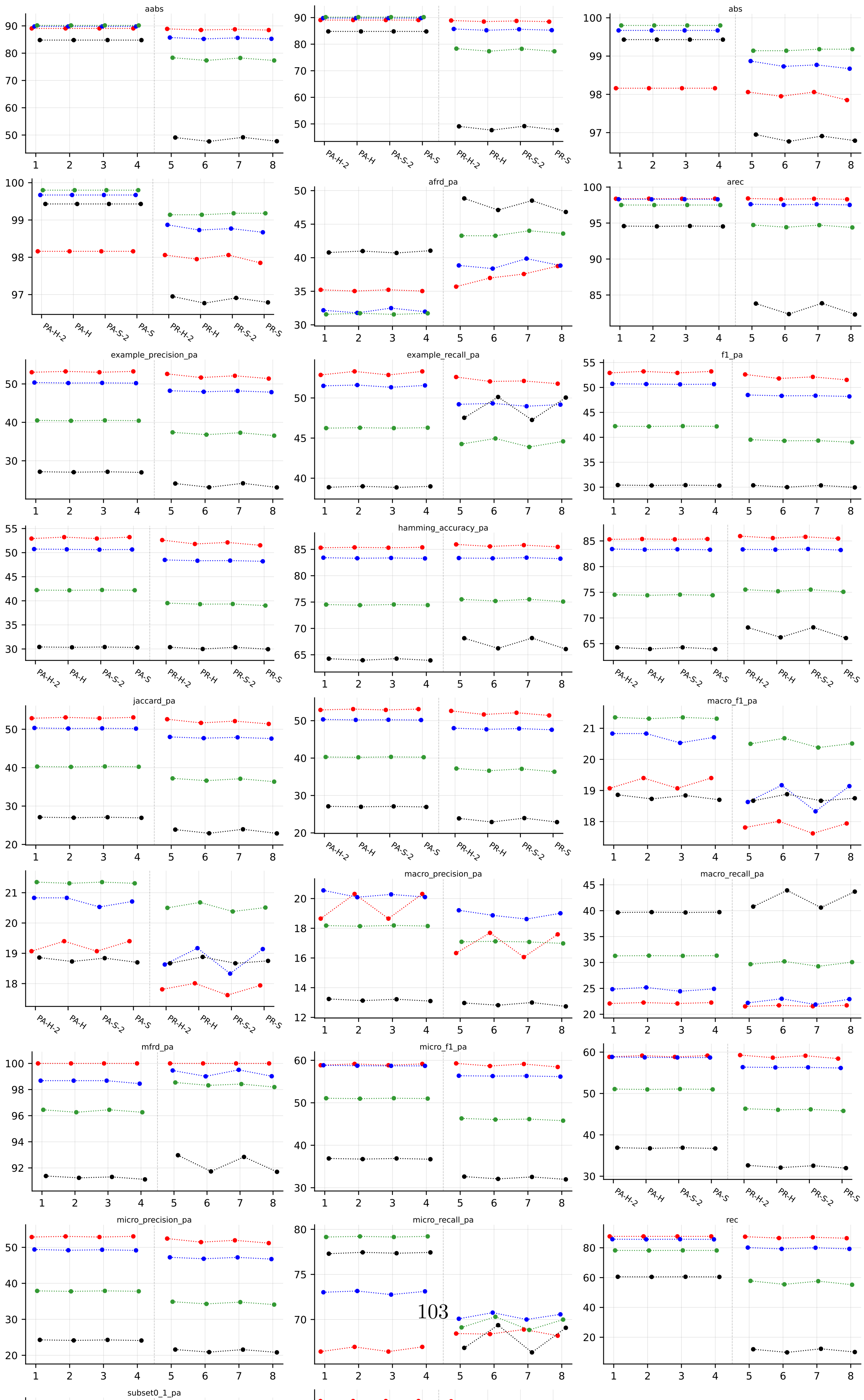
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macro_recall_pa
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micro_recall_pa
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PR-S-2
PR-S

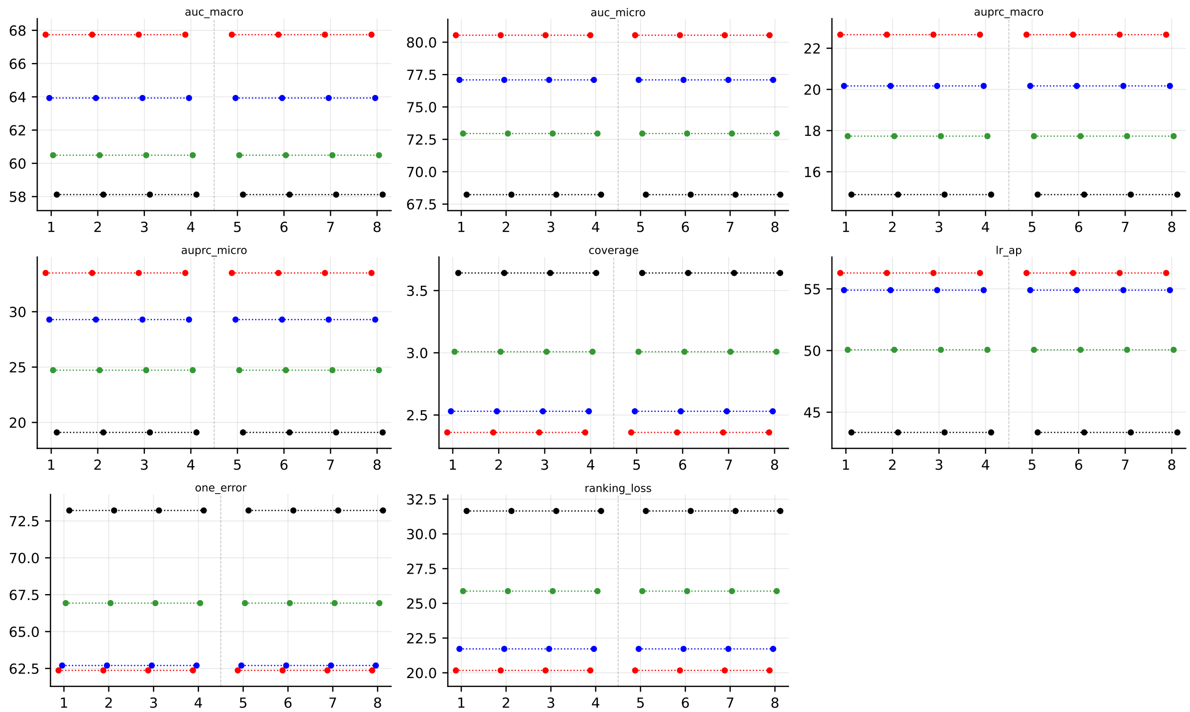


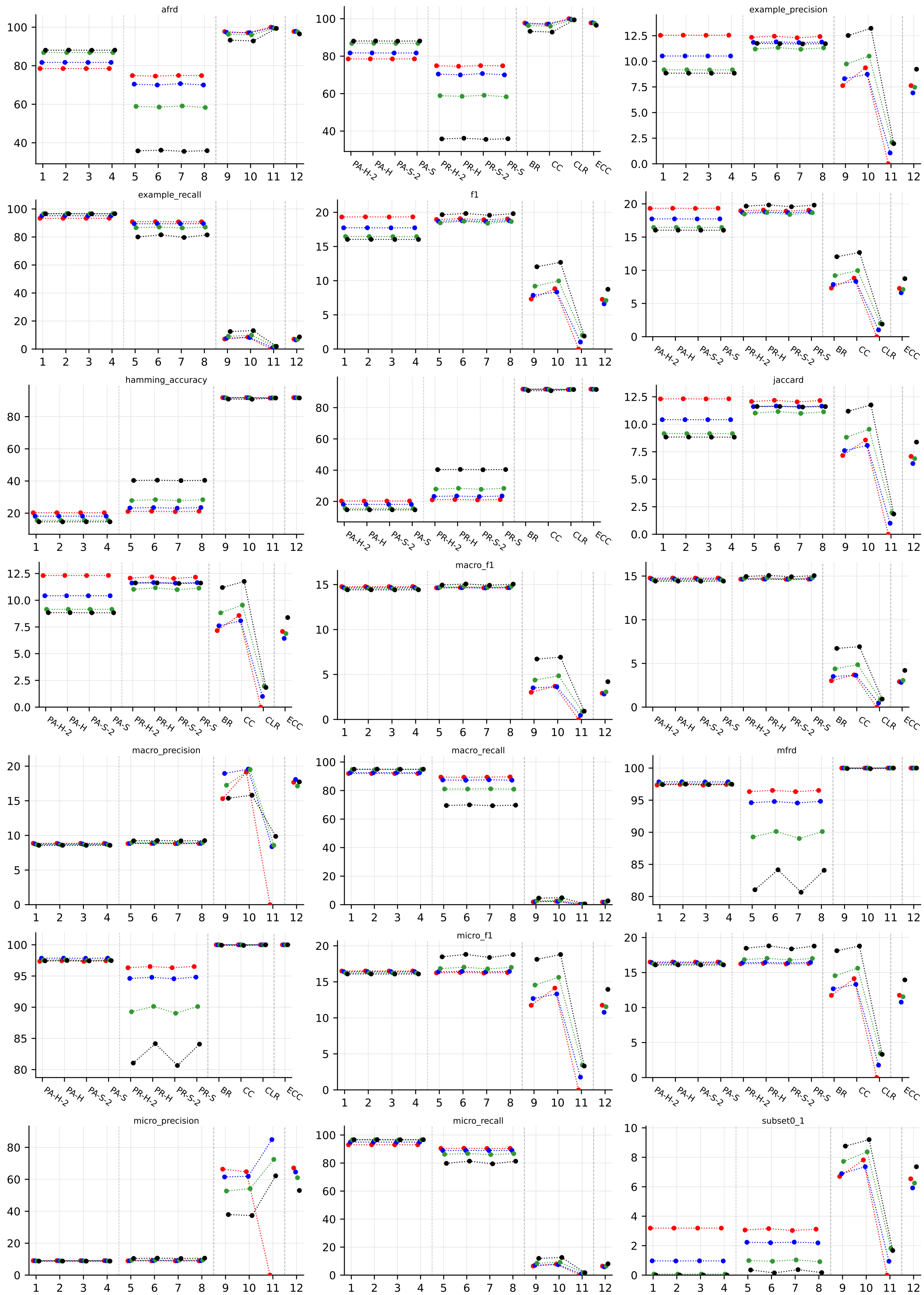
Table 31: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `PlantPseAAC` (RF base learner).

B.5 HumanPseAAC ($K = 14$)

RF • `HumanPseAAC` • BinaryVector — Standard MLC predictions

RF • `HumanPseAAC` • PartialAbstention — Partial abstention

RF • `HumanPseAAC` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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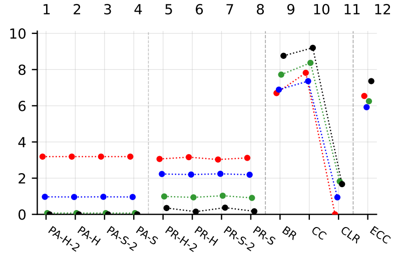

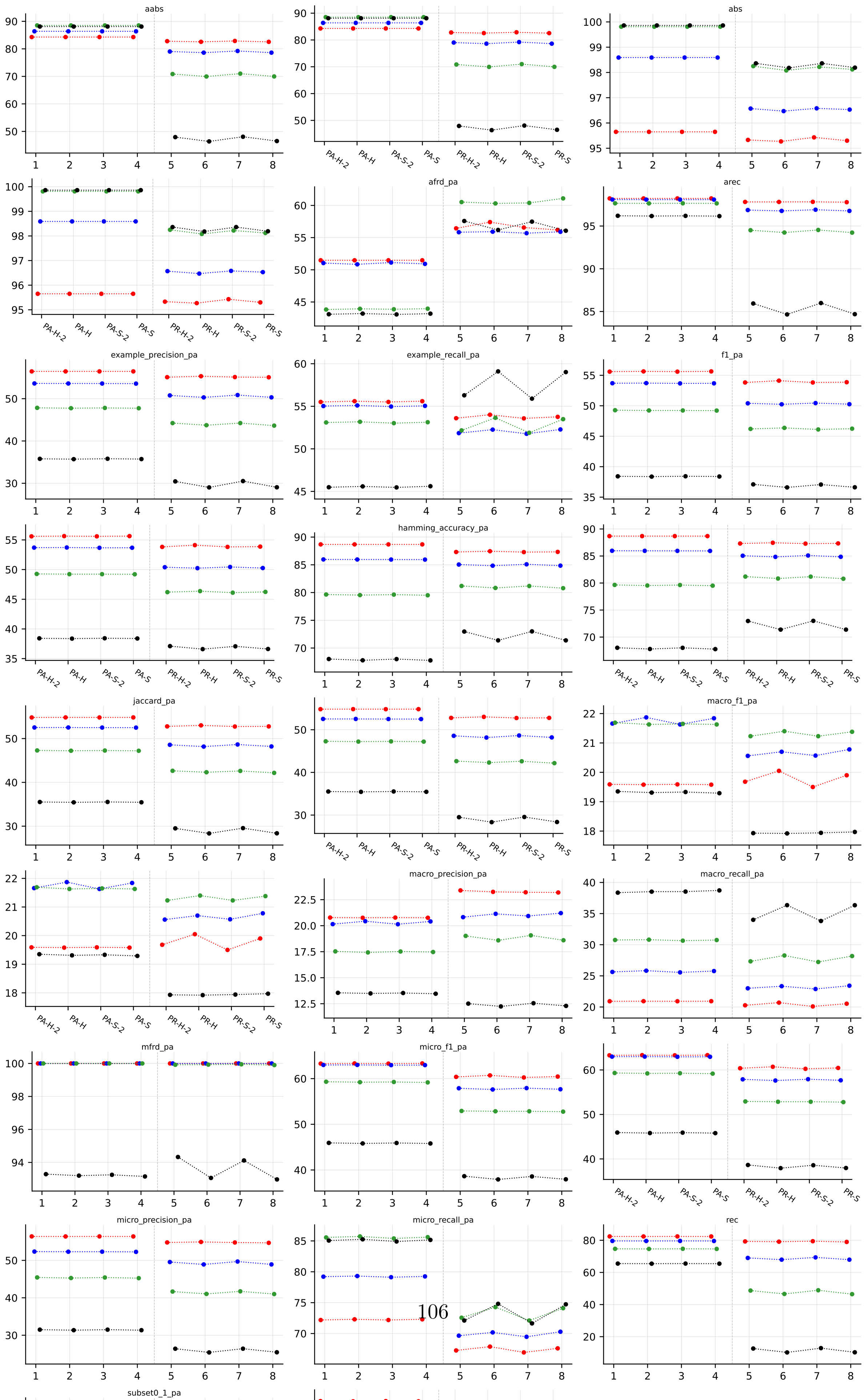
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jaccard_pa
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macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
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PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

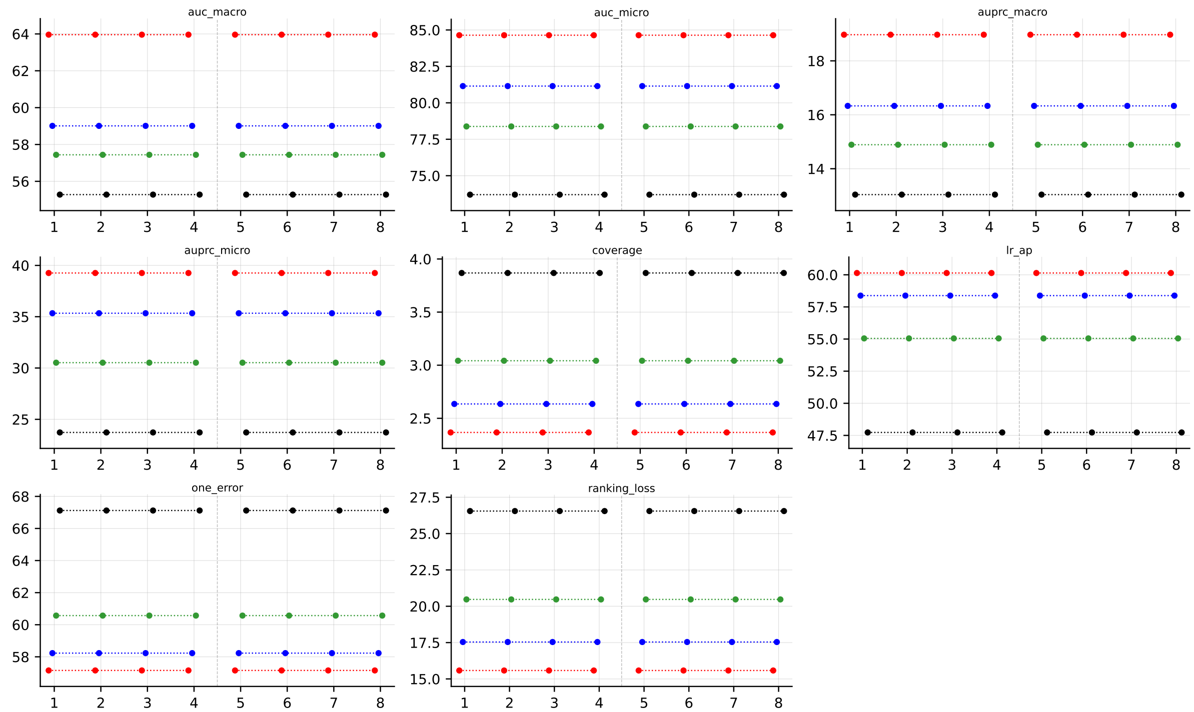


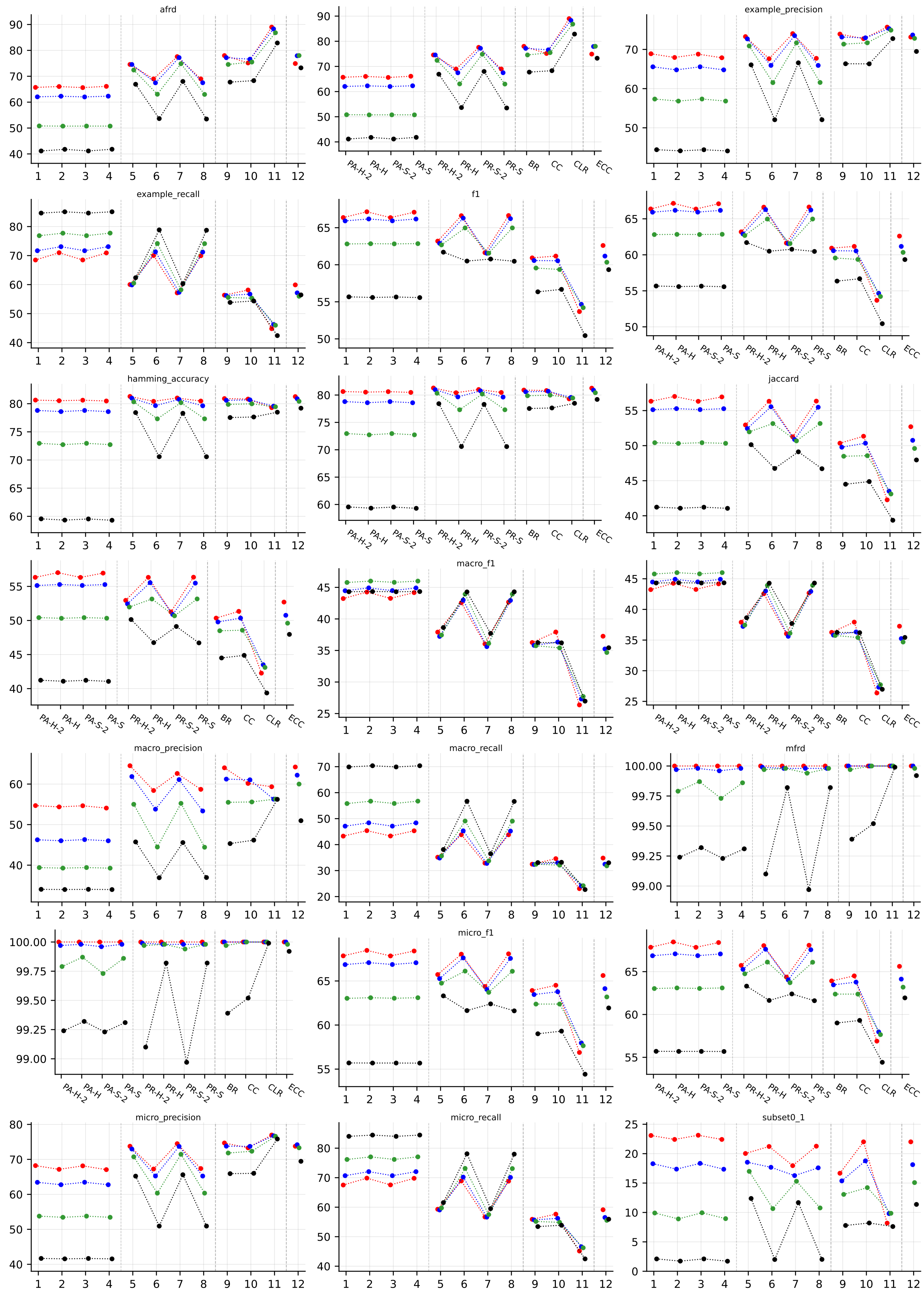
Table 34: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `HumanPseAAC` (RF base learner).

B.6 Yeast ($K = 14$)

RF • `Yeast` • BinaryVector — Standard MLC predictions

RF • `Yeast` • PartialAbstention — Partial abstention

RF • `Yeast` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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jaccard
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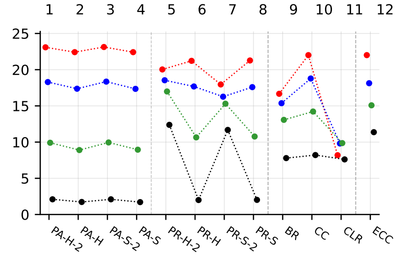
macro_recall
mfrd
micro_f1
micro_precision
micro_recall
subset0_1
PA-H-2 PA-H PA-S-2 PA-S PR-H-2 PR-H PR-S-2 PR-S BR CC CLR ECC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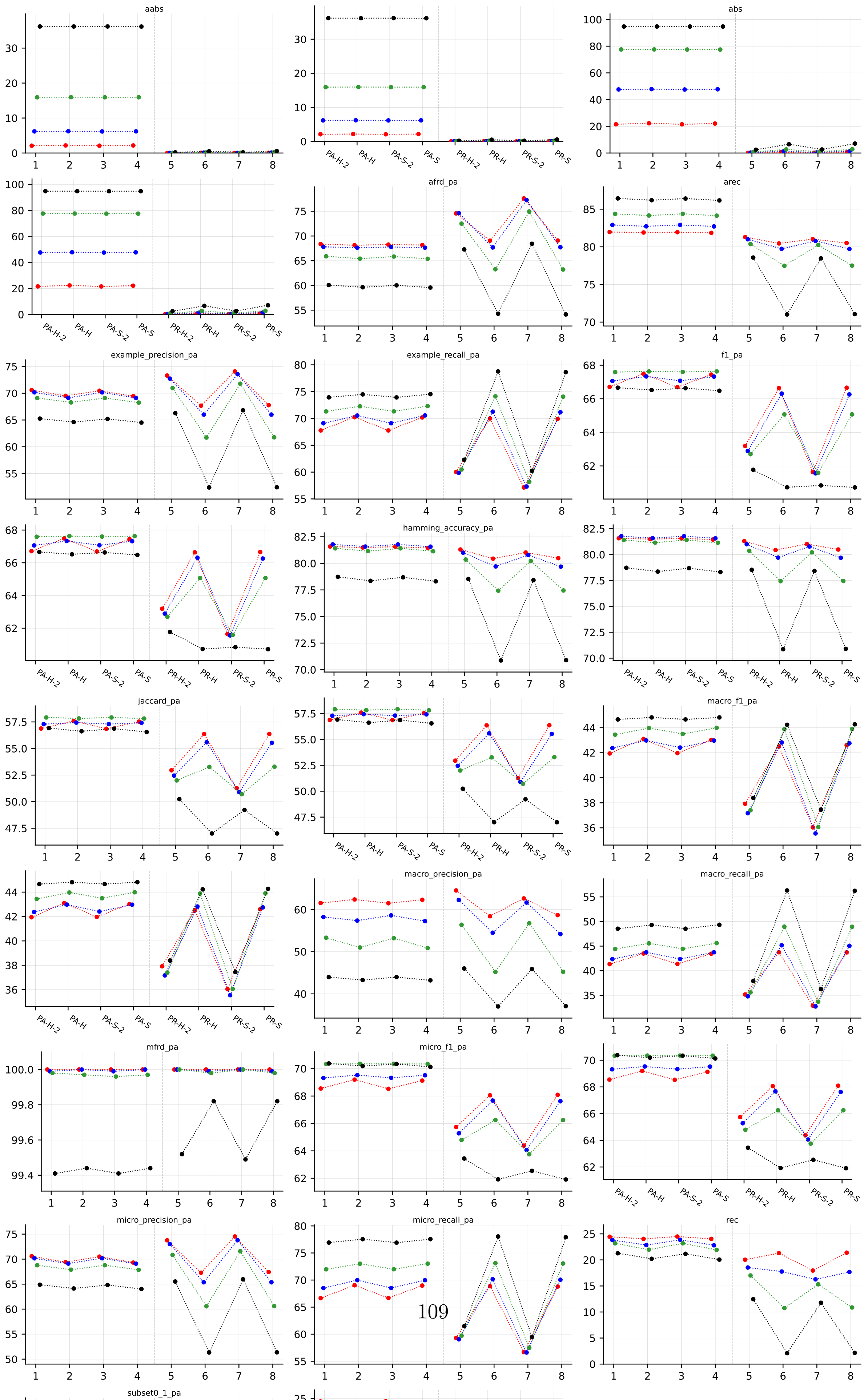
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macro_recall_pa
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micro_precision_pa
micro_recall_pa
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PA-S-2
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PR-H-2
PR-H
PR-S-2
PR-S

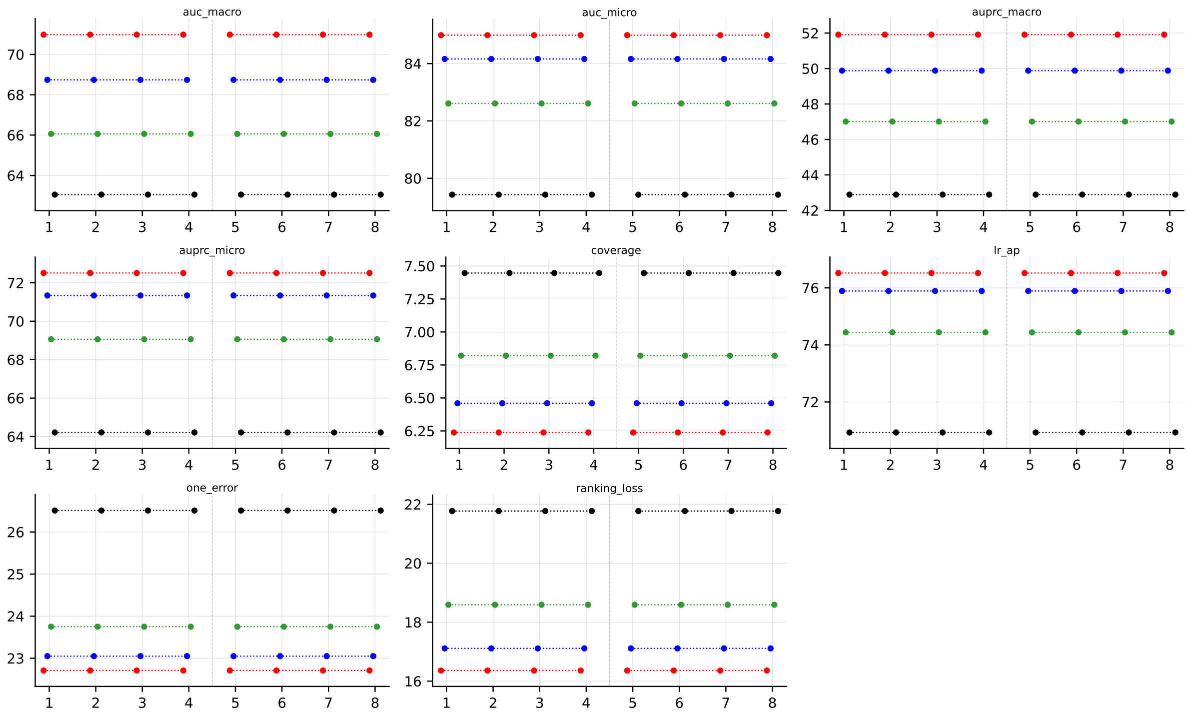


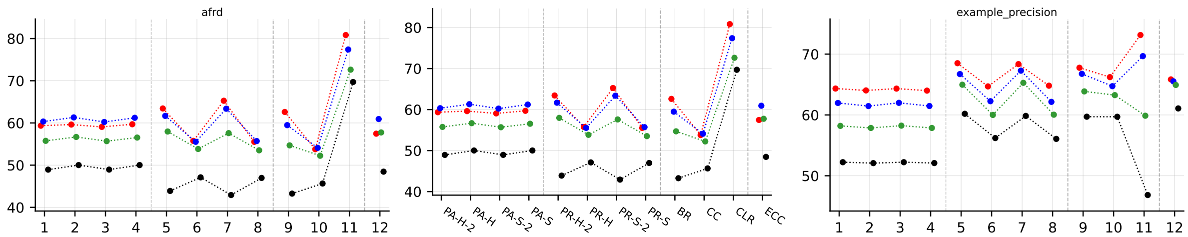
Table 37: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `Yeast` (RF base learner).

B.7 CHD_49 ($K = 6$)

RF • `CHD_49` • BinaryVector — Standard MLC predictions

RF • `CHD_49` • PartialAbstention — Partial abstention

RF • `CHD_49` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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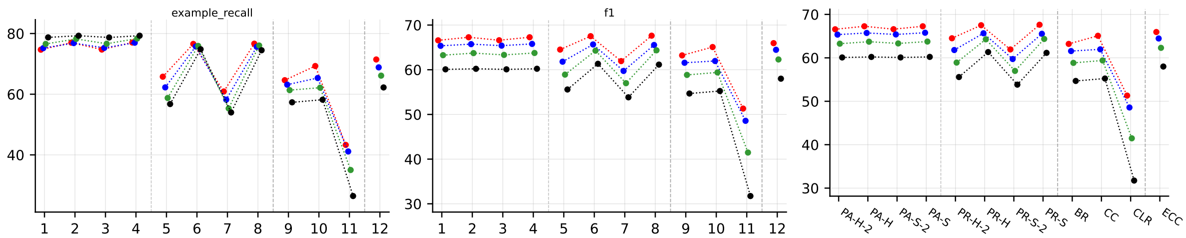
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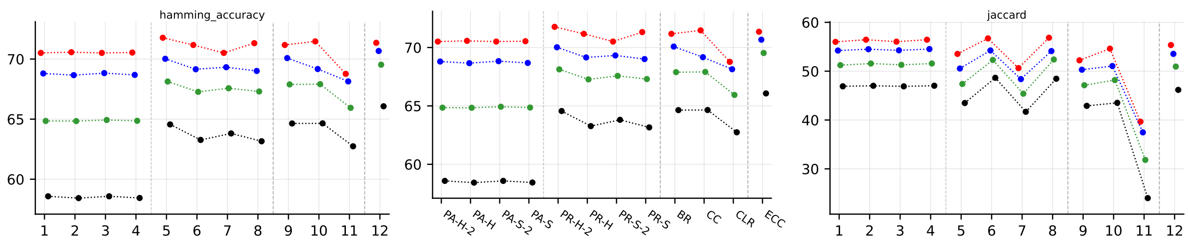
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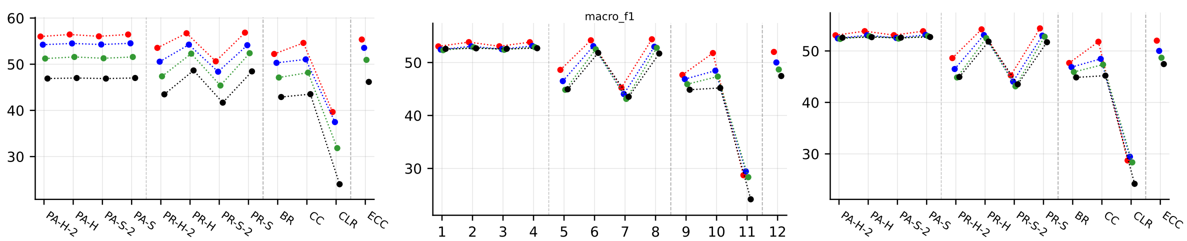
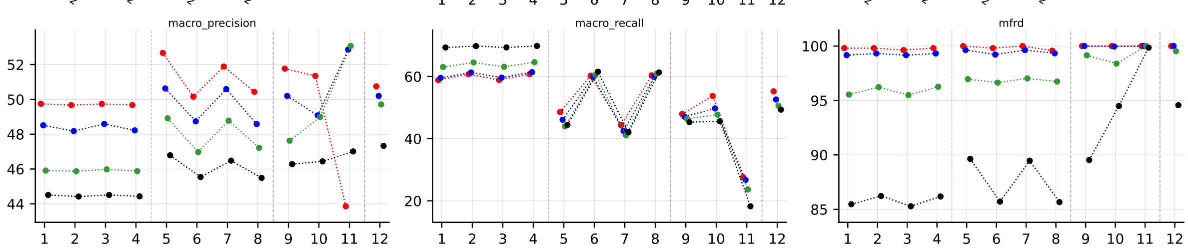
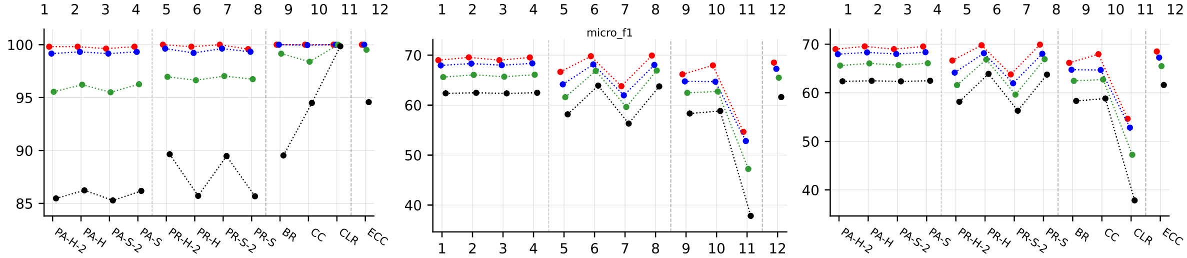
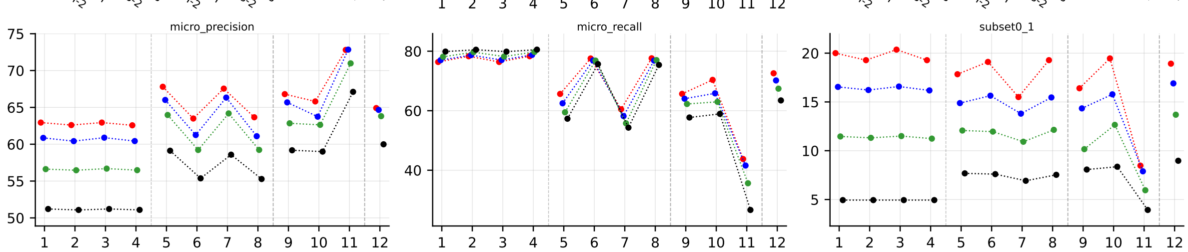

macro_f1


macro_precision
macro_recall
mfrd


micro_f1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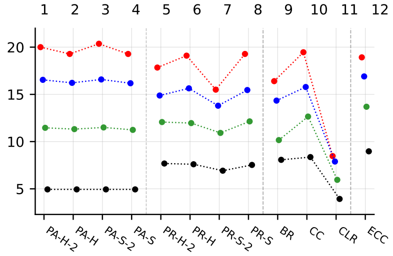

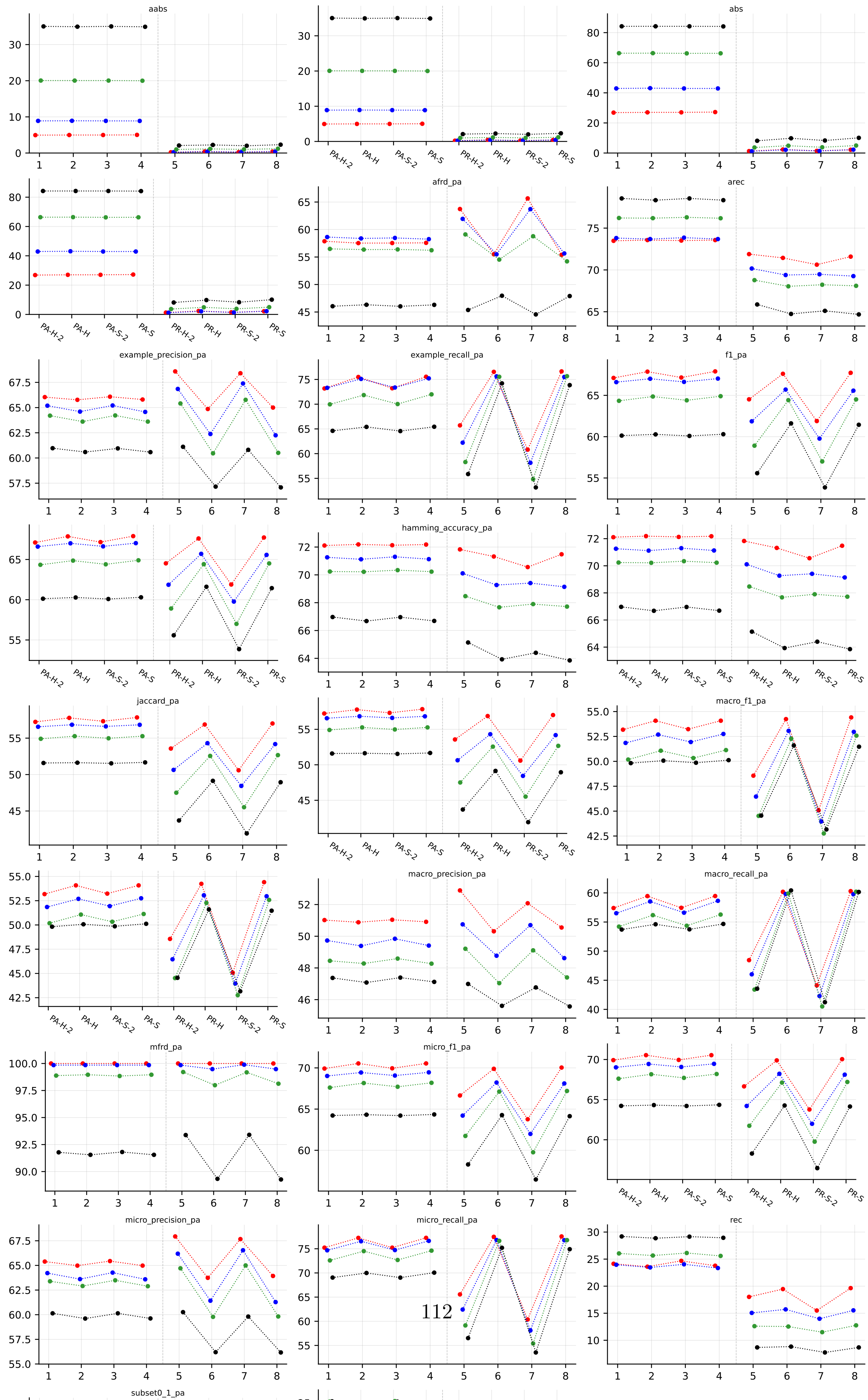
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f1_pa
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macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
rec
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PA-S-2
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PR-H-2
PR-H
PR-S-2
PR-S

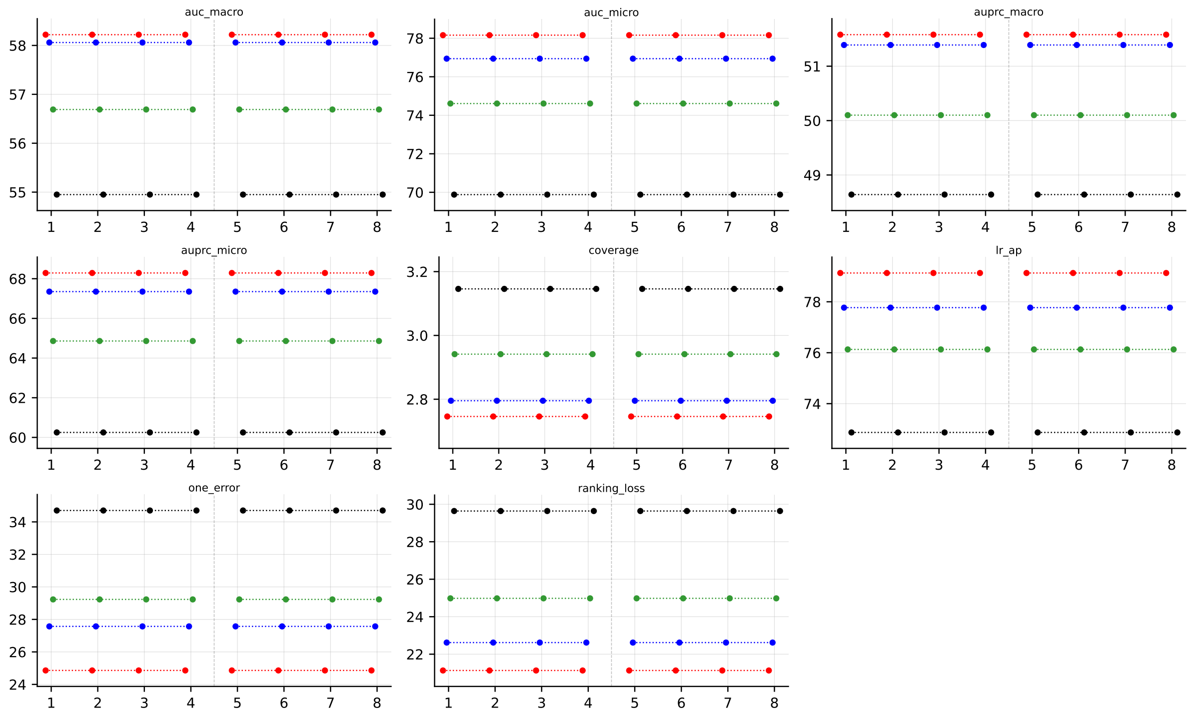


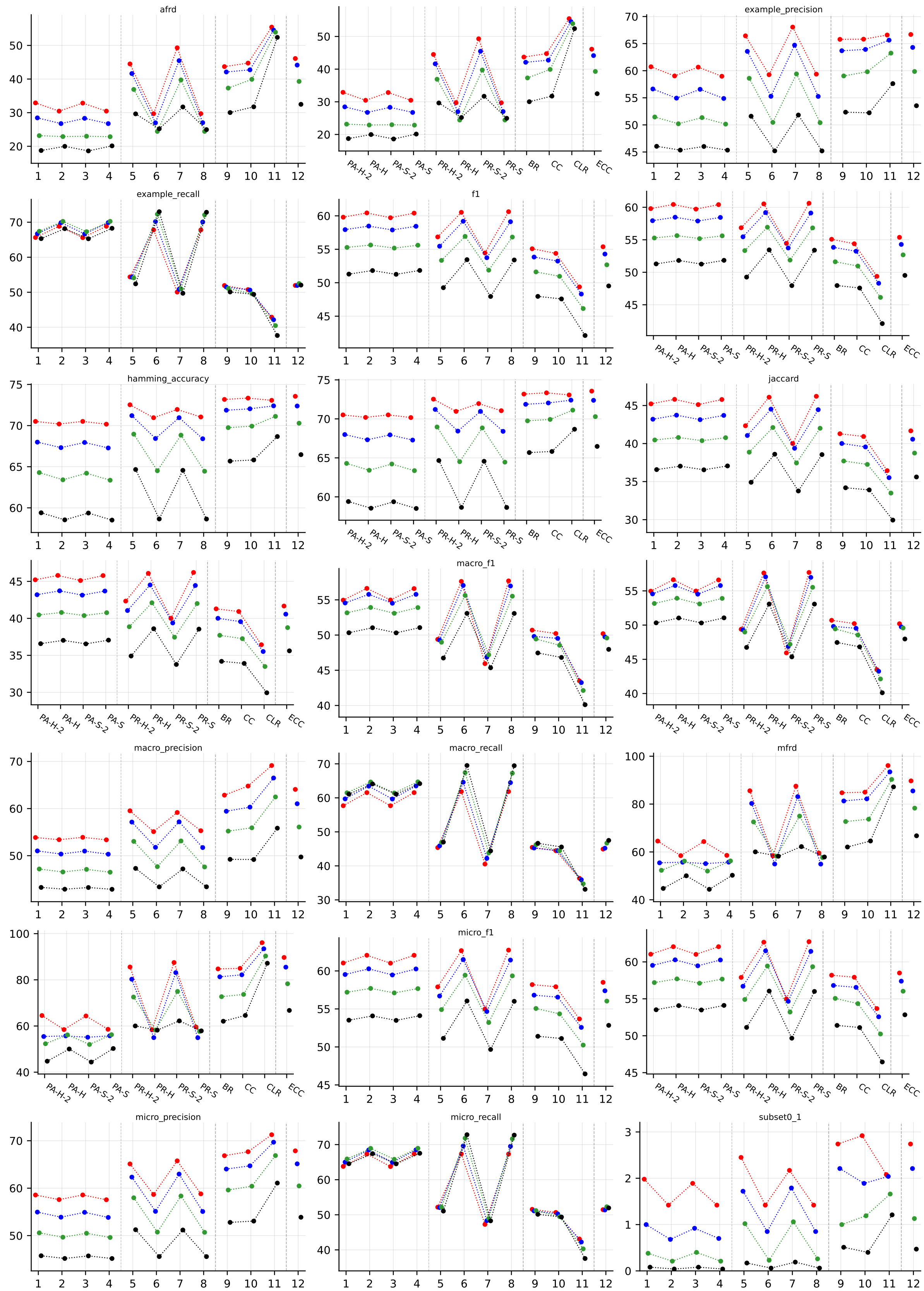
Table 40: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `CHD_49` (RF base learner).

B.8 Water-quality ($K = 14$)

RF • `Water-quality` • BinaryVector — Standard MLC predictions

RF • `Water-quality` • PartialAbstention — Partial abstention

RF • `Water-quality` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)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jaccard
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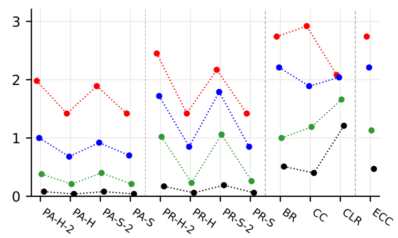
macro_precision
macro_recall
mfrd
micro_f1
micro_precision
micro_recall
subset0_1
PA-H-2
PA-H
PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S
BR
CC
CLR
ECC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PA-S-2
PA-S
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PR-H
PR-S-2
PR-S
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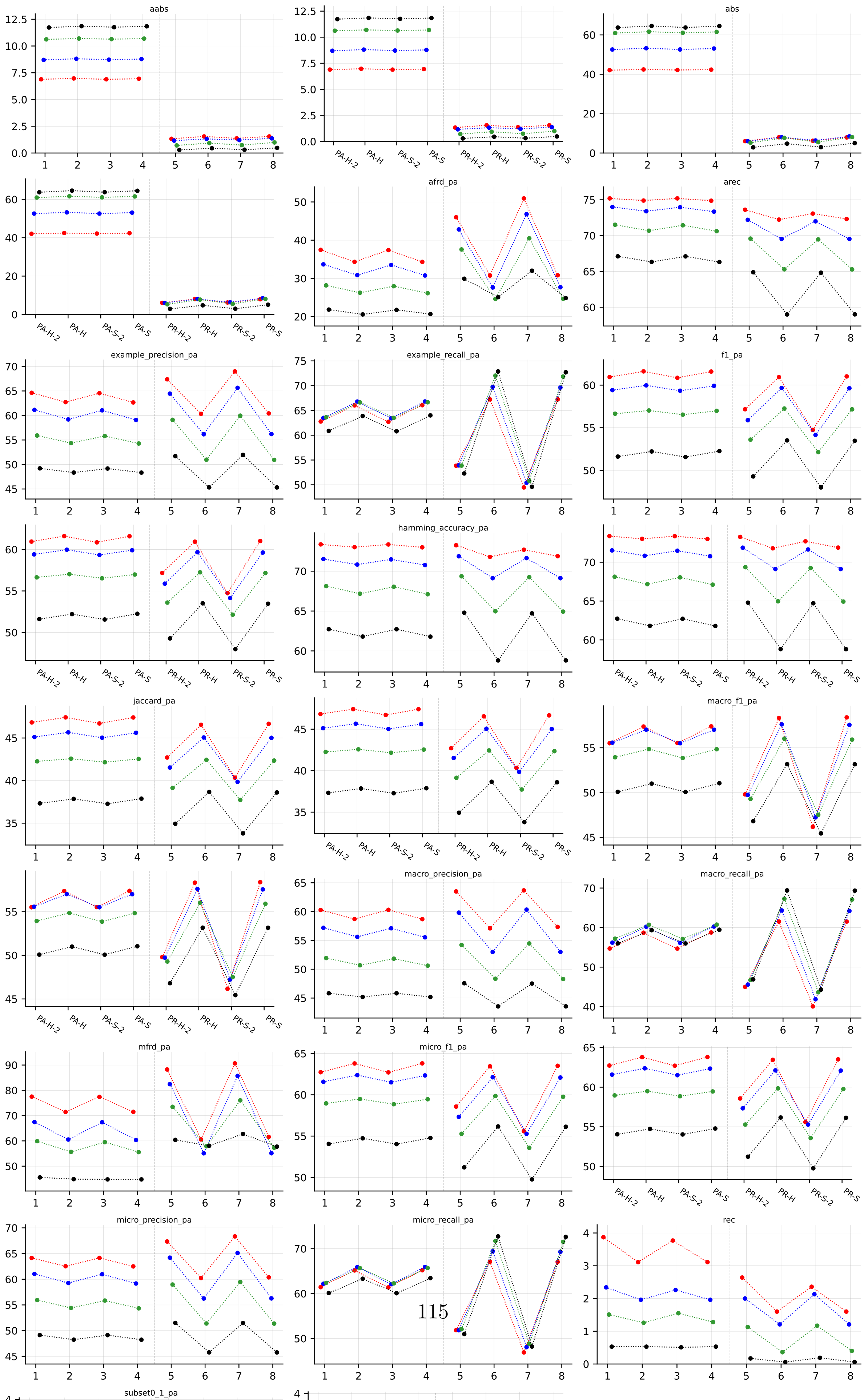
CC
CLR
ECC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macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
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PA-S-2
PA-S
PR-H-2
PR-H
PR-S-2
PR-S

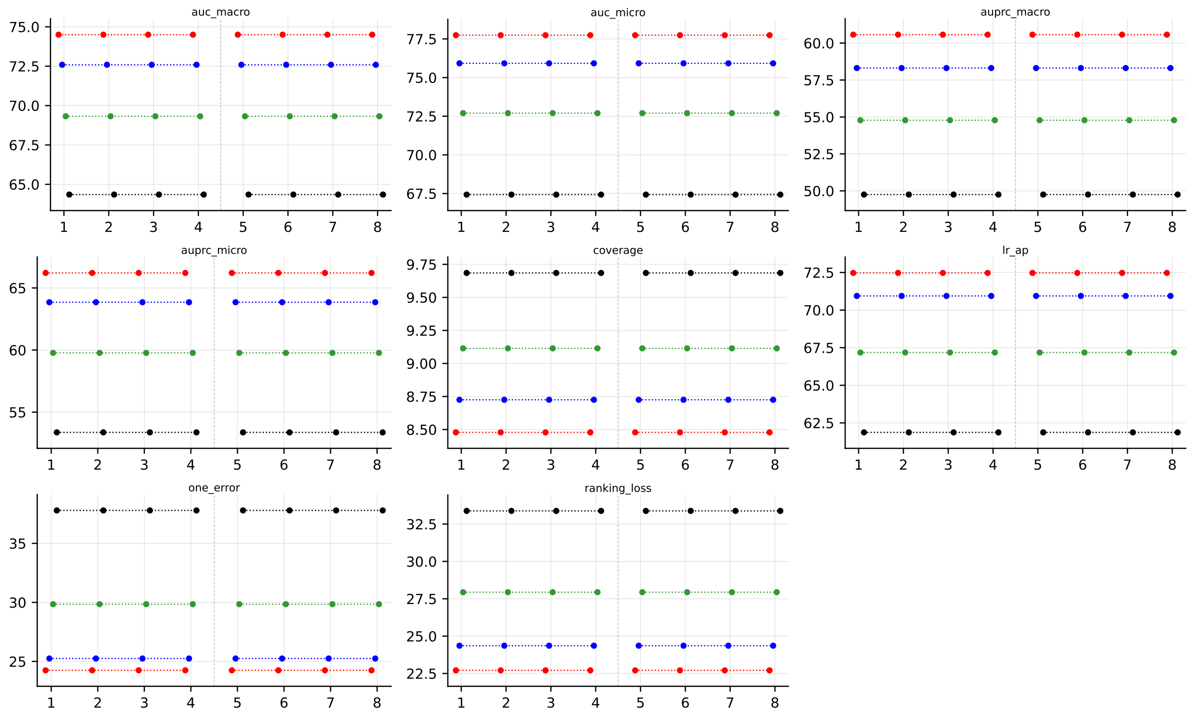


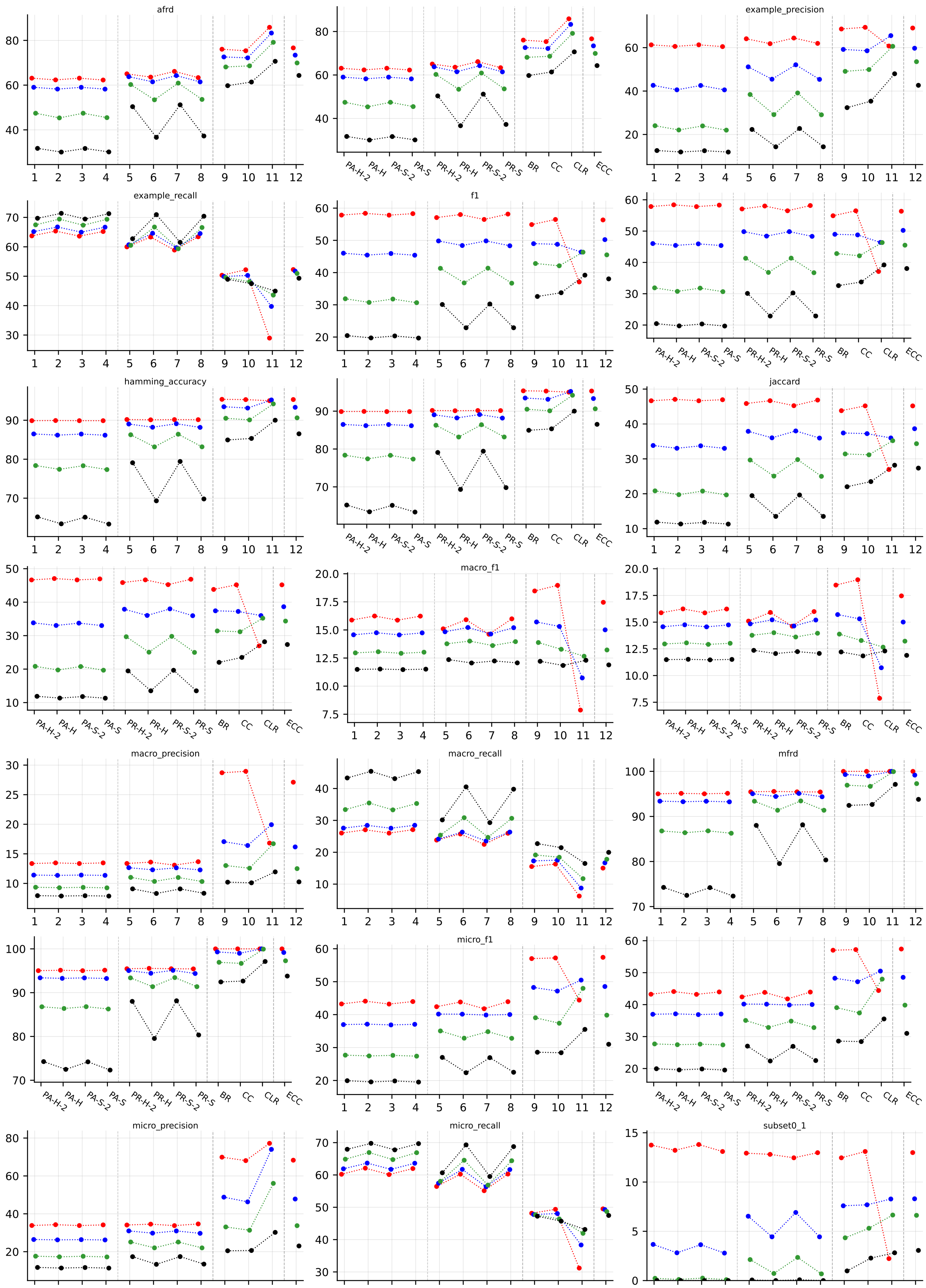
Table 43: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `Water-quality` (RF base learner).

B.9 enron ($K = 53$)

RF • `enron` • BinaryVector — Standard MLC predictions

RF • `enron` • PartialAbstention — Partial abstention

RF • `enron` • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)



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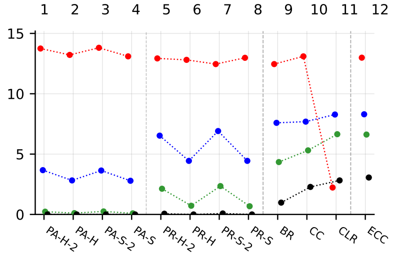
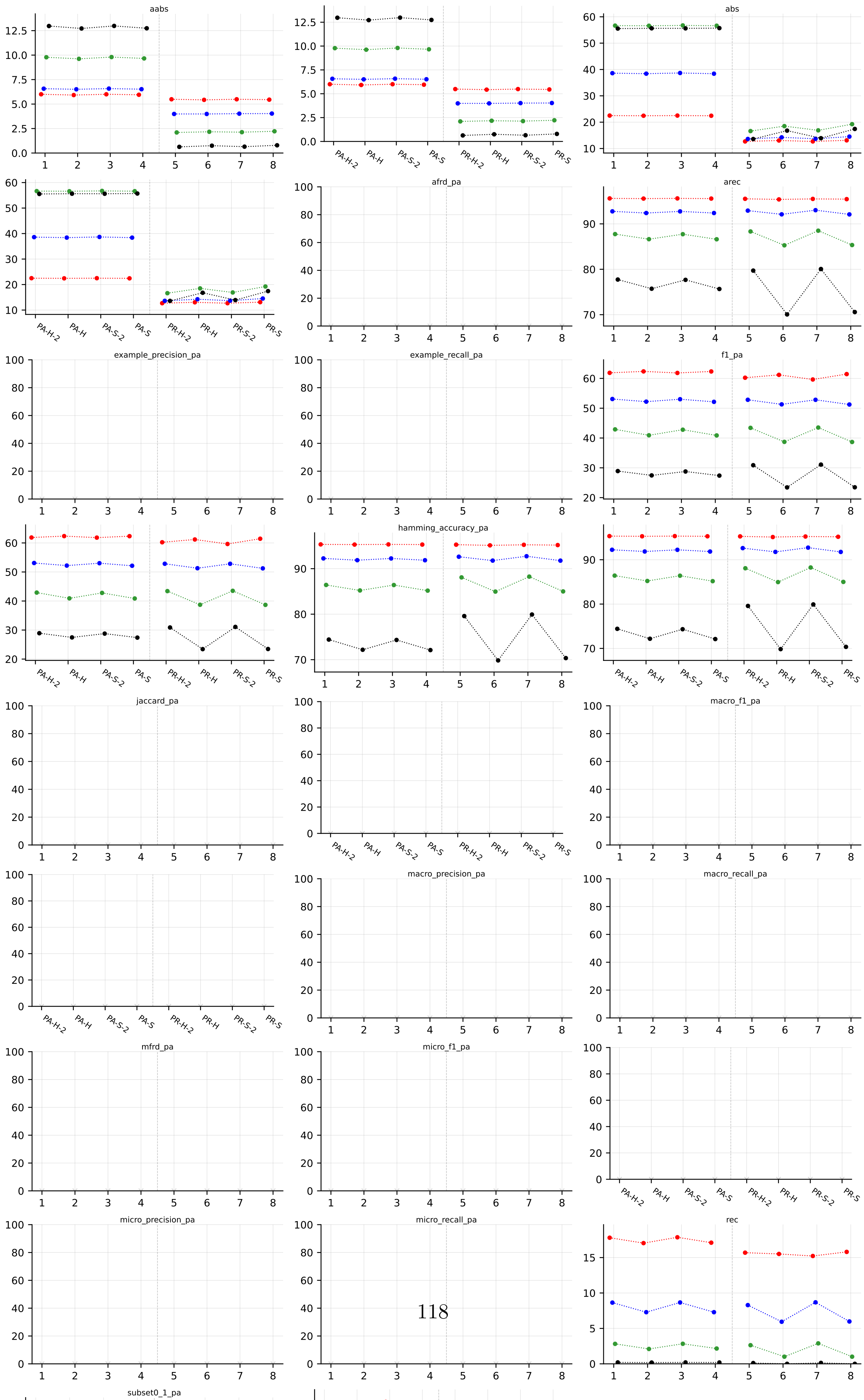
macro_precision
macro_recall
mfrd
micro_f1
micro_precision
micro_recall
subset0_1




aabs
abs
afrd_pa
arec
example_precision_pa
example_recall_pa
f1_pa
hamming_accuracy_pa
jaccard_pa
macro_f1_pa
macro_precision_pa
macro_recall_pa
mfrd_pa
micro_f1_pa
micro_precision_pa
micro_recall_pa
rec
subset0_1_pa

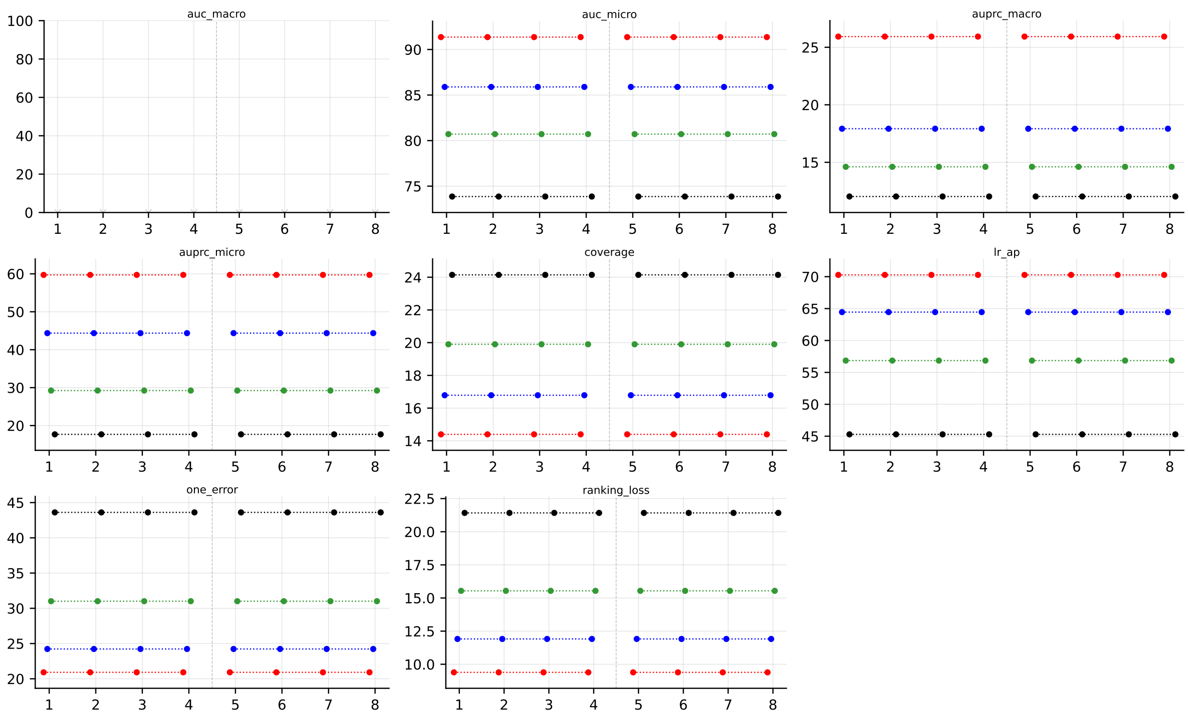


Table 46: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `enron` (RF base learner).

C LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

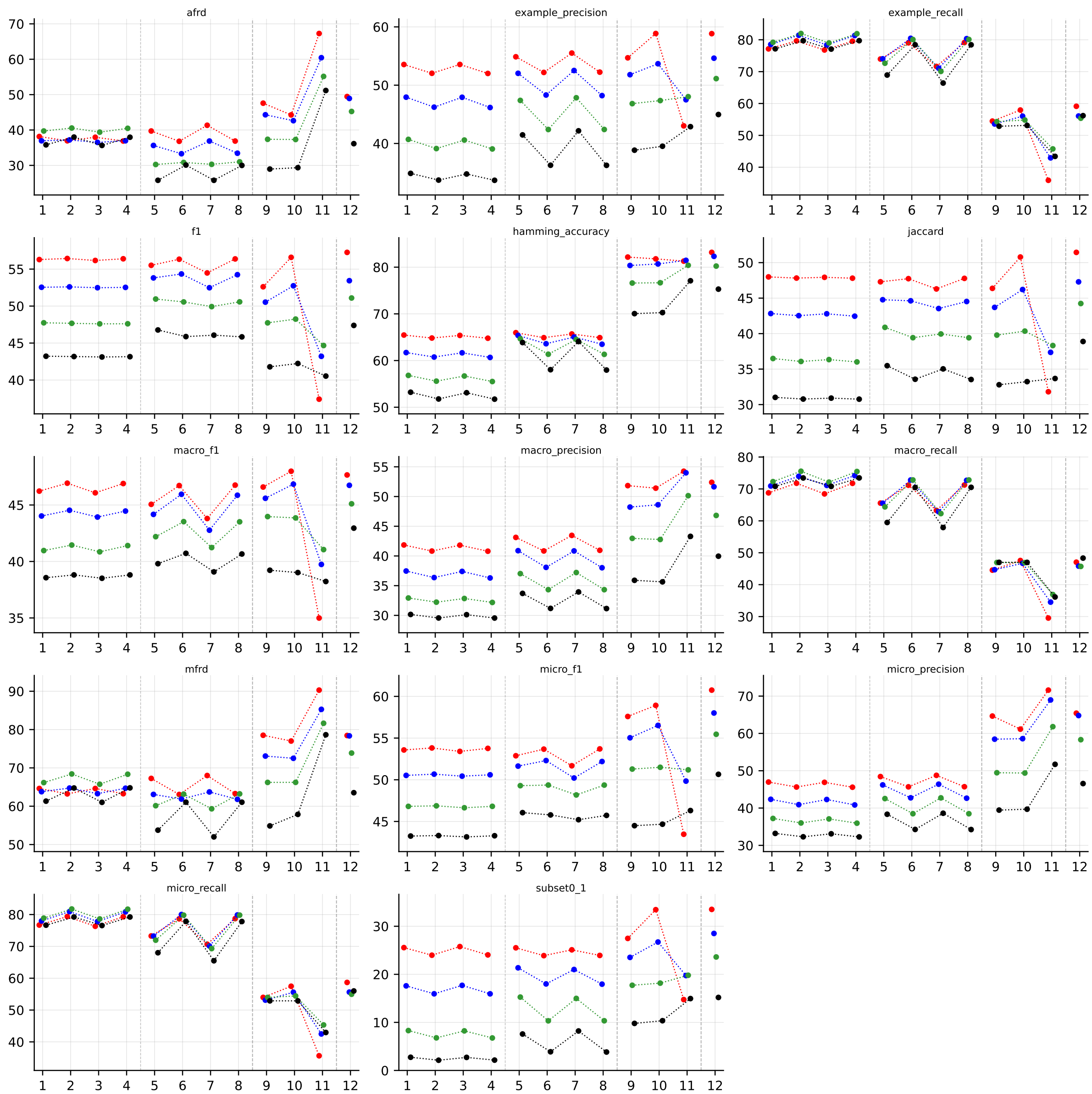


Table 47: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

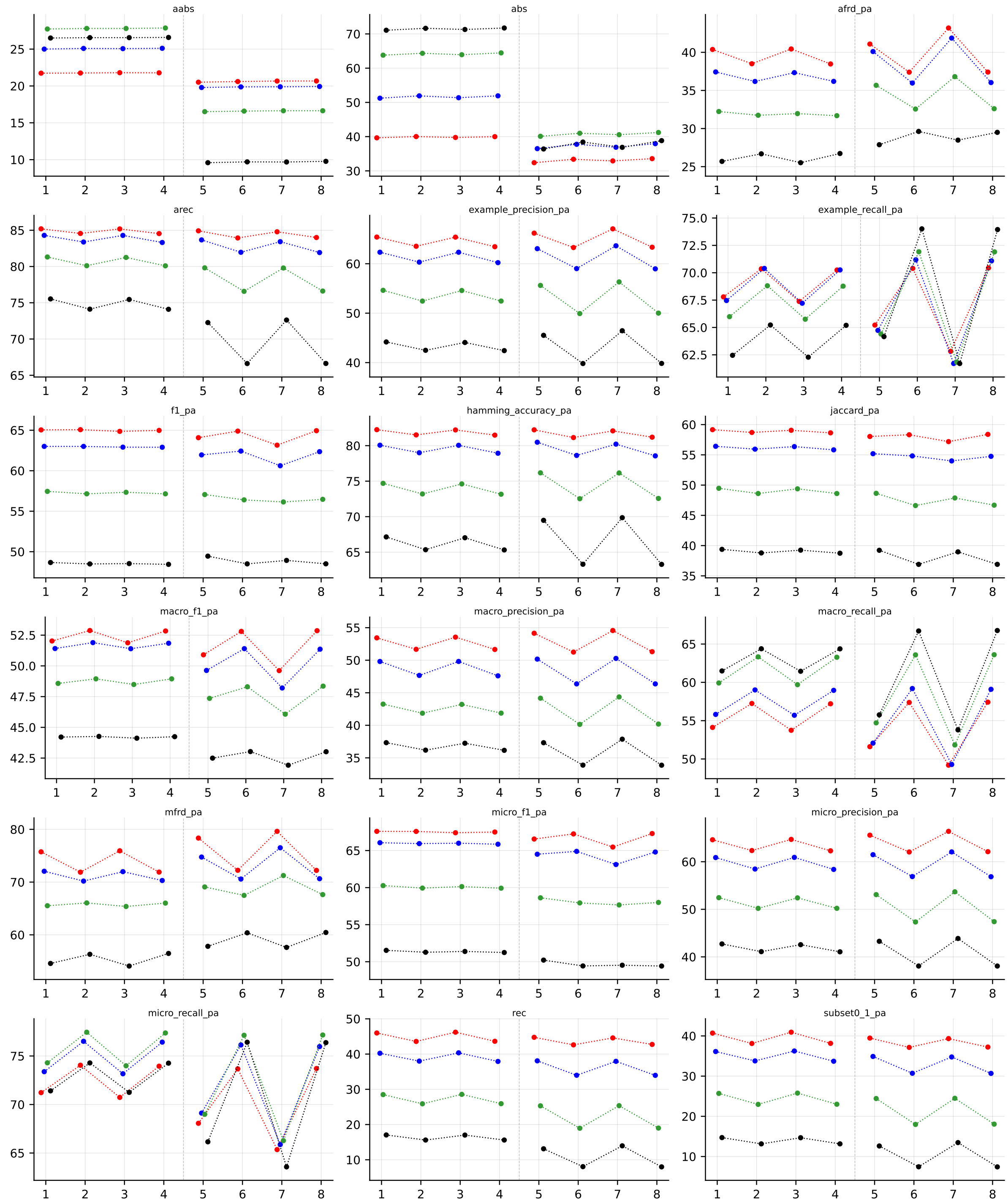


Table 48: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

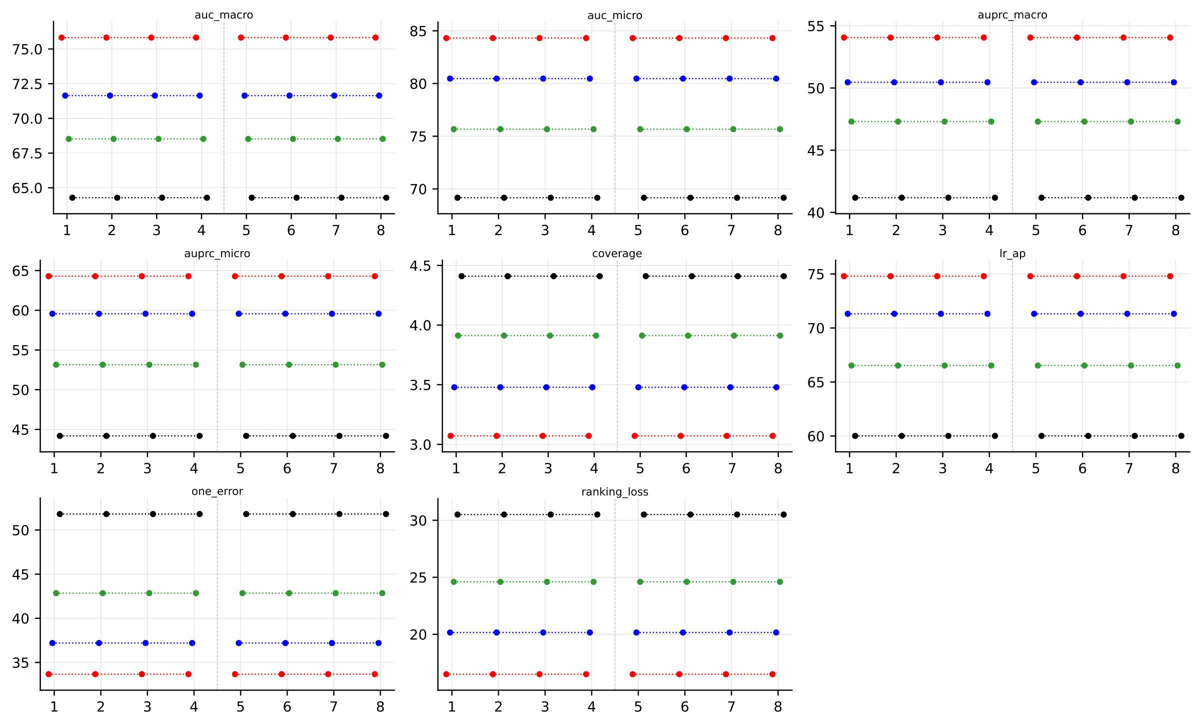


Table 49: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.